

电子电路与系统基础(A2)---动态电路

第7讲：有源滤波

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A班课程内容安排

第一学期：电阻	序号	第二学期：动态
电路定律	1	电容电感
电阻电源	2	一阶时域
电路定理	3	正弦稳态
方程列写	4	一阶时频
网络分析	5	二阶时域
二极管	6	二阶时频
晶体管	7	有源滤波
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期末复习	15	电路抽象

有源滤波 内容

■ 滤波器

- 有频率特性的LTI系统均可视为某种形式的滤波电路
- 狭义的滤波器是指那些符合特定中心频点、特定带宽、特定带内特性（如最大平坦特性，带内信号尽可能接近于无失真传输）、特定带外衰减（带外衰减越大越好）等特定要求的LTI系统

■ 无源滤波器

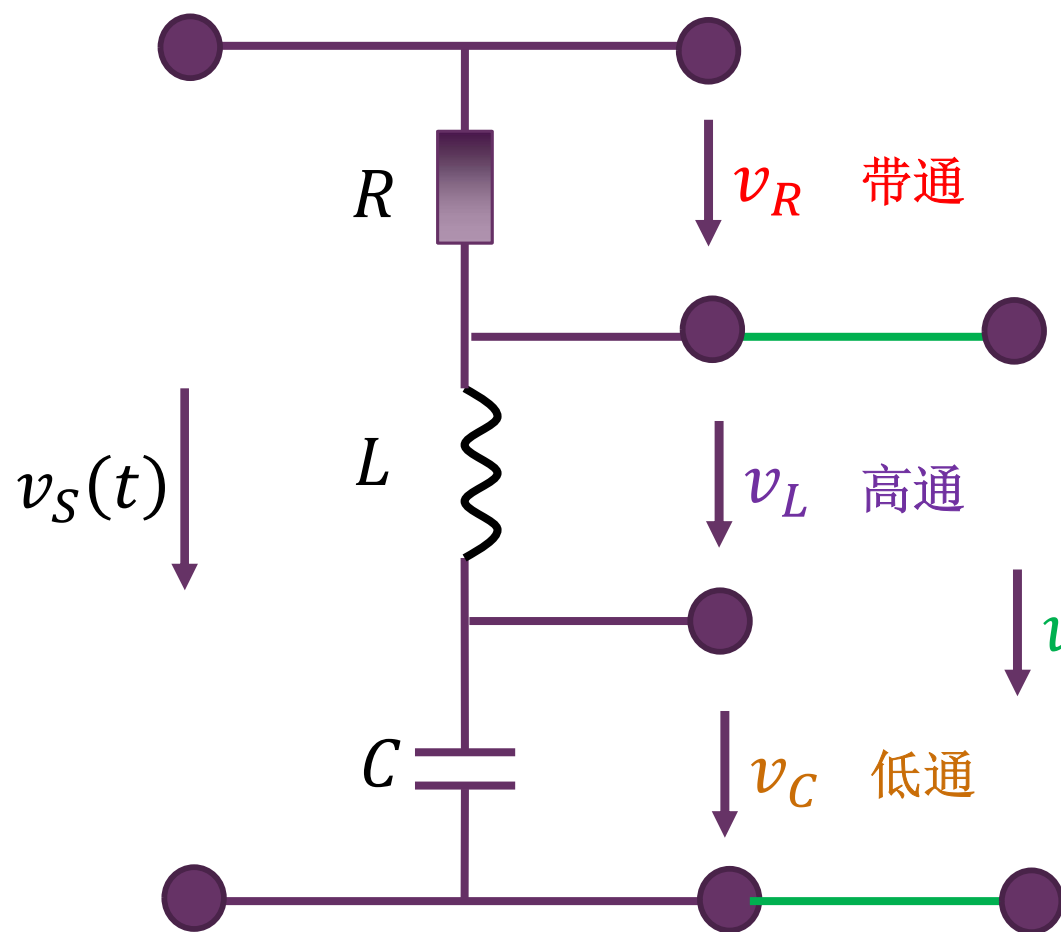
- 由无源RLC、金属波导谐振腔、介质谐振腔、石英晶体等构造的滤波器
 - 一阶RC、一阶RL低通、高通滤波器
 - 二阶RLC滤波器：低通、高通、带通、带阻
 - 高阶LC滤波器：设计流程规范成熟，可以实现各种逼近类型的滤波器，如幅度最大平坦的巴特沃思逼近滤波器，群延时最大平坦的贝塞尔逼近滤波器等，具体设计流程见《通信电路》《信号与系统》课程

■ 有源滤波器

- 有源网络（晶体管、运放、…）参与形成滤波功能的滤波器：本课程也只讲到一阶和二阶。高阶有源滤波器设计可以先设计出LC无源滤波器，再转换为有源滤波器
 - 有源RC滤波器（运放参与）
 - Gm-C滤波器（跨导放大器参与）
 - …对数滤波器（晶体管参与，见后续专业课或有讲解）…

一阶RC/RL滤波器√

串联RLC分压分析：典型的二阶滤波特性



电感高频开路，分担的是输入电压中的高频分量，以电感分压为输出则具高通滤波特性

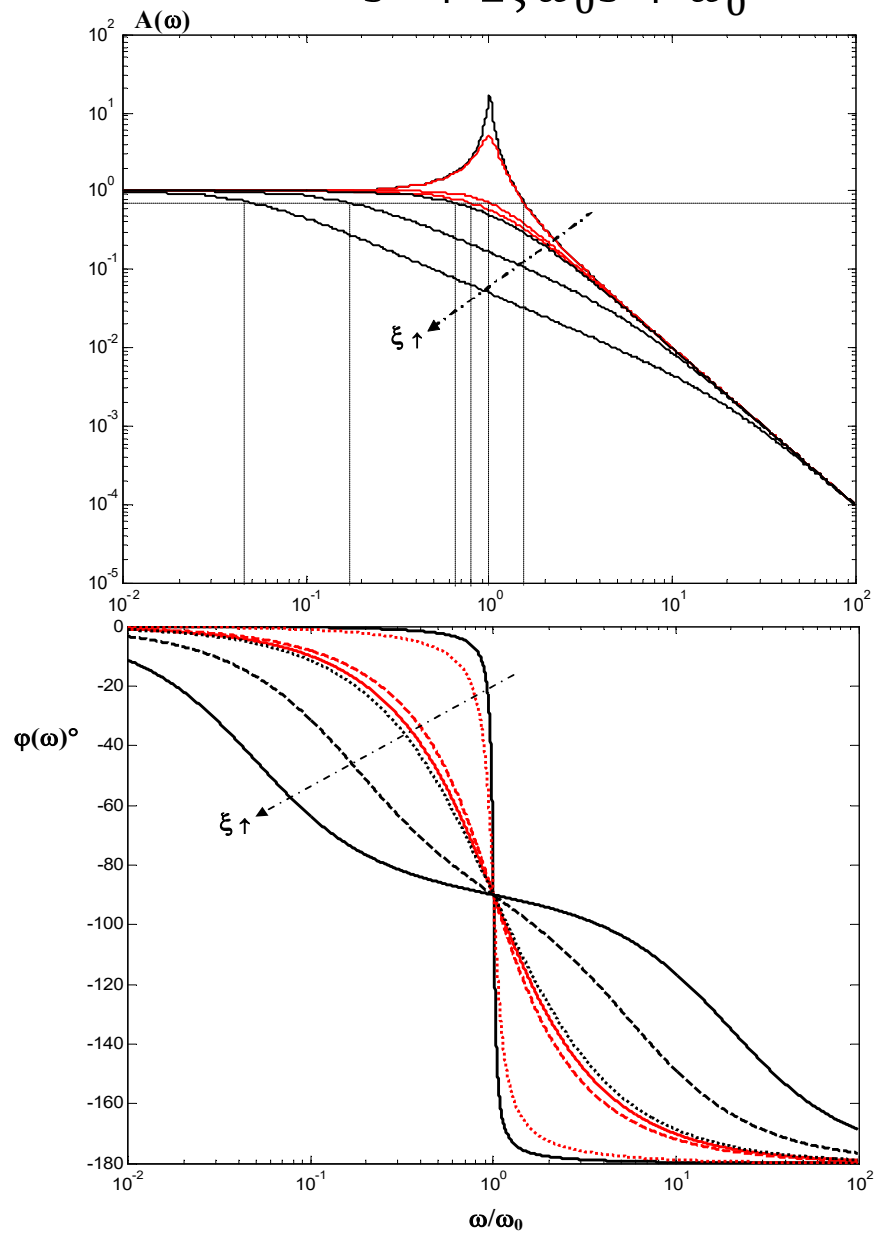
电容低频开路，分担的是输入电压中的低频分量，以电容分压为输出则具低通滤波特性

谐振频点串联**LC**短路，电阻分担的是输入电源中的中间频率分量，以电阻分压为输出则具带通滤波特性

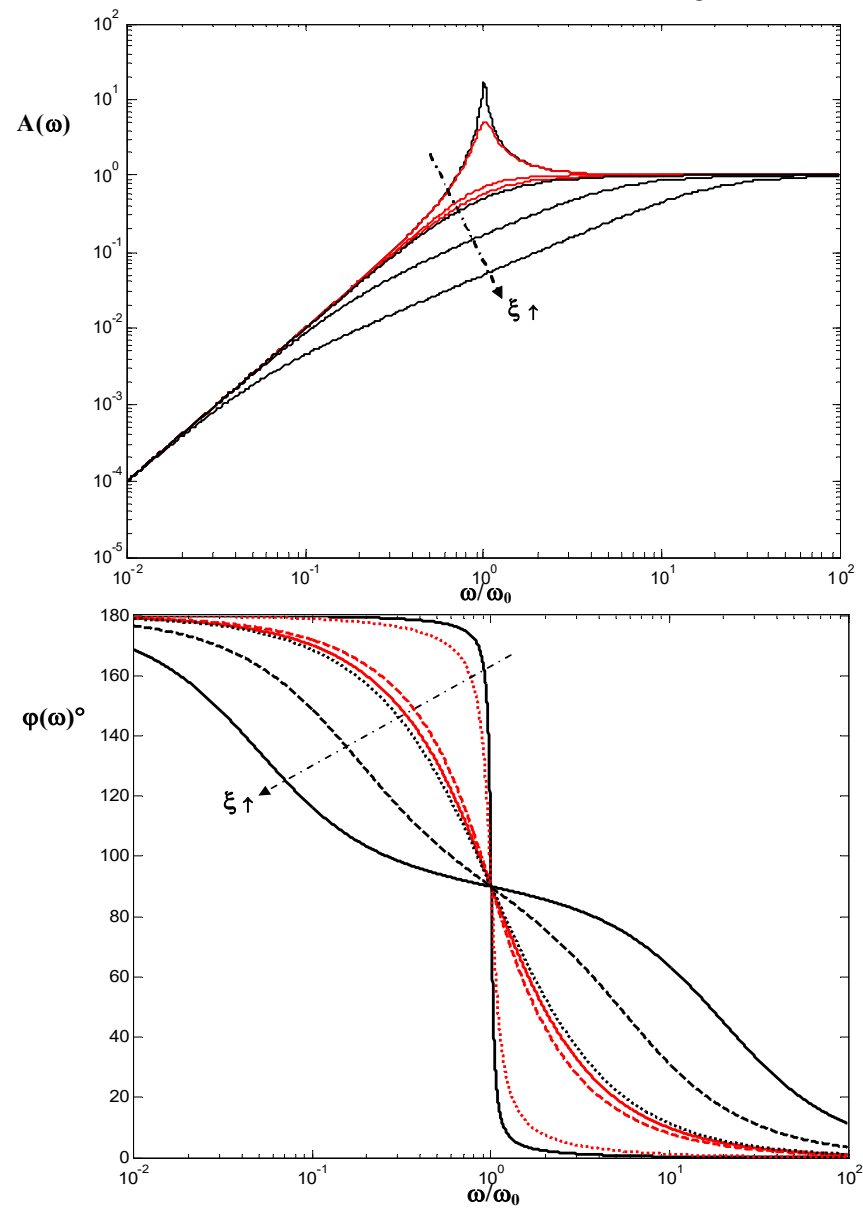
以串联**LC**分压为输出者，则具带阻滤波特性：谐振频点**LC**短路无输出，而低频电容开路、高频电感开路使得串联**LC**均获得全部分压

低通、高通特性回顾

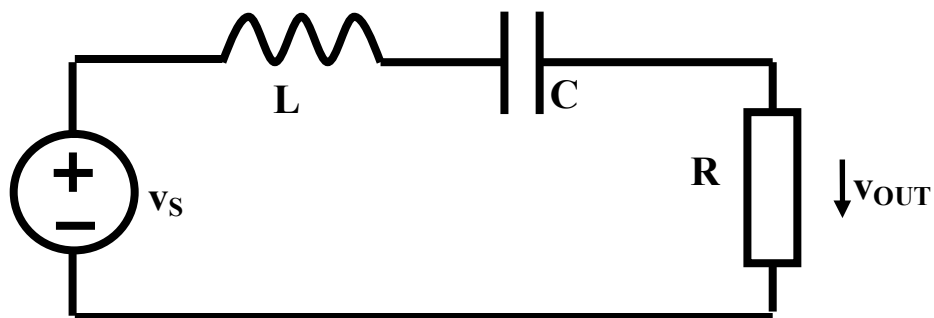
$$H(s) = \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$



$$H(s) = \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$



二阶带通：电阻分压



直观理解：

低频：电容开路，电感短路，信号通不过

高频：电容短路，电感开路，信号通不过

自由振荡频点：电容电抗 $-jZ_0$ 和电感电抗 $+jZ_0$ 抵消，犹如短路，信号通过

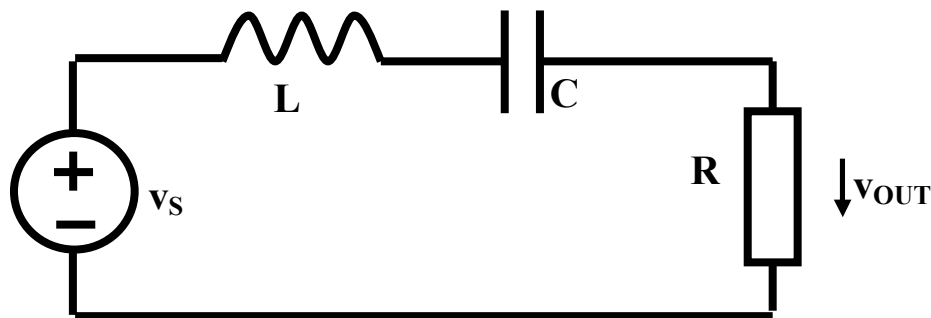
$$H(j\omega) = \frac{\dot{V}_R}{\dot{V}_S} = \frac{R}{R + j\omega L + \frac{1}{j\omega C}} = \frac{j\omega RC}{(j\omega)^2 LC + j\omega RC + 1}$$

$$\stackrel{j\omega \rightarrow s}{=} \frac{sRC}{s^2 LC + sRC + 1} = \frac{s \frac{R}{L}}{s^2 + s \frac{R}{L} + \frac{1}{LC}} = H_0 \frac{2\xi \omega_0 s}{s^2 + 2\xi \omega_0 s + \omega_0^2}$$

二阶带通传函典型形式

$$H_0 = 1 \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad \xi = \frac{R}{2Z_0} = \frac{R}{2} \sqrt{\frac{C}{L}}$$

考察带通习惯用Q值

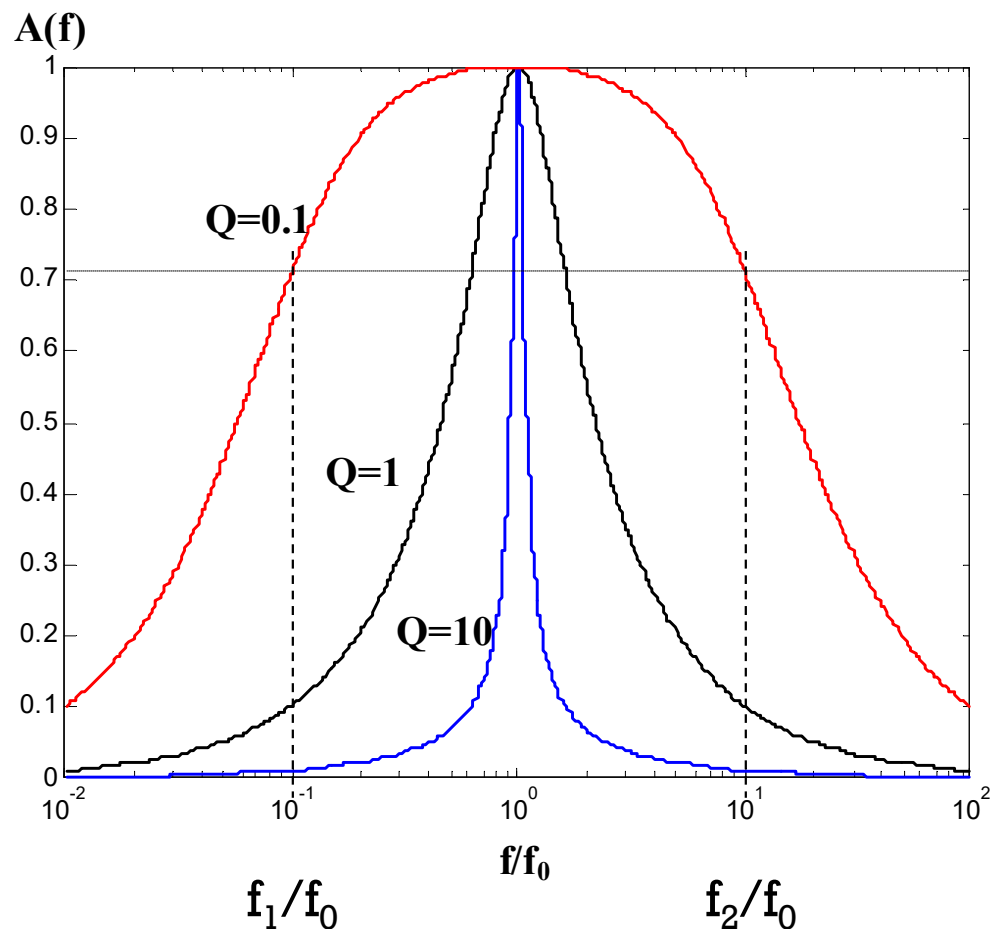


$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{Z_0}{R} = \frac{\omega_0 L}{R} = \frac{1}{2\xi}$$

品质因数：系统储能与耗能之比

$$\begin{aligned} H(j\omega) &= \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2} = \frac{\frac{\omega_0}{Q} s}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2} \stackrel{s=j\omega}{=} \frac{j \frac{\omega_0}{Q} \omega}{-\omega^2 + j \frac{\omega_0}{Q} \omega + \omega_0^2} \\ &= \frac{1}{1 + jQ \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)} = \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}} e^{-j \arctan Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)} = A(\omega) e^{j\varphi(\omega)} \end{aligned}$$

幅频特性：带通选频特性



$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$A(f) = \frac{1}{\sqrt{1 + Q^2 \left(\frac{f}{f_0} - \frac{f_0}{f} \right)^2}}$$

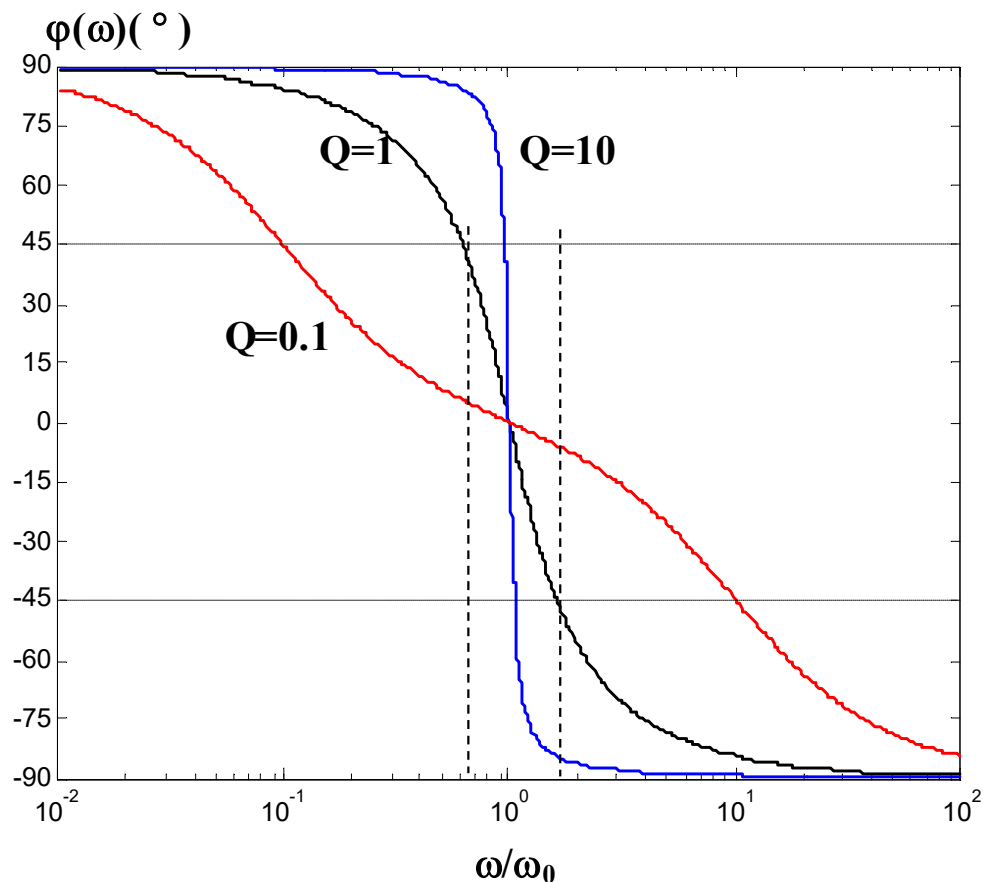
$$\text{3dB通频带: } Q \left(\frac{f}{f_0} - \frac{f_0}{f} \right) = \pm 1$$

$$\Rightarrow f_{1,2} = \dots$$

$$\Rightarrow BW_{3dB} = f_2 - f_1 = \frac{f_0}{Q}$$

$$f_0 = \sqrt{f_1 f_2}$$

相频特性



$$BW_{3dB} \cdot \tau_{g0} = \frac{f_0}{Q} \cdot \frac{2Q}{2\pi f_0} = \frac{1}{\pi} = 0.32$$

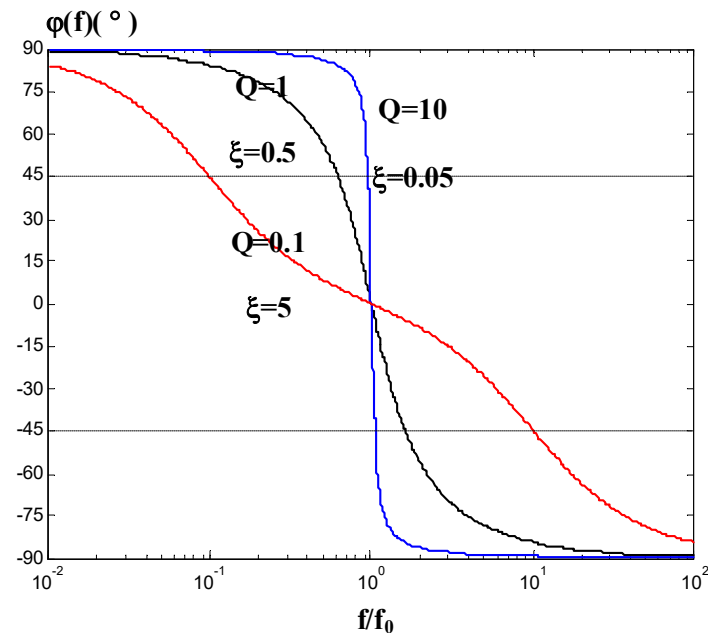
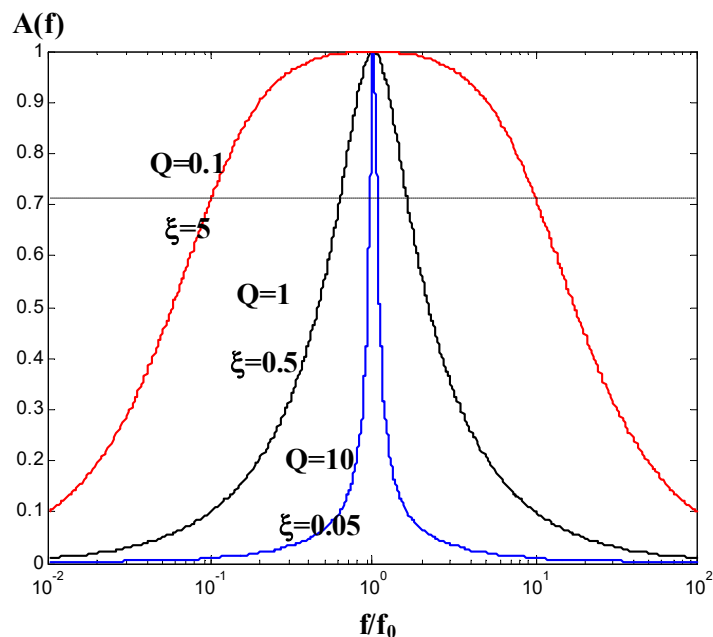
带宽越窄，信号延时越大：带通、低通均成立的结论

$$\varphi(\omega) = -\arctan Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

$$\tau_{g0} = - \left. \frac{d\varphi_i}{d\omega} \right|_{\omega=\omega_0} = \frac{2Q}{\omega_0}$$

$$\begin{aligned} \varphi(\omega) &\approx -Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \\ &= -Q \left(\frac{(\omega - \omega_0)(\omega + \omega_0)}{\omega_0 \omega} \right) \\ &\approx -2Q \frac{(\omega - \omega_0)}{\omega_0} = -(\omega - \omega_0) \tau_{g0} \end{aligned}$$

谐振频点的疑问？



$$H(j\omega) = \frac{\dot{V}_R}{\dot{V}_S} = A(\omega)e^{j\phi(\omega)}$$

$$v_S(t) = V_{Sm} \cos \omega t$$

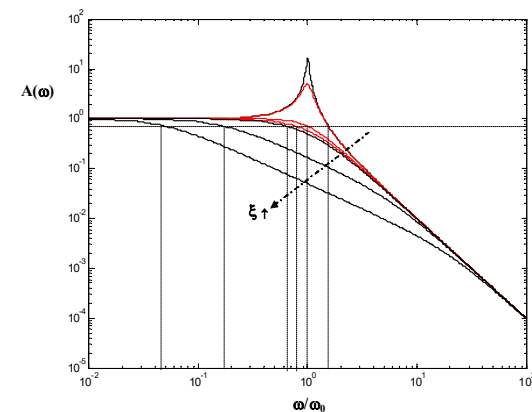
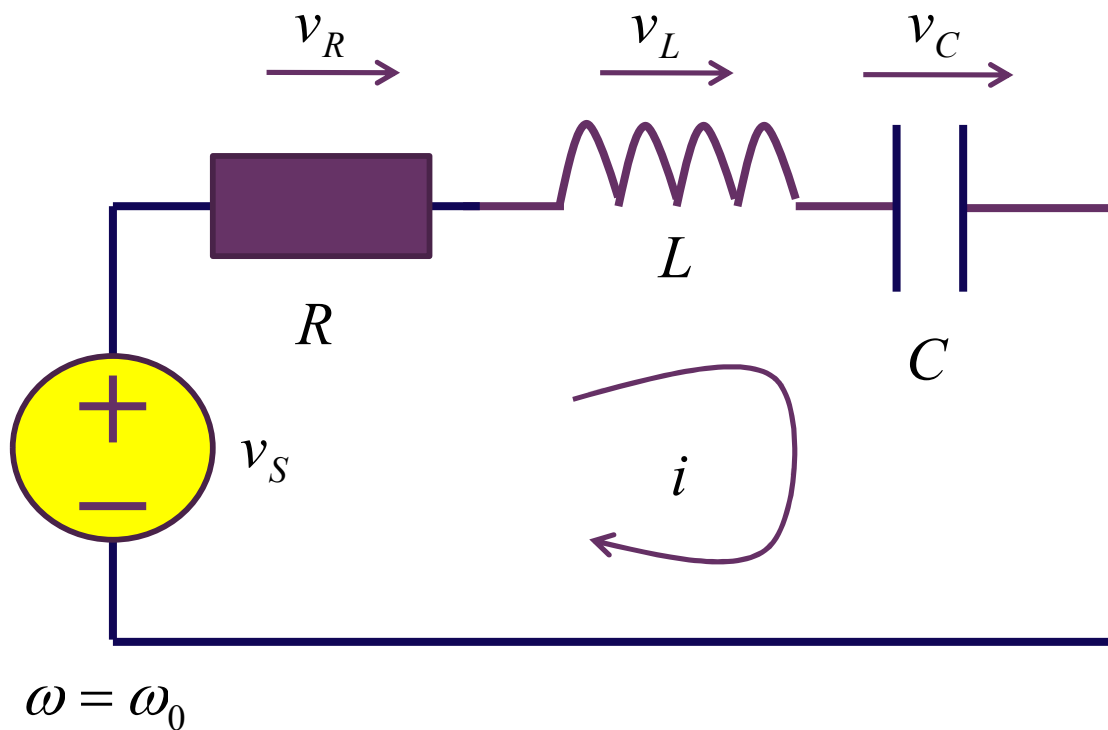
$$V_{R\infty}(t) = V_{Sm} A(\omega) \cos(\omega t + \phi(\omega))$$

$$v_S(t) = V_{Sm} \cos \omega_0 t$$

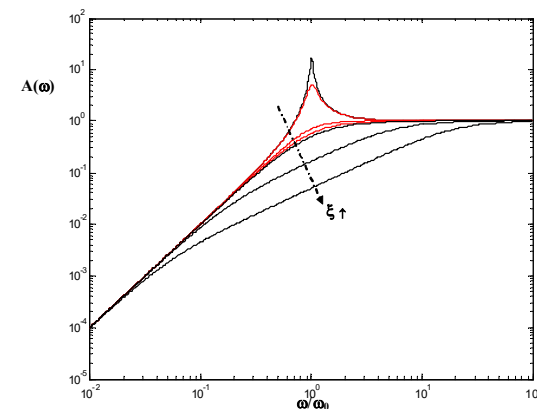
$$v_R(t) = V_{Sm} \cos \omega_0 t$$

显然：在谐振频点位置，电阻获得全部输入电压？？？
难道电容、电感没有分压？.....电阻电路的观点

串联谐振为电压谐振



$$A(\omega_0) = \frac{1}{2\xi} = Q$$



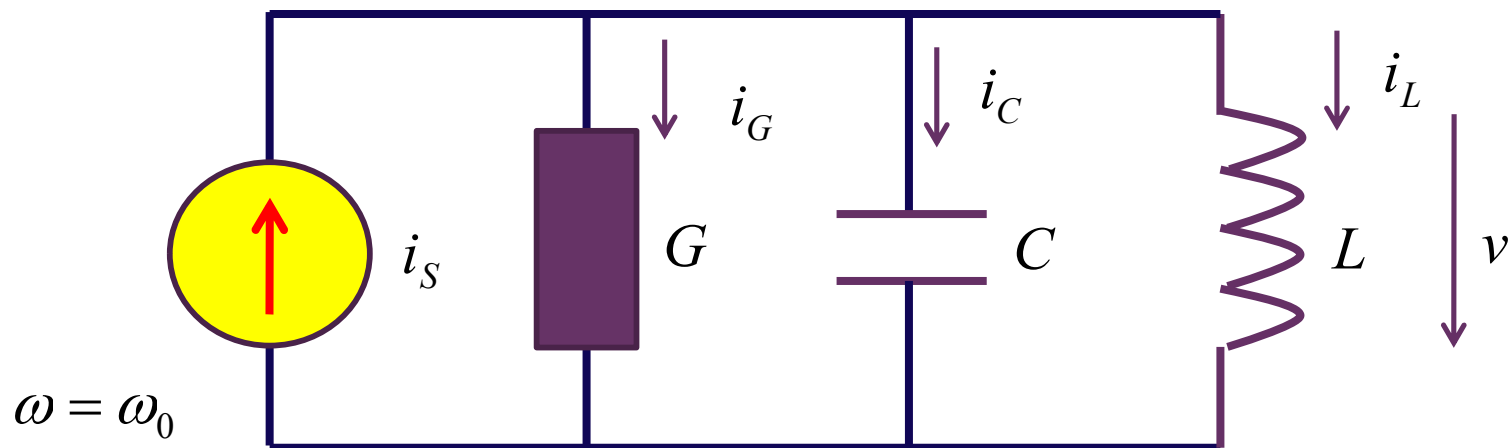
$$\dot{V}_R = \dot{V}_S$$

$$\dot{V}_L = j\omega_0 L \dot{I} = j \frac{\omega_0 L}{R} \dot{V}_S = j \frac{Z_0}{R} \dot{V}_S = jQ \dot{V}_S$$

$$\dot{I} = \frac{\dot{V}_R}{R} = \frac{\dot{V}_S}{R}$$

$$\dot{V}_C = \frac{\dot{I}}{j\omega_0 C} = -j \frac{1}{\omega_0 CR} \dot{V}_S = -j \frac{Z_0}{R} \dot{V}_S = -jQ \dot{V}_S$$

并联谐振为电流谐振



$$\dot{I}_G = \dot{I}_S$$

$$\dot{I}_C = j\omega_0 C \dot{V} = j \frac{\omega_0 C}{G} \dot{I}_S = j \frac{Y_0}{G} \dot{I}_S = jQ \dot{I}_S$$

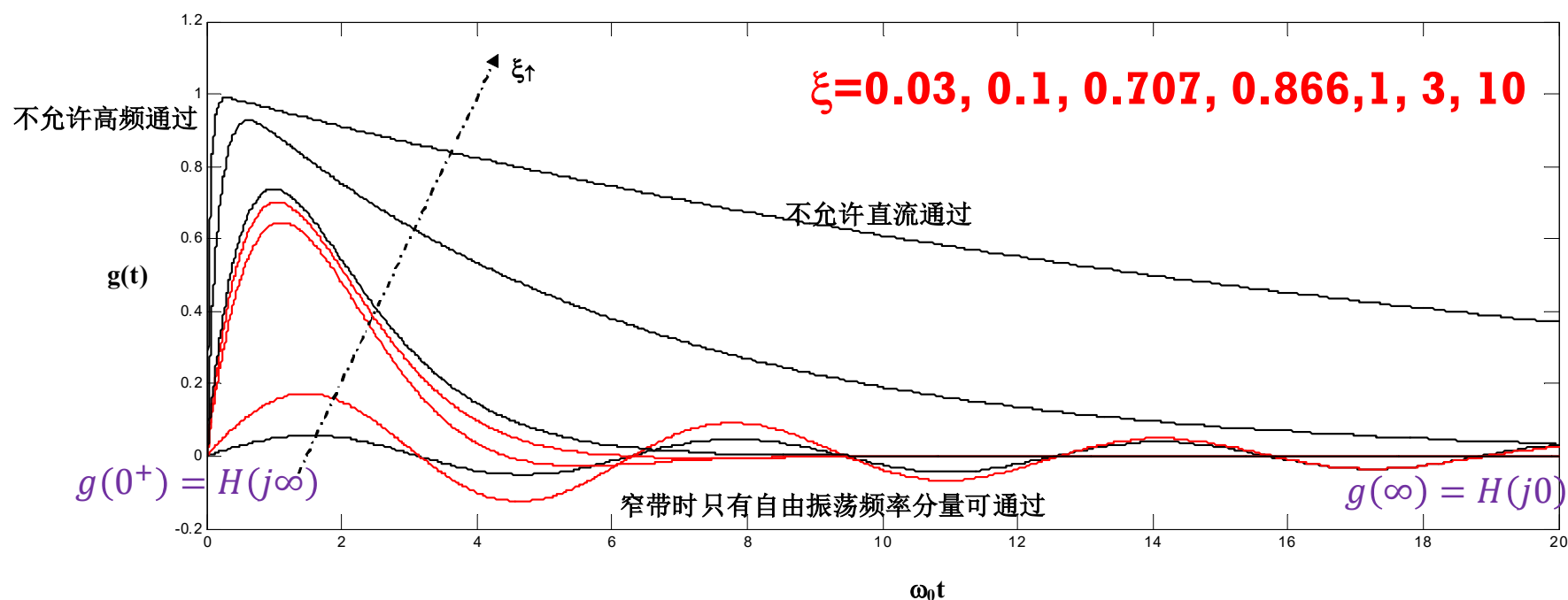
$$\dot{V} = \frac{\dot{I}_G}{G} = \frac{\dot{I}_S}{G}$$

$$\dot{I}_L = \frac{\dot{V}}{j\omega_0 L} = -j \frac{1}{\omega_0 L G} \dot{I}_S = -j \frac{Y_0}{G} \dot{I}_S = -jQ \dot{I}_S$$

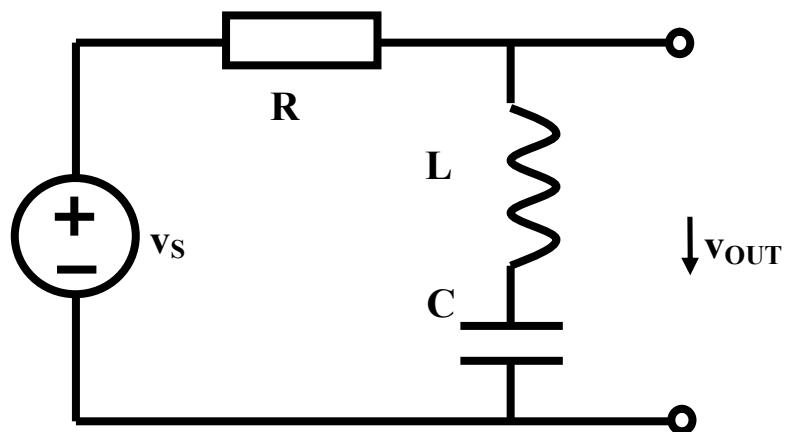
冲激响应和阶跃响应

$$h(t) = 2\xi\omega_0 e^{-\xi\omega_0 t} \left(\cos \sqrt{1-\xi^2} \omega_0 t - \frac{\xi}{\sqrt{1-\xi^2}} \sin \sqrt{1-\xi^2} \omega_0 t \right) \cdot U(t)$$

$$g(t) = \frac{2\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$



二阶带阻：LC分压



直观理解：

低频：电容开路，电感短路，信号全过

高频：电容短路，电感开路，信号全过

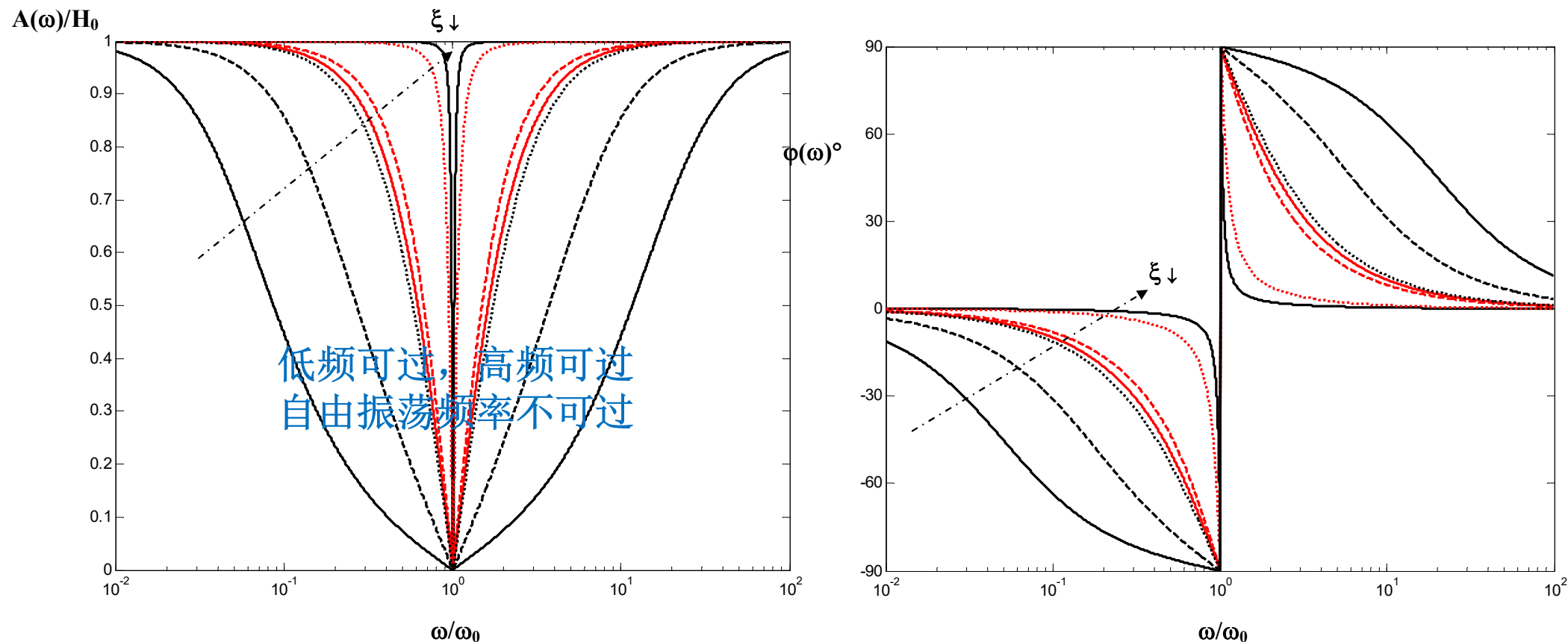
自由振荡频点：电容电抗和电感电抗抵消，犹如短路，输出为**0**，信号通不过

$$H(j\omega) = \frac{\dot{V}_L + \dot{V}_C}{\dot{V}_S} = H_0 \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} \Big|_{s=j\omega}$$

二阶带阻滤波器传函的一般形式

$$A(\omega) = H_0 \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\xi\omega_0\omega)^2}} \quad \varphi(\omega) = \begin{cases} -\arctan \frac{2\xi\omega_0\omega}{\omega_0^2 - \omega^2} & \omega < \omega_0 \\ \pi - \arctan \frac{2\xi\omega_0\omega}{\omega_0^2 - \omega^2} & \omega > \omega_0 \end{cases}$$

幅频特性、相频特性



在谐振频点，信号通不过，传函为**0**，该位置出现相位**180°**跳变

只有传函为**0**的点允许相位出现**180°**跳变，其他位置相位均连续

时域特性

$$H_{BS}(s)\Big|_{s=j\omega} = \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} + \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = H_{HP}(s) + H_{LP}(s)$$

电感分压 电容分压

$$h_{HP}(t) = \delta(t) + \omega_0 e^{-\xi\omega_0 t} \left[-2\xi \cos \sqrt{1-\xi^2} \omega_0 t + \frac{2\xi^2 - 1}{\sqrt{1-\xi^2}} \sin \left(\sqrt{1-\xi^2} \omega_0 t \right) \right] \cdot U(t) \quad \text{电感分压}$$

$$h_{LP}(t) = \frac{\omega_0}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \left(\sqrt{1-\xi^2} \omega_0 t \right) \cdot U(t) \quad \text{电容分压}$$

$$h_{BS}(t) = \delta(t) - 2\xi\omega_0 e^{-\xi\omega_0 t} \left[\cos \sqrt{1-\xi^2} \omega_0 t - \frac{\xi}{\sqrt{1-\xi^2}} \sin \left(\sqrt{1-\xi^2} \omega_0 t \right) \right] \cdot U(t) = \delta(t) - h_{BP}(t)$$

电感分压+电容分压=电源电压-电阻分压

$$H_{BS}(s)\Big|_{s=j\omega} = \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = 1 - \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2} = 1 - H_{BP}(s)$$

说明什么？可以直接利用时域冲激响应和频域传递函数的一一对应关系！

寻找对应关系：一阶RC

一阶**RC**电路：电容分压，低通

$$H_{LP,1}(s) = \frac{1}{1+s\tau} = \frac{\omega_0}{s+\omega_0}$$

$$h_{LP,1}(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}} \cdot U(t) = \omega_0 e^{-\omega_0 t} \cdot U(t)$$

$$\frac{\omega_0}{s+\omega_0} \Leftrightarrow \omega_0 e^{-\omega_0 t} \cdot U(t)$$

无论**RC**、**RL**，还是其他物理系统，只要是一阶低通，必有如是时频对应关系

$$\frac{1}{s+\omega_0} \Leftrightarrow e^{-\omega_0 t} \cdot U(t)$$

一阶**RC**电路：电阻分压，高通

$$H_{HP,1}(s) = \frac{s\tau}{1+s\tau} = \frac{s}{s+\omega_0} = 1 - \frac{\omega_0}{s+\omega_0}$$

$$h_{HP,1}(t) = \delta(t) - \frac{1}{\tau} e^{-\frac{t}{\tau}} \cdot U(t) = \delta(t) - \omega_0 e^{-\omega_0 t} \cdot U(t)$$

无需三要素，直接用时频对应关系即可
可以免除考察实际电路，只需知道输入到输出的传递函数即可

$$1 \Leftrightarrow \delta(t)$$

直通电路

寻找时频关系：二阶RLC

电容分压，二阶低通

$$H_{LP,2}(s) = \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$h_{LP,2}(t) = \frac{\omega_0}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

无需五要素，直接用时频对应关系即可
可以免除考察实际二阶电路，只需知道
输入到输出的传递函数即可，只要传递
函数一样，内部电路形式不重要

$$\begin{array}{c} \frac{\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} \\ \Downarrow \\ \frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t) \end{array}$$

寻找时频关系：二阶高通

电感分压，二阶高通

$$h_{HP,2}(t) = \underline{\underline{\delta(t)}} - 2\xi\omega_0 \underbrace{e^{-\xi\omega_0 t} \cos \sqrt{1-\xi^2} \omega_0 t \cdot U(t)}_{\leftarrow} - (1-2\xi^2)\omega_0 \underbrace{\frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)}_{\leftarrow}$$

$$H_{HP,2}(s) = \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = 1 - \frac{2\xi\omega_0 s + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$= \underline{\underline{1}} - 2\xi\omega_0 \underbrace{\frac{s + \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}}_{\leftarrow} - (1-2\xi^2)\omega_0 \underbrace{\frac{\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}}_{\leftarrow}$$

$$\frac{s + \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} \Leftrightarrow e^{-\xi\omega_0 t} \cos \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

时频对应关系验证：二阶带通

电阻分压，二阶带通

$$H_{BP,2}(s) = \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2} = 2\xi\omega_0 \frac{s + \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} - 2\xi^2\omega_0 \frac{\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$h_{BP,2}(t) = 2\xi\omega_0 e^{-\xi\omega_0 t} \cos \sqrt{1-\xi^2} \omega_0 t \cdot U(t) - 2\xi^2\omega_0 \frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

$$\frac{s + \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} \Leftrightarrow e^{-\xi\omega_0 t} \cos \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

$$\frac{\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} \Leftrightarrow \frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

时频对应表

$h(t)$	$H(s)$	
$\delta(t)$	1	直通
$U(t)$	$\frac{1}{s}$	
$e^{-\omega_0 t} \cdot U(t)$	$\frac{1}{s + \omega_0}$	负实根
$t^n e^{-\omega_0 t} \cdot U(t)$	$\frac{n!}{(s + \omega_0)^{n+1}}$	负实重根
$e^{-\xi \omega_0 t} \cos \sqrt{1 - \xi^2} \omega_0 t \cdot U(t)$	$\frac{s + \xi \omega_0}{s^2 + 2\xi \omega_0 s + \omega_0^2}$	共轭复根
$e^{-at} \cos bt \cdot U(t)$	$\frac{s + a}{(s + a)^2 + b^2}$	
$\frac{1}{\sqrt{1 - \xi^2}} e^{-\xi \omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t \cdot U(t)$	$\frac{\omega_0}{s^2 + 2\xi \omega_0 s + \omega_0^2}$	
$e^{-at} \sin bt \cdot U(t)$	$\frac{b}{(s + a)^2 + b^2}$	

一阶低通
$$h_{LP,1}(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}} \cdot U(t) = \omega_0 e^{-\omega_0 t} \cdot U(t)$$

$$g_{LP,1}(t) = \int_{-\infty}^t h_{LP,1}(t) dt = \dots = \left(1 - e^{-\frac{t}{\tau}}\right) \cdot U(t)$$

频域积分运算为 **1/s**，频域分解后，对应到时域

$$G_{LP,1}(s) = \frac{1}{s} H_{LP,1}(s) = \frac{1}{s} \cdot \frac{1}{1 + s\tau} = \frac{1}{s} \cdot \frac{\omega_0}{s + \omega_0} = \frac{1}{s} - \frac{1}{s + \omega_0}$$

积分运算

$$g_{LP,1}(t) = U(t) - e^{-\omega_0 t} \cdot U(t) = \left(1 - e^{-\frac{t}{\tau}}\right) \cdot U(t)$$

一阶高通
$$G_{HP,1}(s) = \frac{1}{s} H_{HP,1}(s) = \frac{1}{s} \cdot \frac{s\tau}{1 + s\tau} = \frac{1}{s + \omega_0}$$

$$\frac{1}{s} \Leftrightarrow U(t)$$

$$g_{HP,1}(t) = e^{-\omega_0 t} \cdot U(t) = e^{-\frac{t}{\tau}} \cdot U(t)$$

信号时频对应关系

阶跃响应

二阶低通

$$G_{LP,2}(s) = \frac{1}{s} H_{LP,2}(s) = \frac{1}{s} \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = \frac{1}{s} - \frac{s + 2\xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$= \frac{1}{s} - \frac{s + \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} - \xi \frac{\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$g_{LP,2}(t) = \left(1 - e^{-\xi\omega_0 t} \left(\cos \sqrt{1-\xi^2} \omega_0 t + \frac{\xi}{\sqrt{1-\xi^2}} \sin \sqrt{1-\xi^2} \omega_0 t \right) \right) \cdot U(t)$$

二阶带阻

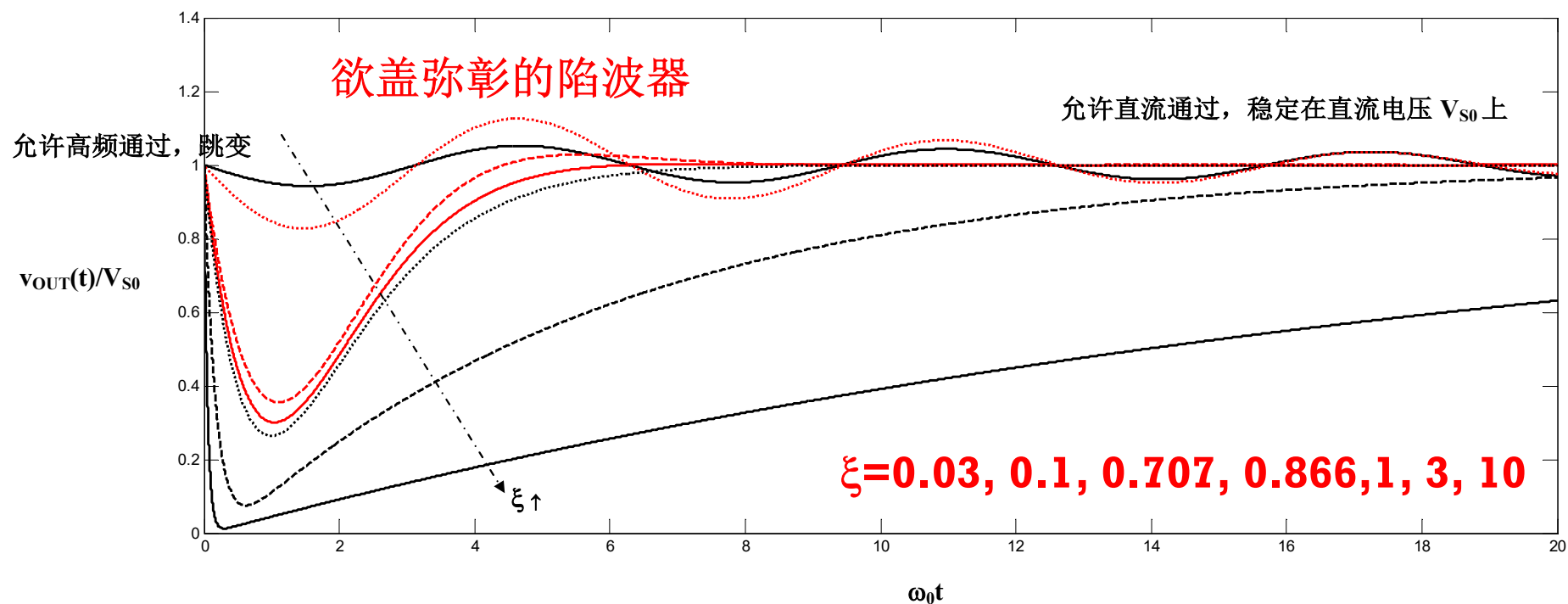
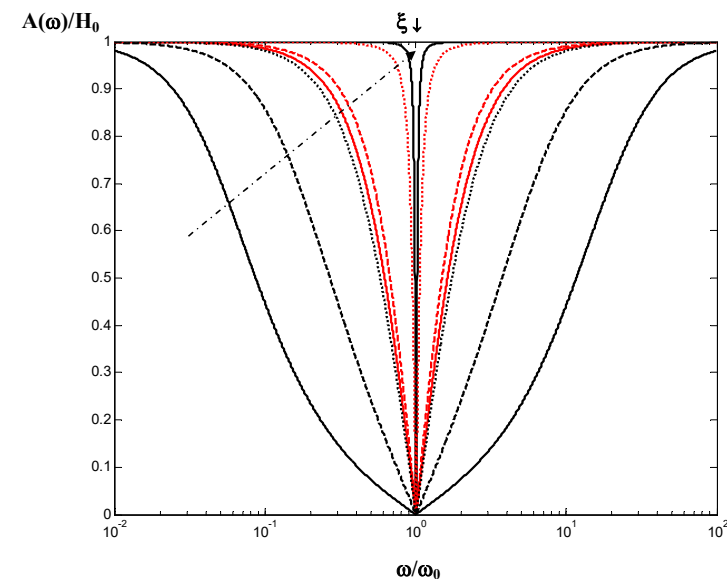
$$G_{BS,2}(s) = \frac{1}{s} H_{BS,2}(s) = \frac{1}{s} \cdot \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = \frac{1}{s} - 2\xi \frac{\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$g_{BS,2}(t) = U(t) - \frac{2\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

无需五要素法，直接根据时频对应关系给冲激响应和阶跃响应
和真实系统的实际构成无关，仅和端口传递特性有关
端口抽象后将电路分析转化为信号的数学分析

二阶带阻阶跃响应

$$g_{BS,2}(t) = U(t) - \frac{2\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$



有源滤波

（只讲大原理，实际电路操作细节见后续专业课）

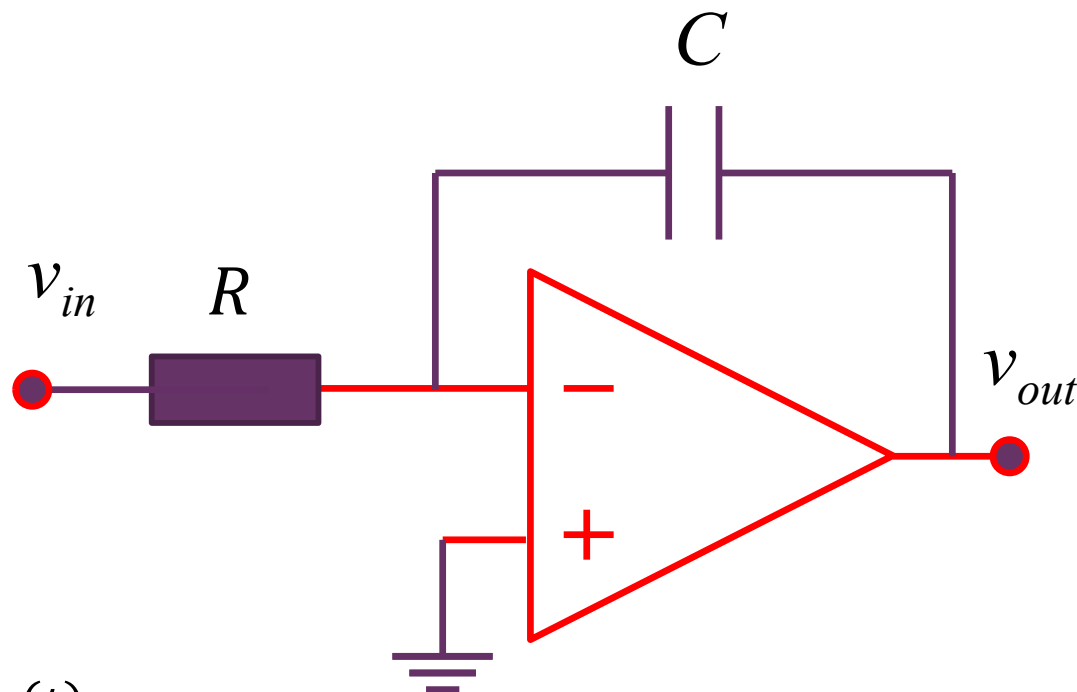
- 集成滤波器设计中电感难以实现，多用有源滤波器实现滤波功能
 - 电感替代法：先设计LC滤波器，之后
 - 用有源积分器（有源器件+电容）替代电感的积分功能
 - 用回旋器将电容回旋等效为电感
 - 将高阶滤波器传递函数分解为一阶、二阶滤波器传递函数之积，只需设计一阶有源滤波器和二阶有源滤波器（单向网络），之后级联即可实现高阶滤波功能（单向网络的传递函数等于级联单向网络分传递函数之积）
 - 一阶、二阶有源RC滤波器
 - 其他有源滤波器（具体电路实现见后续专业课或研究生课程）
 - Gm-C滤波器、电流型对数滤波器、...

RC理想积分电路

$$\frac{V_{in}(s) - 0}{R} = sC(0 - V_{out}(s))$$

$$V_{out}(s) = -\frac{V_{in}(s)}{sRC}$$

$$v_{out}(t) = -\frac{1}{RC} \int v_{in}(t) dt = -\frac{1}{C} \int \frac{v_{in}(t)}{R} dt$$



只要运放是理想运放，就可以实现理想积分，就可用于替代电感的积分功能

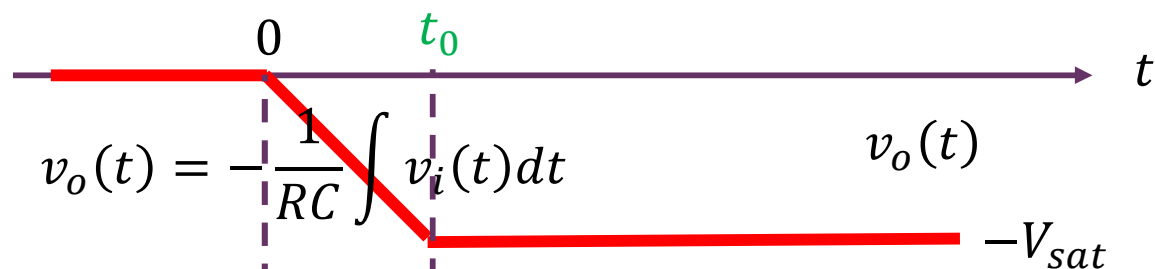
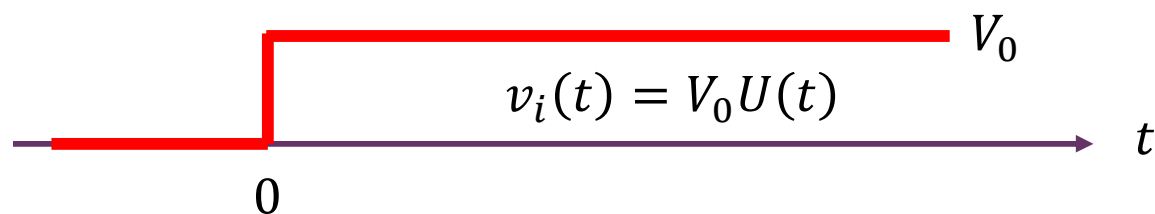
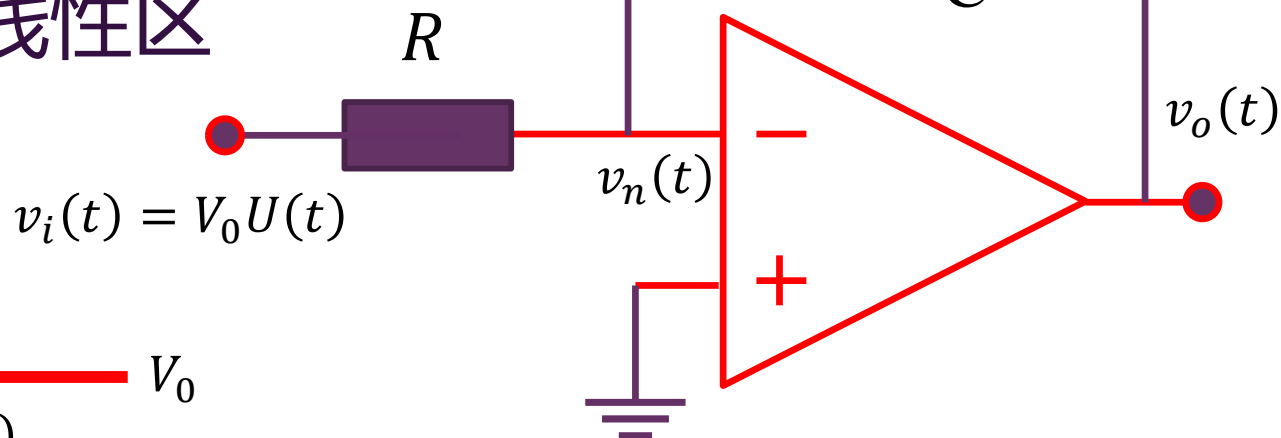
$$i_L(t) = \frac{1}{L} \int v_L(t) dt$$

将电路方程中的电感电流乘以一个电阻量纲的系数即可转化为电压，之后用**RC**理想积分电路实现电压积分即可：用电容积分功能替代电感积分功能

$$v_{out}(t) = Ri_L(t) = \frac{R}{L} \int v_L(t) dt = \frac{1}{R \frac{L}{R^2}} \int v_{in}(t) dt = \frac{1}{RC} \int v_{in}(t) dt$$

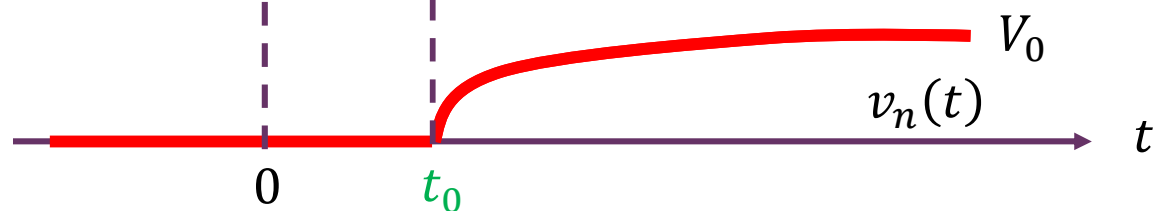
如何用电压替代电流，后续专业课讨论，...（**LC**滤波器可以用有源**RC**滤波器替代）

有源积分电路的限制 运放必须工作在线性区 (作业2.4)



只要运放输出电压在正负饱和电压之间，实现的就是理想积分功能

运放进入饱和区，虚短特性不再，积分功能消失

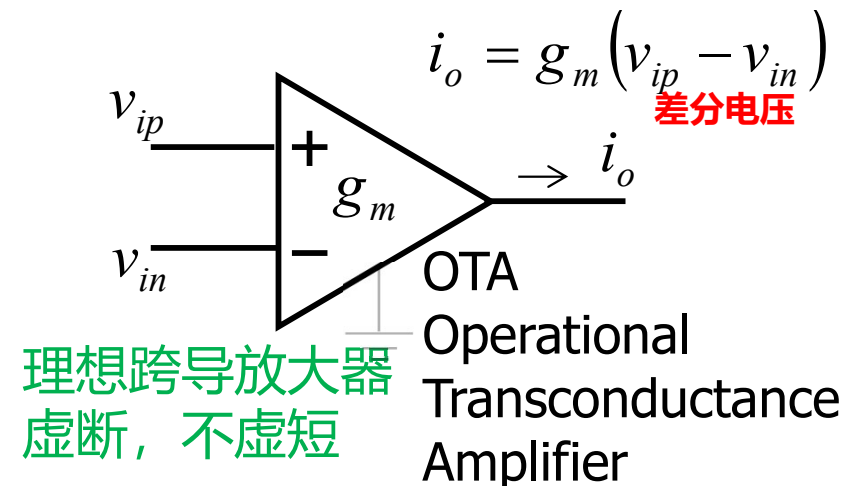
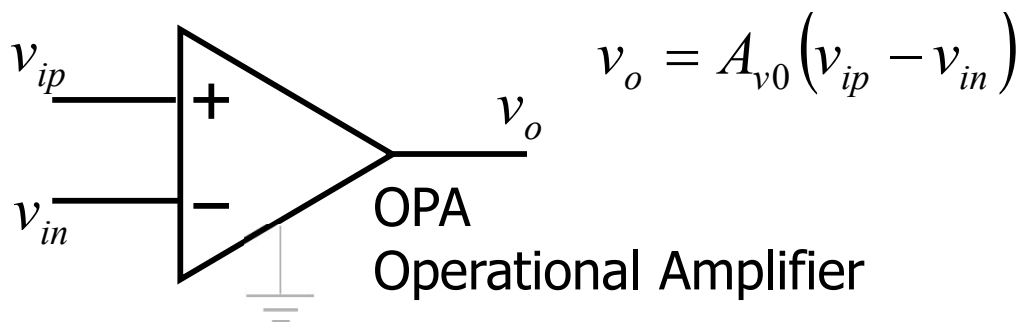


$$v_o(t) = \begin{cases} 0 & t < 0 \\ -V_0 \frac{t}{RC} & 0 < t < t_0 \\ -V_{sat} & t > t_0 \end{cases}$$

$$t_0 = \frac{V_{sat}}{V_0} RC$$

$$v_n(t) = \begin{cases} 0 & t < t_0 \\ V_0 (1 - e^{-\frac{t-t_0}{RC}}) & t > t_0 \end{cases}$$

有源积分用有源器件：除了OPA外 还有OTA最常用



线性区工作：电压放大器

理想压控压源
理想压控流源

理想运放线性区模型

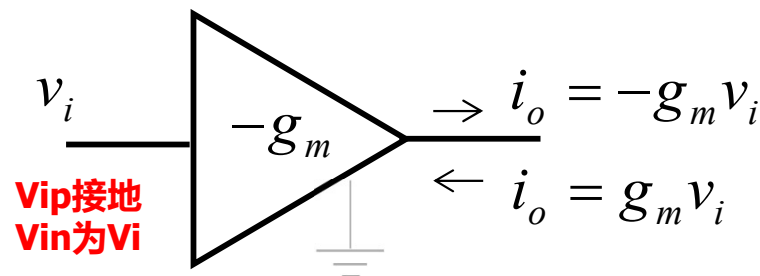
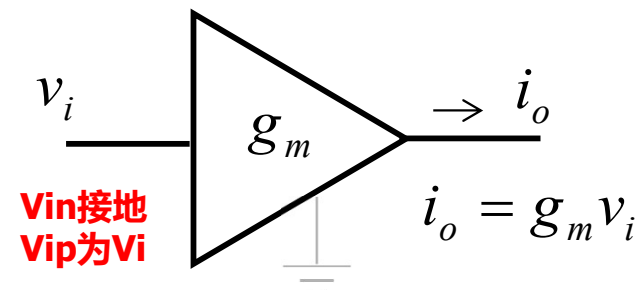
$$A_{v0} \rightarrow \infty \quad v_{ip} = v_{in}$$

虚短

$$A_{v0}R_{id} \rightarrow \infty \quad i_{ip} = i_{in} = 0$$

虚断

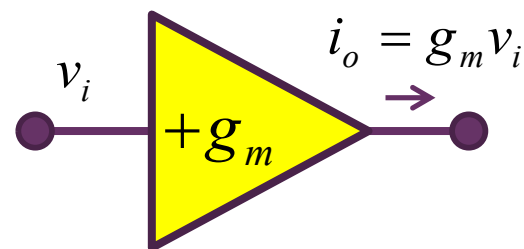
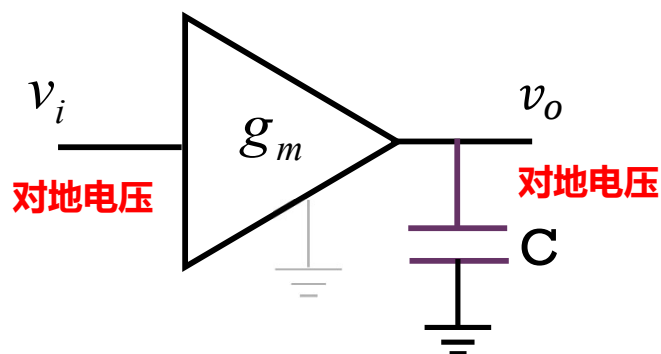
采用负反馈闭环结构使其工作在线性区



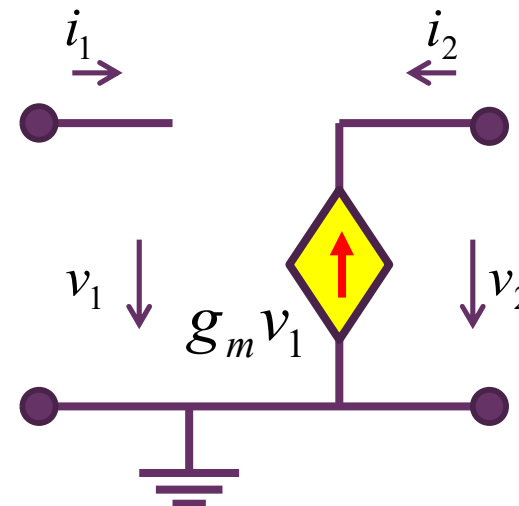
开环结构：高频低功耗应用

默认地可能不画，但存在公共地和输入输出端点构成输入输出端口

Gm-C积分器



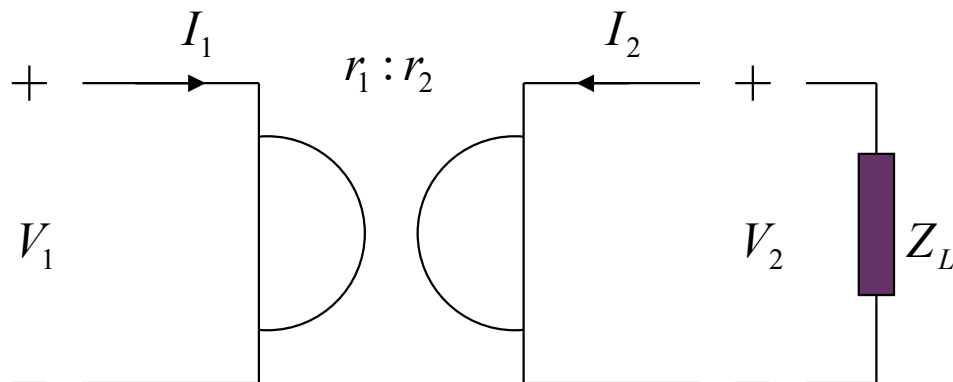
有默认
地端点
作为端
口的一个端点



$$v_o(t) = v_c(t) = \frac{1}{C} \int i_c(t) dt = \frac{1}{C} \int g_m v_i(t) dt = \frac{g_m}{C} \int v_i(t) dt$$

能够实现理想积分，就可实现理想电感的理想积分功能：在积分功能上替代电感的**Gm-C**滤波器、有源**RC**滤波器设计，见后续专业课讨论

将电容回旋成电感



$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & -r_{m1} \\ r_{m2} & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & g_{m2} \\ -g_{m1} & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$r_{m2} = r_{m1} = r_m$: 理想回旋器无损无源

$$\begin{aligned} p_{\Sigma} &= v_1 i_1 + v_2 i_2 \\ &= -r_{m1} i_2 i_1 + r_{m2} i_1 i_2 \stackrel{r_{m1}=r_{m2}}{=} 0 \end{aligned}$$

$$Z_{in} = \frac{\dot{V}_1}{\dot{I}_1} = \frac{-r_{m1} \dot{I}_2}{\dot{V}_2 / r_{m2}} = r_{m1} r_{m2} \frac{1}{-\dot{V}_2 / \dot{I}_2} = \frac{r_{m1} r_{m2}}{Z_L}$$

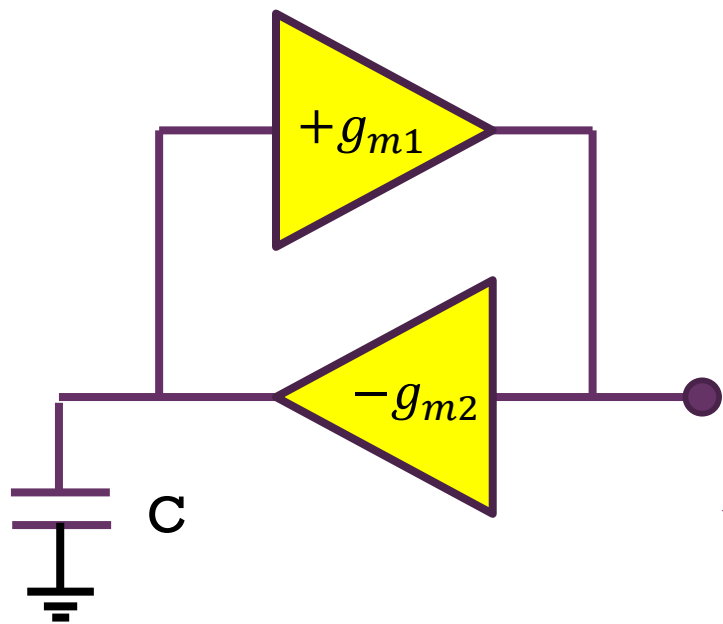
$$Z_L = \frac{1}{j\omega C} \Rightarrow Z_{in} = j\omega r_{m1} r_{m2} C = j\omega L_{eq}$$

$$\Rightarrow L_{eq} = r_{m1} r_{m2} C$$

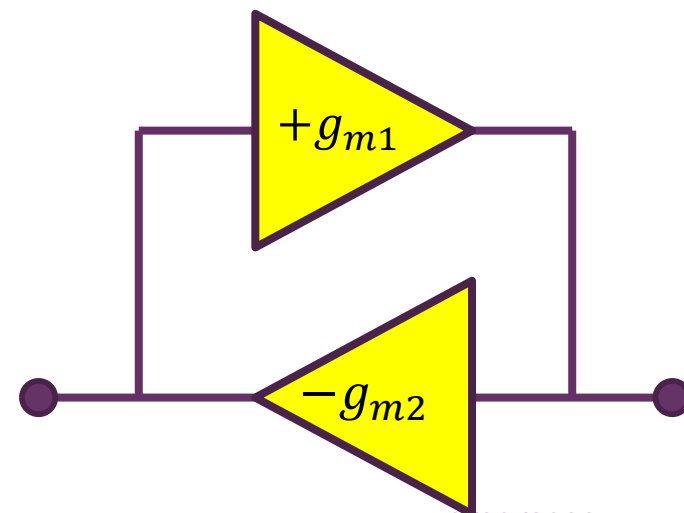
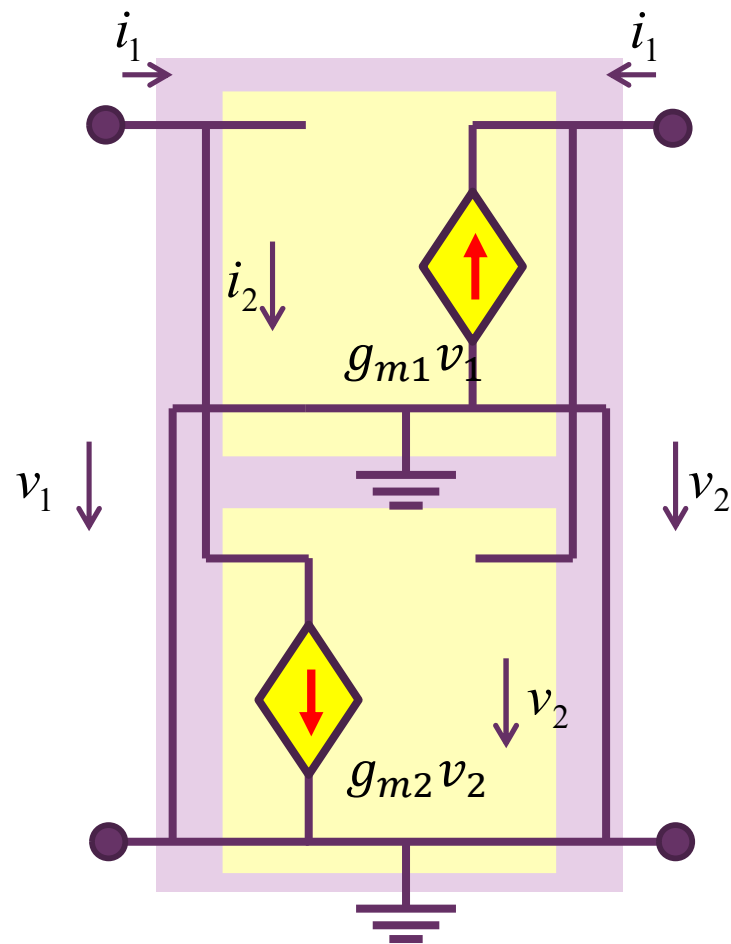
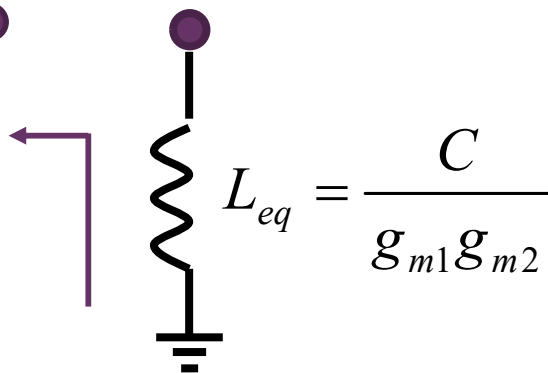
$$L_{eq} = \frac{C}{g_{m1} g_{m2}}$$

集成电路中的等效电感

$$\mathbf{y} = \begin{bmatrix} 0 & g_{m2} \\ -g_{m1} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -g_{m1} & 0 \end{bmatrix} + \begin{bmatrix} 0 & g_{m2} \\ 0 & 0 \end{bmatrix}$$



作业：接
地电感和
浮地电感

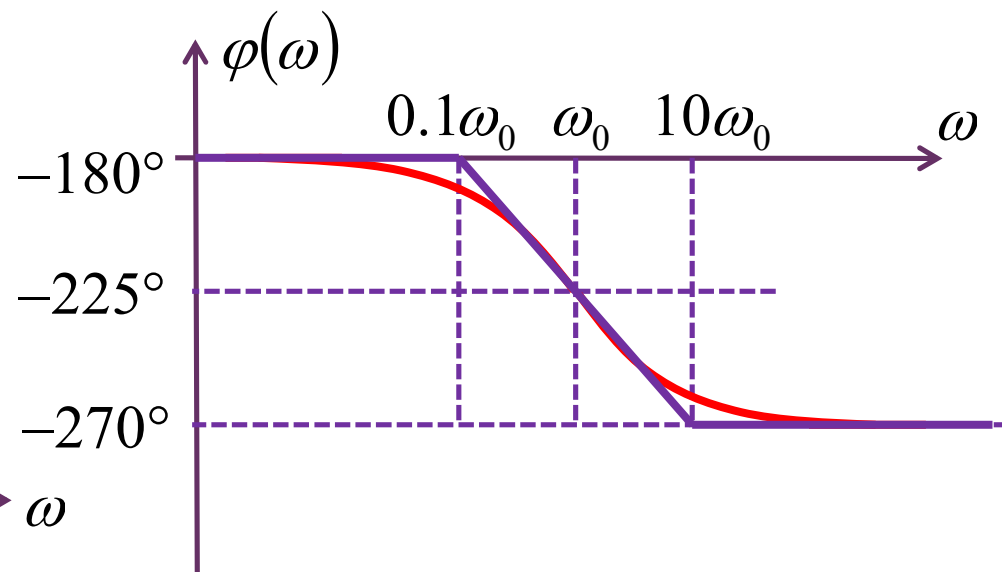
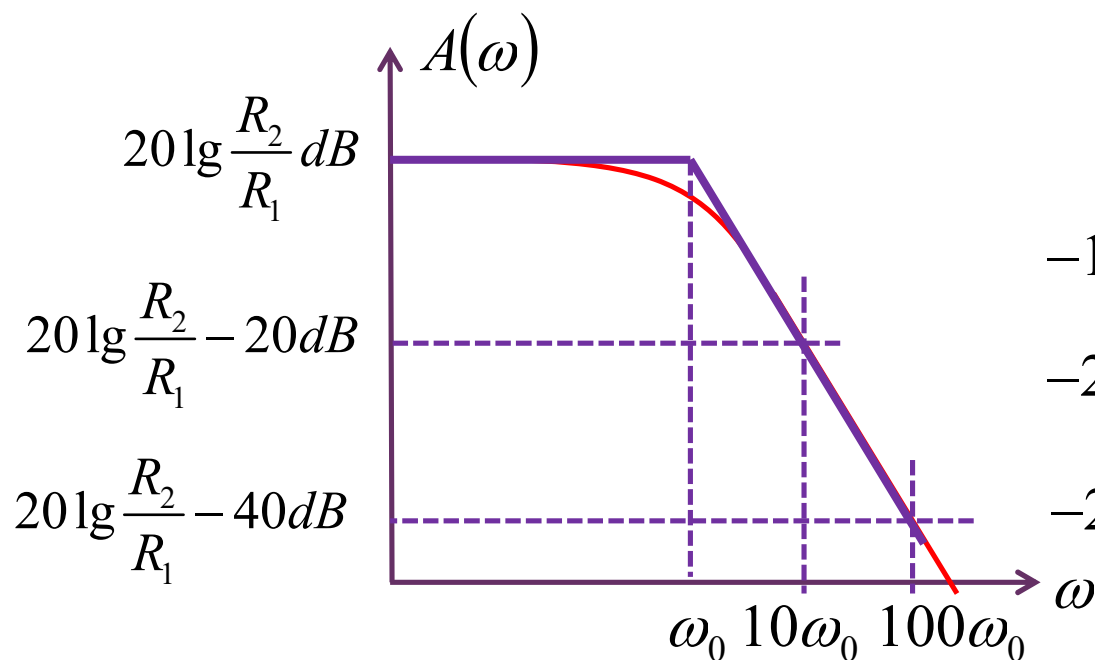
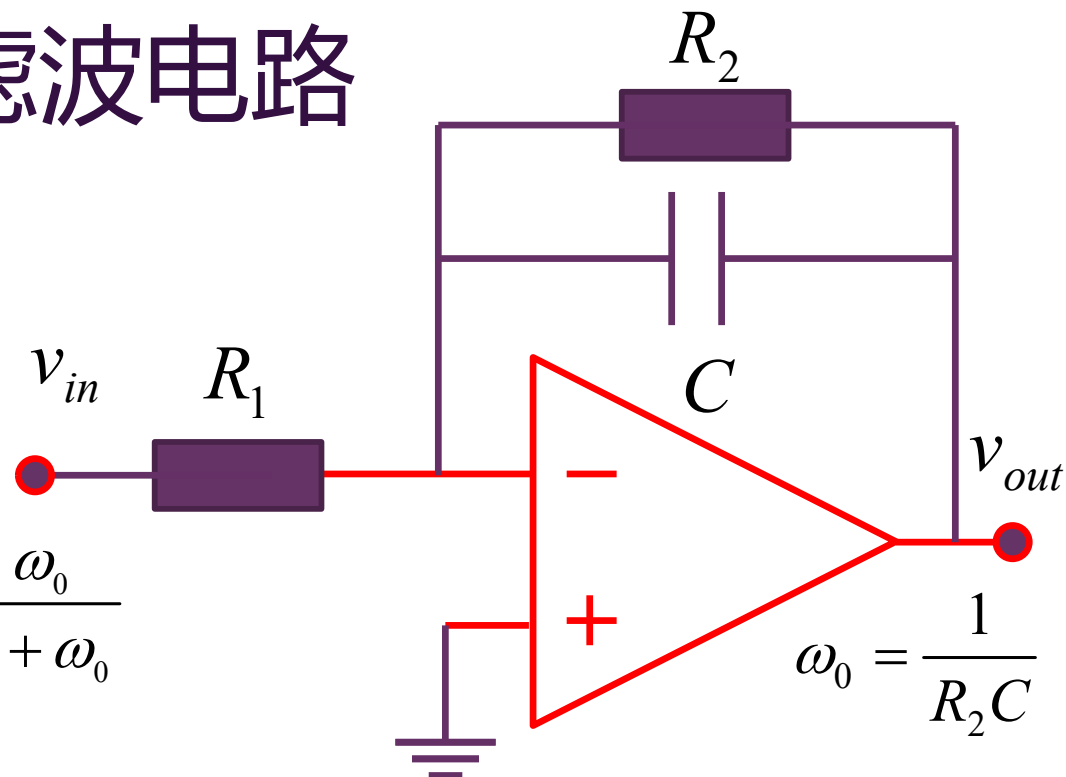


一阶节

一阶有源RC低通滤波电路

$$\frac{V_{in}(s)}{R_1} = \frac{0 - V_{out}(s)}{Z_2(s)} = -V_{out}(s) \left(\frac{1}{R_2} + sC \right)$$

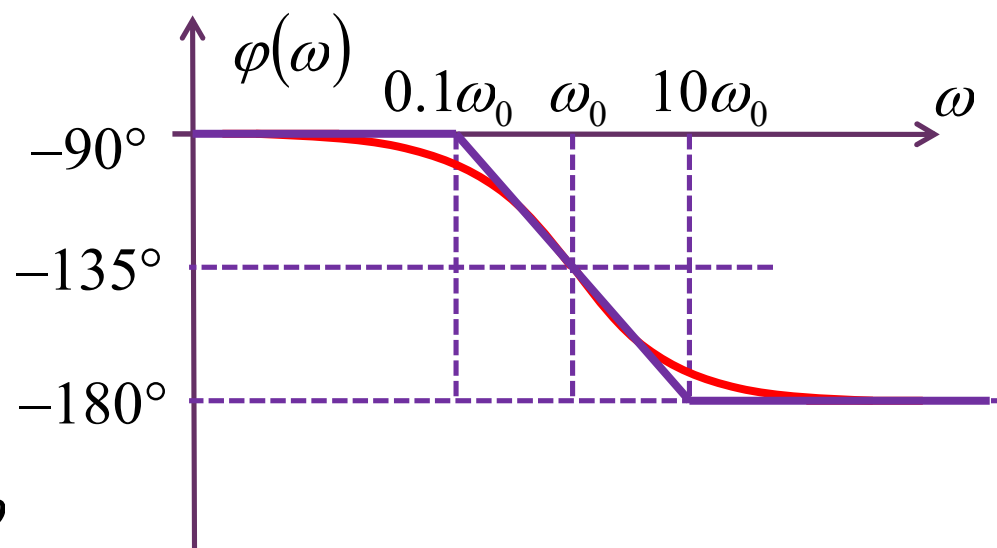
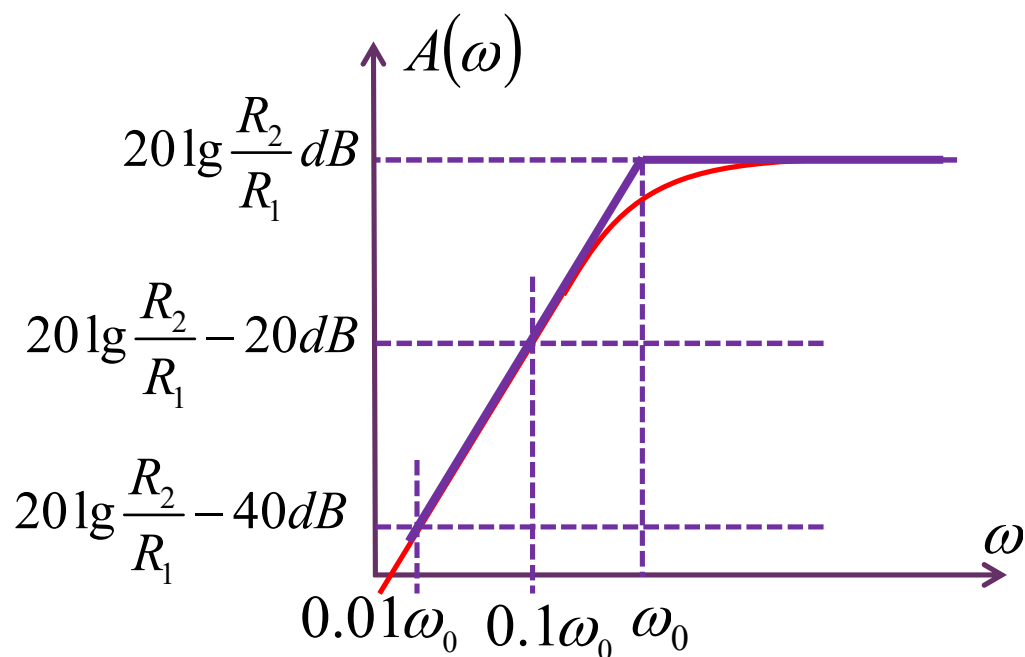
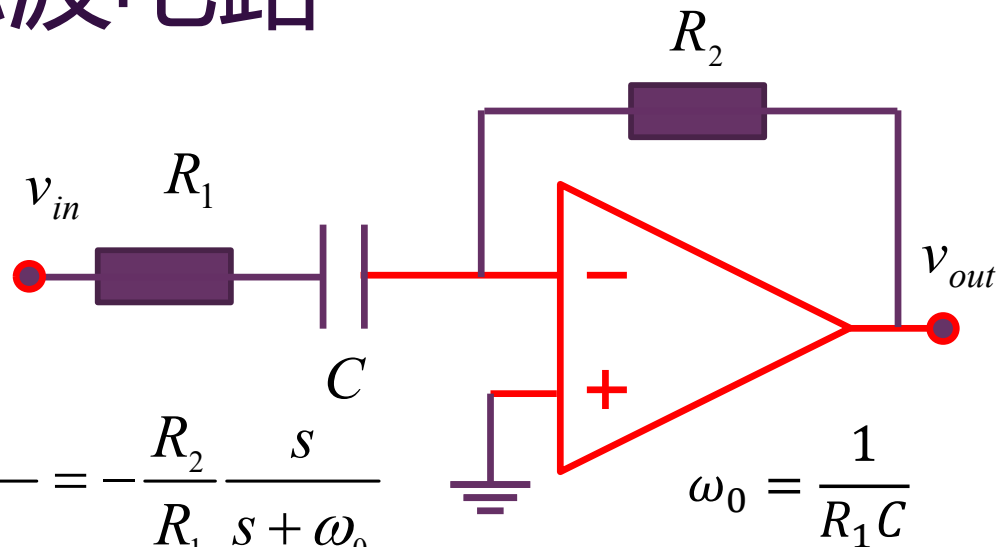
$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_2}{R_1} \frac{1}{(sR_2C + 1)} = -\frac{R_2}{R_1} \frac{\omega_0}{s + \omega_0}$$



一阶有源RC高通滤波电路

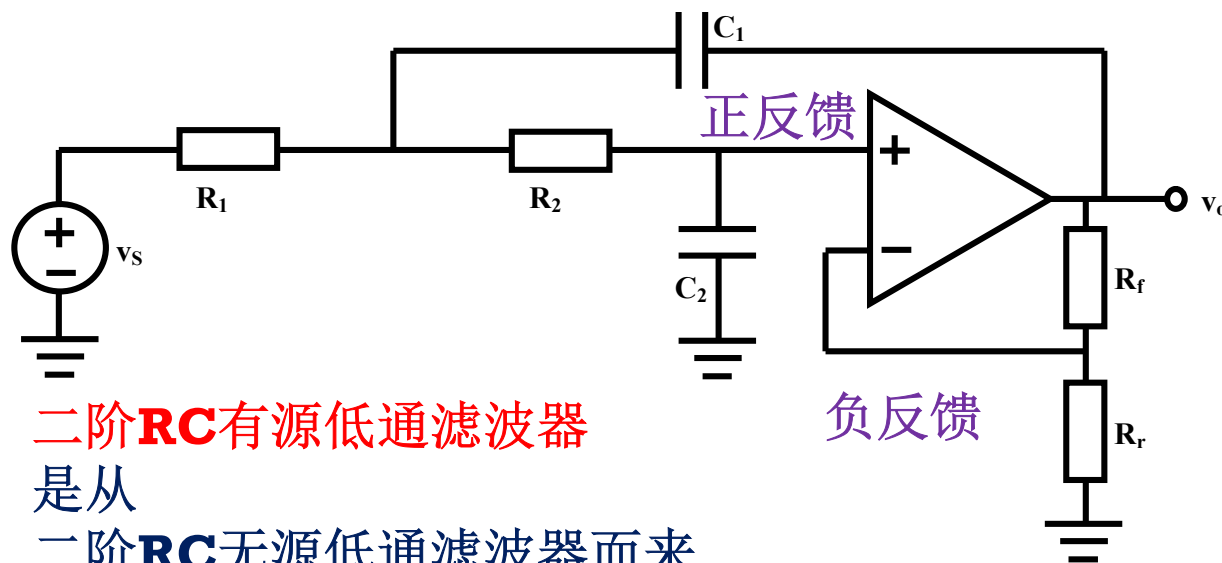
$$\frac{V_{in}(s)}{R_1 + \frac{1}{sC}} = \frac{0 - V_{out}(s)}{R_2}$$

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_2}{R_1 + \frac{1}{sC}} = -\frac{R_2}{R_1} \frac{1}{1 + \frac{1}{sR_1C}} = -\frac{R_2}{R_1} \frac{s}{s + \omega_0}$$



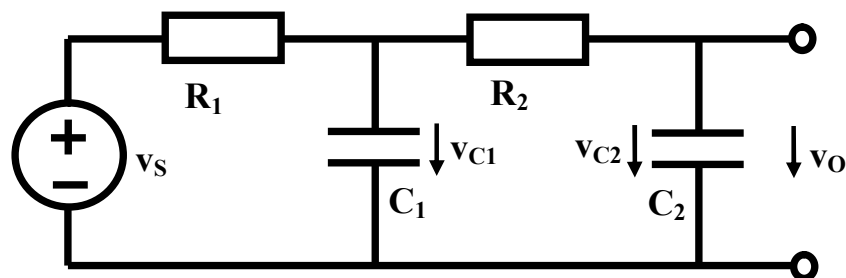
二阶节

二阶RC低通滤波电路



二阶**RC**有源低通滤波器
是从
二阶**RC**无源低通滤波器而来

只有负反馈才能确保运放工作在线性区（下半学期讨论），这里负反馈大于正反馈，整体看仍然是负反馈，因而可用虚短和虚断进行分析

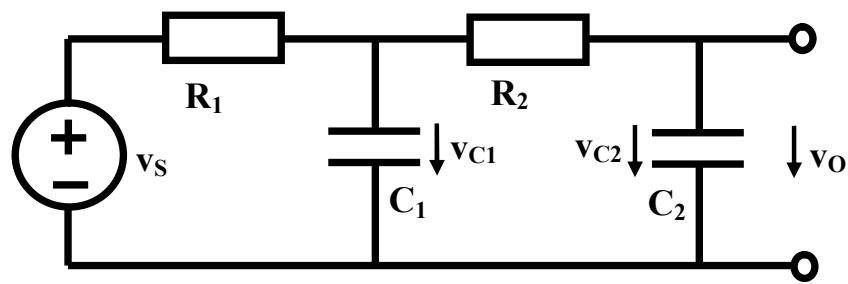


二阶**RC**无源低通滤波器

无源二阶**RC**低通的阻尼系数恒大于**1**（过阻尼），当我们需要小于**1**的阻尼系数（欠阻尼）时，可以通过有源电路，同时提供正反馈，则可等效出负电阻，用于抵偿正电阻的消耗，则可使得阻尼减小

$$H(s) = \frac{V_o}{V_s} = \frac{1}{1 + s(R_1 C_1 + R_1 C_2 + R_2 C_2) + s^2 R_1 C_1 R_2 C_2} = H_0 \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

二阶无源RC低通滤波电路分析



$$\frac{\dot{V}_o}{\dot{V}_s} = \frac{1}{A} = \frac{1}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2) + j\omega R_1 C_2}$$

$$\begin{aligned} H(s) &= \frac{1}{(1 + sR_1C_1)(1 + sR_2C_2) + sR_1C_2} \\ &= \frac{1}{s^2 R_1 C_1 R_2 C_2 + s(R_1 C_1 + R_2 C_2 + R_1 C_2) + 1} \\ &= \frac{\frac{1}{R_1 C_1 R_2 C_2}}{s^2 + \left(\frac{1}{R_2 C_2} + \frac{1}{R_1 C_1} + \frac{1}{R_2 C_1} \right) s + \frac{1}{R_1 C_1 R_2 C_2}} \\ &= \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} \end{aligned}$$

$$\begin{aligned} \mathbf{ABCD} &= \begin{bmatrix} 1 & R_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ j\omega C_1 & 1 \end{bmatrix} \begin{bmatrix} 1 & R_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ j\omega C_2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 + j\omega R_1 C_1 & R_1 \\ j\omega C_1 & 1 \end{bmatrix} \begin{bmatrix} 1 + j\omega R_2 C_2 & R_2 \\ j\omega C_2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2) + j\omega R_1 C_2 & \dots \\ \dots & \dots \end{bmatrix} \end{aligned}$$

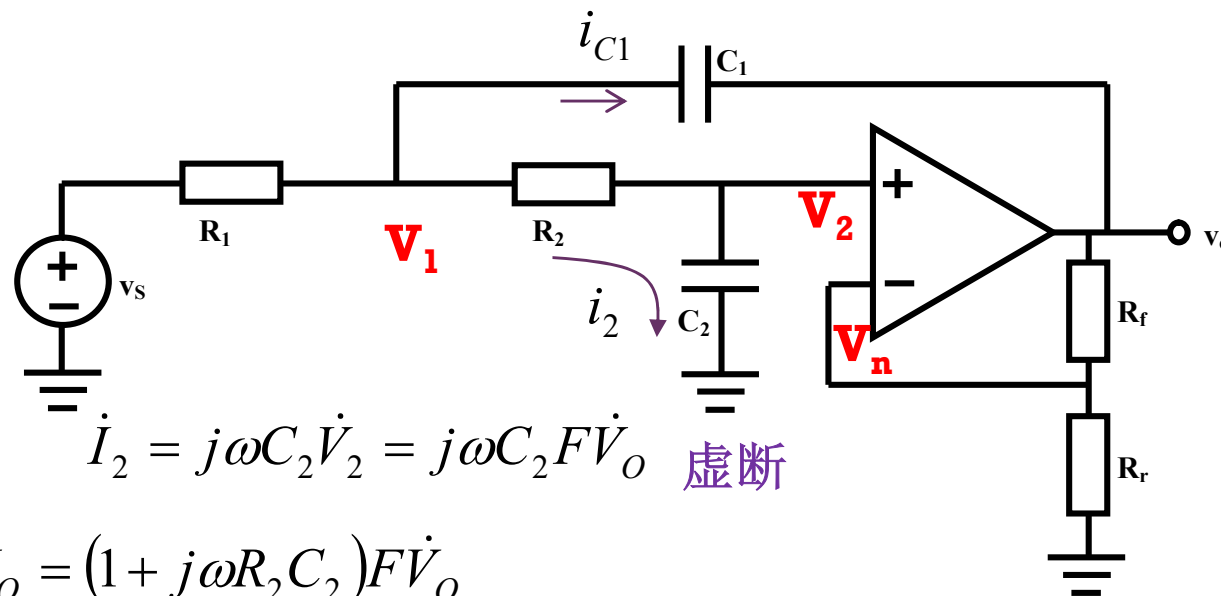
$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{R_1 C_1 R_2 C_2}} \\ \xi &= 0.5 \left(\sqrt{\frac{R_2 C_2}{R_1 C_1}} + \sqrt{\frac{R_1 C_1}{R_2 C_2}} + \sqrt{\frac{R_1 C_2}{R_2 C_1}} \right) \\ &\geq 0.5 \left(2 + \sqrt{\frac{R_1 C_2}{R_2 C_1}} \right) = 1 + 0.5 \sqrt{\frac{R_1 C_2}{R_2 C_1}} > 1 \end{aligned}$$

无源**RC**只能实现阻尼系数大于1的二阶滤波，但这种滤波器又可分解为两个一阶滤波传递函数相乘（级联），滤波器分解级联实现方案需要的是欠阻尼的二阶节

二阶有源RC低通滤波电路分析：倒推法

$$\dot{V}_n = \frac{R_r}{R_r + R_f} \dot{V}_o = F_n \dot{V}_o \quad \text{虚断}$$

$$\dot{V}_2 = \dot{V}_n = F_n \dot{V}_o \quad \text{虚短}$$



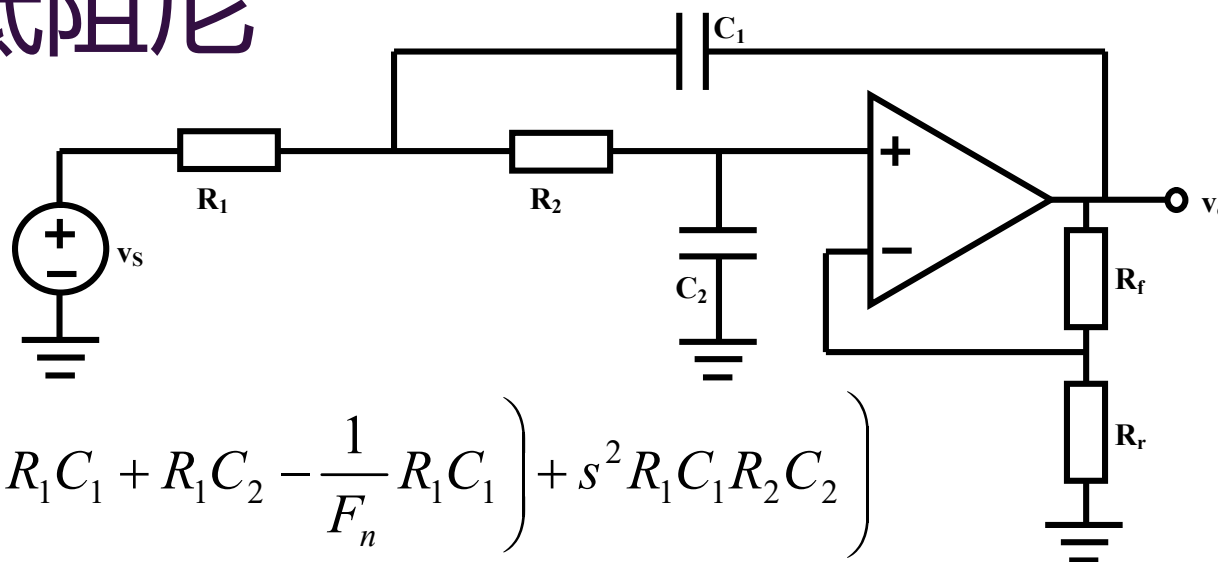
$$\dot{I}_2 = j\omega C_2 \dot{V}_2 = j\omega C_2 F_n \dot{V}_o \quad \text{虚断}$$

$$\dot{V}_1 = \dot{V}_2 + \dot{I}_2 R_2 = F_n \dot{V}_o + j\omega R_2 C_2 F_n \dot{V}_o = (1 + j\omega R_2 C_2) F_n \dot{V}_o$$

$$\dot{I}_{C1} = j\omega C_1 (\dot{V}_1 - \dot{V}_o) = j\omega C_1 \dot{V}_o ((1 + j\omega R_2 C_2) F_n - 1)$$

$$\begin{aligned} \dot{V}_s &= \dot{V}_1 + (\dot{I}_{C1} + \dot{I}_2) R_1 = (1 + j\omega R_2 C_2) F_n \dot{V}_o + (j\omega C_1 \dot{V}_o ((1 + j\omega R_2 C_2) F_n - 1) + j\omega C_2 F_n \dot{V}_o) R_1 \\ &= F_n \dot{V}_o \left(1 + j\omega R_2 C_2 + j\omega R_1 C_1 + j\omega R_1 C_1 j\omega R_2 C_2 - \frac{j\omega R_1 C_1}{F_n} + j\omega R_1 C_2 \right) \\ &= F_n \dot{V}_o \left(1 + s \left(R_2 C_2 + R_1 C_1 + R_1 C_2 - \frac{1}{F_n} R_1 C_1 \right) + s^2 R_1 C_1 R_2 C_2 \right) \end{aligned}$$

用正反馈降低阻尼



$$\dot{V}_S = F_n \dot{V}_O \left(1 + s \left(R_2 C_2 + R_1 C_1 + R_1 C_2 - \frac{1}{F_n} R_1 C_1 \right) + s^2 R_1 C_1 R_2 C_2 \right)$$

$$H(s)_{s=j\omega} = \frac{\dot{V}_O}{\dot{V}_S} = \frac{1}{F_n} \frac{1}{1 + s \left(R_2 C_2 + R_1 C_1 + R_1 C_2 - \frac{1}{F_n} R_1 C_1 \right) + s^2 R_1 C_1 R_2 C_2}$$

$$= \frac{1}{F} \frac{1}{1 + 2\xi \frac{s}{\omega_0} + \left(\frac{s}{\omega_0} \right)^2}$$

$$F = \frac{R_r}{R_r + R_f}$$

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}}$$

$$\xi = 0.5 \left(\sqrt{\frac{R_2 C_2}{R_1 C_1}} + \sqrt{\frac{R_1 C_1}{R_2 C_2}} + \sqrt{\frac{R_1 C_2}{R_2 C_1}} - \frac{1}{F_n} \sqrt{\frac{R_1 C_1}{R_2 C_2}} \right)$$

正反馈导致阻尼降低，但正反馈不能过大，过大可导致负阻尼，变成振荡器

二阶有源RC高通滤波电路

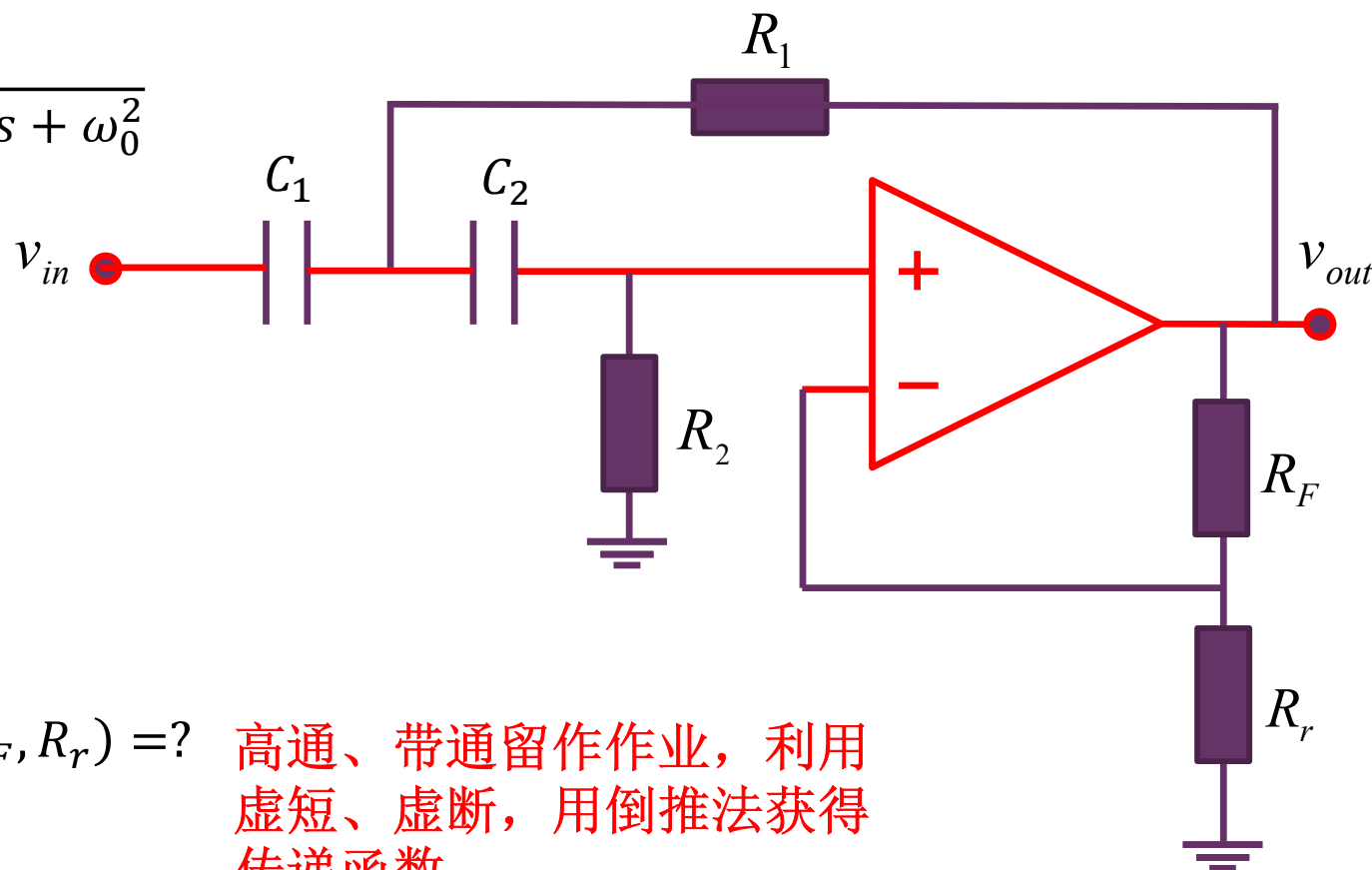
从二阶无源**RC**高通滤波器转换而来，解决其阻尼系数无法小于**1**的问题

$$H(s) = H_0 \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$H_0 = 1 + \frac{R_F}{R_r}$$

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}}$$

$\xi = \xi(R_1, C_1, R_2, C_2, R_F, R_r) = ?$ 高通、带通留作作业，利用虚短、虚断，用倒推法获得传递函数



第07讲 有源滤波 内容小结

- 无源的RC滤波器是有损网络，可用于信号的滤波处理，一般不会用于带内信号的功率传输
 - 而且有些特性的高阶滤波器无法实现：阻尼系数大于1
- 无源的LC滤波器是无损网络，可用于带内信号及其功率的有效传输和带外信号的滤除
 - LC滤波器设计有成熟规范的操作流程
- 集成滤波器设计往往基于LC滤波器设计，只是用电容的积分功能替代电感的积分功能，以解决集成电路内部难以集成高质量电感的问题
 - 有源滤波器设计其实是**积分器**设计，或**回旋器**设计，或**一阶节**、**二阶节**设计，…

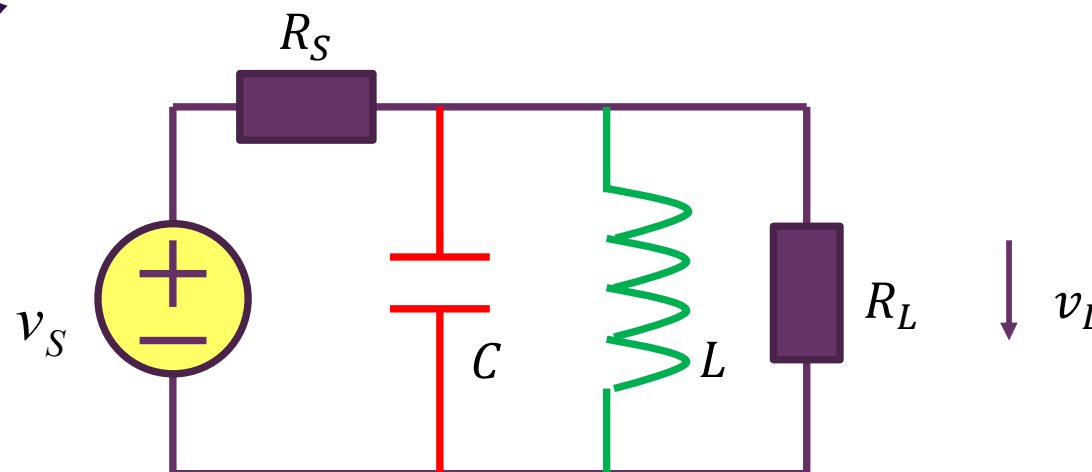
本讲内容在教材中的章节位置

- P780-807: 二阶滤波器
- P199: 理想回旋器
- P702: 一阶RC有源滤波器
- P788: 二阶RC有源滤波器

作业讲解

作业6.3 带通滤波

以上为课堂授课内容
以下为录像自学内容



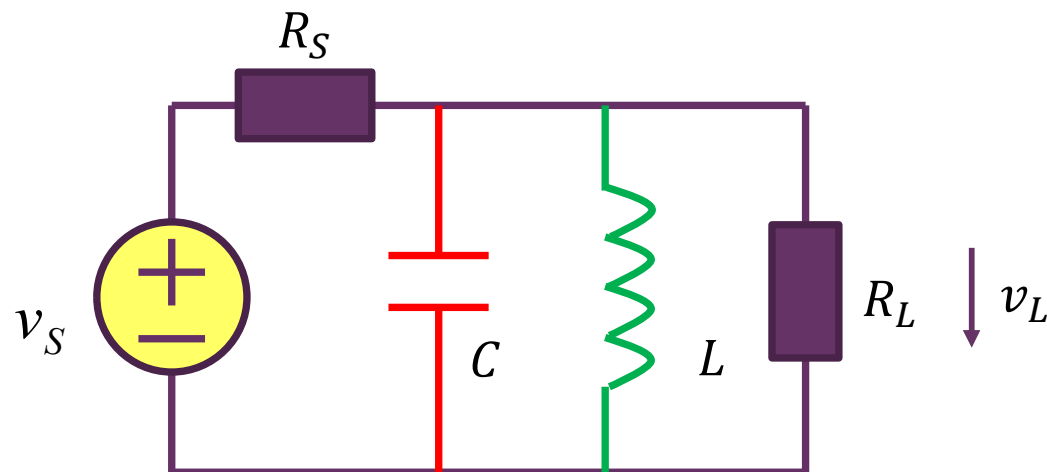
■ 频域分析

- 给出传递函数表达式
- 说明这是一个典型带通滤波器传递函数，并给出相应的Q值、自由振荡频率 ω_0 ，3dB带宽 BW_{3dB}

■ 时域分析

- 用五要素法给出该系统的单位冲激响应和单位阶跃响应

频域传递函数

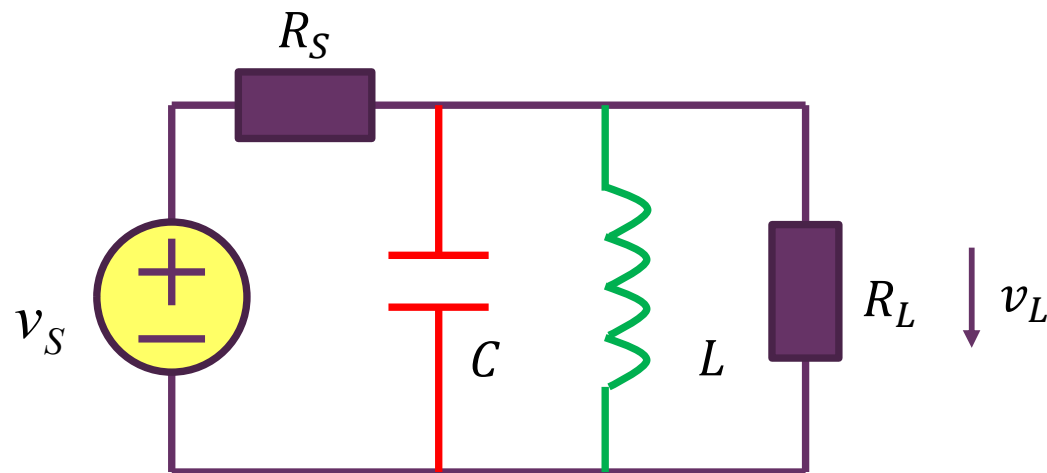


$$\begin{aligned}
 H(j\omega) &= \frac{\dot{V}_L(j\omega)}{\dot{V}_S(j\omega)} \stackrel{s=j\omega}{=} \frac{\frac{1}{\frac{1}{R_L} + sC + \frac{1}{sL}}}{R_S + \frac{1}{\frac{1}{R_L} + sC + \frac{1}{sL}}} = \frac{R_L}{R_S + R_L + sCR_S R_L + \frac{R_L}{sL} R_S} \\
 &= \frac{R_L}{R_S + R_L} \frac{sL}{s^2 LC \frac{R_S R_L}{R_S + R_L} + sL + \frac{R_S R_L}{R_S + R_L}} = \frac{R_L}{R_S + R_L} \frac{\frac{s}{C(R_S || R_L)}}{s^2 + \frac{s}{C(R_S || R_L)} + \frac{1}{LC}} = H_0 \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2}
 \end{aligned}$$

$$H_0 = \frac{R_L}{R_S + R_L} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad \xi = \frac{1}{2(R_S || R_L)} \sqrt{\frac{L}{C}} = \frac{G_S + G_L}{2Y_0}$$

直接写传函

1、直流电感短路，输出为0；高频电容短路，输出为0；在谐振频点，并联LC相当于开路，信号有输出，故而这是一个典型的二阶带通滤波器，显然它将具有如下的典型传函形式



$$H(s) = \frac{\dot{V}_L(s)}{\dot{V}_S(s)} = H_0 \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

2、其中， H_0 为中心频点（谐振频点）上的传递系数：谐振频点上，并联LC开路，传递系数为分压系数

$$H_0 = H(j\omega_0) = \eta = \frac{R_L}{R_S + R_L}$$

ω_0 为中心频点（谐振频点）， ξ 为并联谐振腔的阻尼系数

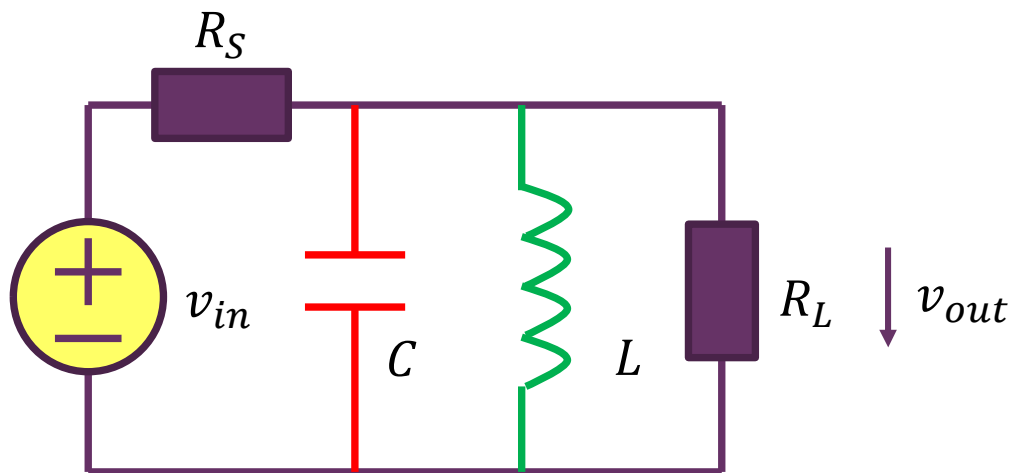
$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \xi = \frac{G}{2Y_0} = \frac{G_S + G_L}{2Y_0} = \frac{1}{2(R_S || R_L)} \sqrt{\frac{L}{C}}$$

3、二阶带通滤波器的3dB带宽为

$$BW_{3dB} = \frac{f_0}{Q} = 2\xi f_0 = 2 \cdot \frac{1}{2(R_S || R_L)} \sqrt{\frac{L}{C}} \cdot \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi(R_S || R_L)C}$$

冲激响应

冲激响应为零状态响应



$\frac{V_0}{\omega_0} \delta(t)$ 确保公式推导每一步量纲的正确性

$$v_{out}(0^+) = v_C(0^+) = \frac{V_0}{R_S Y_0} = \frac{G_S}{G_S + G_L} 2\xi V_0 = 2\xi \eta V_0$$

$$\begin{aligned} \frac{d}{dt} v_{out}(0^+) &= \frac{d}{dt} v_C(0^+) = \frac{i_C(0^+)}{C} \\ &= \frac{i_{in}(0^+) - i_S(0^+) - i_L(0^+) - i_{out}(0^+)}{C} \\ &= \frac{0 - G_S v_C(0^+) - i_L(0^+) - G_L v_C(0^+)}{C} = -\frac{G_S + G_L}{C} \frac{V_0}{R_S Y_0} \\ &= -4\xi^2 \omega_0 \eta V_0 \end{aligned}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \xi = \frac{G}{2Y_0} = \frac{G_S + G_L}{2\sqrt{\frac{C}{L}}}$$

$$v_C(0^-) = 0 \quad i_L(0^-) = 0$$

冲激激励加载瞬间，电容高频短路，电感高频开路，于是所有冲激电压全部加载 R_S 电阻上，产生的冲激电流流过电容（短路），电容电压跳变

$$\begin{aligned} v_C(0^+) &= v_C(0^-) + \frac{1}{C} \int_{0^-}^{0^+} \frac{V_0}{R_S \omega_0} \delta(t) dt \\ &= 0 + \frac{V_0}{R_S \omega_0 C} \int_{0^-}^{0^+} \delta(t) dt = \frac{V_0}{R_S Y_0} \end{aligned}$$

电感两端没有冲激电压，电感电流不会跳变

$$i_L(0^+) = i_L(0^-) = 0$$

$t > 0$ 后激励消失，电容储能最终将被电阻耗光，故而

$$v_{out\infty}(t) = 0$$

单位冲激响应

$$v_{in}(t) = \frac{V_0}{\omega_0} \delta(t)$$

$$v_{out\infty}(t) = 0$$

$$v_{out}(0^+) = 2\xi\eta V_0$$

$$\frac{d}{dt}v_{out}(0^+) = -4\xi^2\omega_0\eta V_0$$

$$v_{out}(t) = v_{out\infty}(t) + (V_{o0} - V_{o\infty 0})e^{-\xi\omega_0 t} \cos \sqrt{1 - \xi^2} \omega_0 t \\ + \left(\frac{\dot{V}_{o0} - \dot{V}_{o\infty 0}}{\xi\omega_0} + V_{o0} - V_{o\infty 0} \right) \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t$$

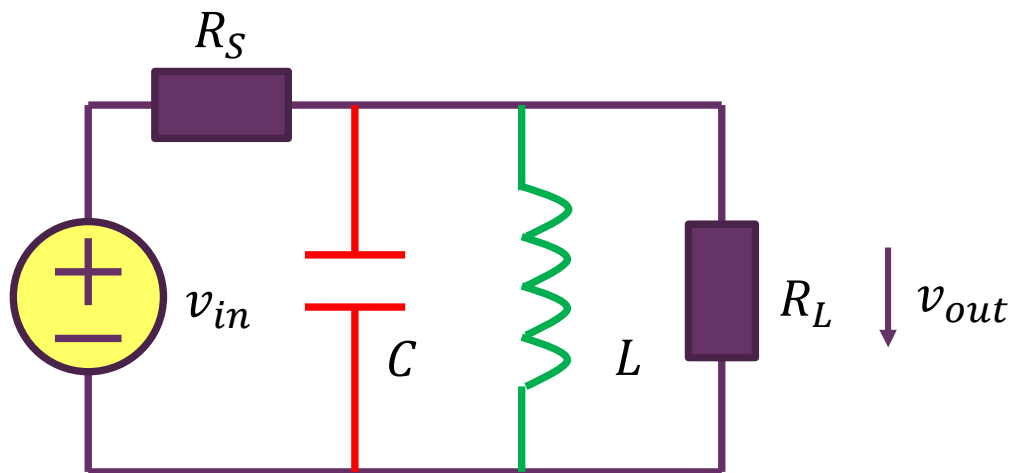
$$= 0 + (2\xi\eta V_0 - 0)e^{-\xi\omega_0 t} \cos \sqrt{1 - \xi^2} \omega_0 t \\ + \left(\frac{-4\xi^2\omega_0\eta V_0 - 0}{\xi\omega_0} + 2\xi\eta V_0 - 0 \right) \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t$$

$$= 2\xi\eta V_0 e^{-\xi\omega_0 t} \left(\cos \sqrt{1 - \xi^2} \omega_0 t - \frac{\xi}{\sqrt{1 - \xi^2}} \sin \sqrt{1 - \xi^2} \omega_0 t \right)$$

$$h(t) = \eta \cdot 2\xi\omega_0 e^{-\xi\omega_0 t} \left(\cos \sqrt{1 - \xi^2} \omega_0 t - \frac{\xi}{\sqrt{1 - \xi^2}} \sin \sqrt{1 - \xi^2} \omega_0 t \right) \\ = \eta \cdot \frac{2\xi\omega_0}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \cos \left(\sqrt{1 - \xi^2} \omega_0 t + \arctan \frac{\xi}{\sqrt{1 - \xi^2}} \right) \quad (t \geq 0)$$

阶跃响应

阶跃响应为零状态响应



$V_0 U(t)$

确保公式推导每一步量纲的正确性

$$v_{out}(0^+) = v_C(0^+) = 0$$

$$\begin{aligned} \frac{d}{dt} v_{out}(0^+) &= \frac{d}{dt} v_C(0^+) = \frac{i_C(0^+)}{C} \\ &= \frac{i_{in}(0^+) - i_S(0^+) - i_L(0^+) - i_{out}(0^+)}{C} \\ &= \frac{G_S V_0 - G_S v_C(0^+) - i_L(0^+) - G_L v_C(0^+)}{C} = \frac{G_S V_0}{C} \\ &= \omega_0 \frac{G_S}{G_S + G_L} \frac{G_S + G_L}{Y_0} V_0 = \eta 2\xi \omega_0 V_0 \end{aligned}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \xi = \frac{G}{2Y_0} = \frac{G_S + G_L}{2\sqrt{\frac{C}{L}}}$$

$$v_C(0^-) = 0 \quad i_L(0^-) = 0$$

阶跃激励加载瞬间，电容高频短路，电感高频开路，于是所有阶跃电压全部加载 R_S 电阻上，产生的阶跃电流流过电容（短路），电容电压不会跳变

$$\begin{aligned} v_C(0^+) &= v_C(0^-) + \frac{1}{C} \int_{0^-}^{0^+} \frac{V_0}{R_S} U(t) dt \\ &= 0 + \frac{V_0}{R_S C} \int_{0^-}^{0^+} U(t) dt = 0 \end{aligned}$$

电感两端没有冲激电压，电感电流不会跳变

$$i_L(0^+) = i_L(0^-) = 0$$

等待足够长时间后，电路变成直流电路，因为电感直流短路，从而

$$v_{out\infty}(t) = 0$$

单位阶跃响应

$$v_{in}(t) = V_0 U(t)$$

$$v_{out\infty}(t) = 0$$

$$v_{out}(0^+) = 0$$

$$\frac{d}{dt} v_{out}(0^+) = 2\xi\omega_0\eta V_0$$

$$v_{out}(t) = v_{out\infty}(t) + (V_{o0} - V_{o\infty 0})e^{-\xi\omega_0 t} \cos \sqrt{1 - \xi^2} \omega_0 t \\ + \left(\frac{\dot{V}_{o0} - \dot{V}_{o\infty 0}}{\xi\omega_0} + V_{o0} - V_{o\infty 0} \right) \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t$$

$$= 0 + (0 - 0)e^{-\xi\omega_0 t} \cos \sqrt{1 - \xi^2} \omega_0 t \\ + \left(\frac{2\xi\omega_0\eta V_0 - 0}{\xi\omega_0} + 0 - 0 \right) \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t$$

$$= 2\eta V_0 \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t$$

$$g(t) = \eta \frac{2\xi}{\sqrt{1 - \xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1 - \xi^2} \omega_0 t \quad (t \geq 0)$$

时频对应关系

Laplace变换对

$$H(s) = \frac{\dot{V}_L(s)}{\dot{V}_S(s)} = \eta \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$G(s) = \frac{1}{s} H(s) = 2\eta \frac{\xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$g(t) = 2\eta \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

$$\frac{\xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} \Leftrightarrow \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

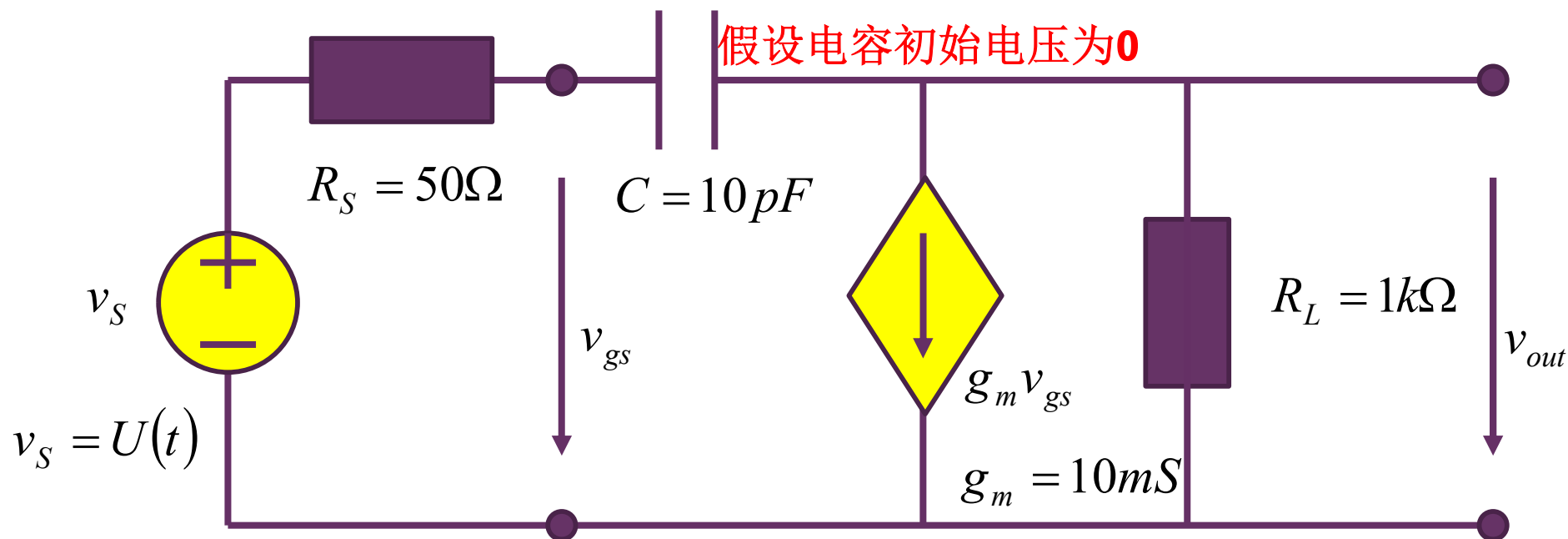
$$H(s) = \frac{\dot{V}_L(s)}{\dot{V}_S(s)} = \eta \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2} = \eta 2\xi\omega_0 \frac{s + \xi\omega_0 - \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$\begin{aligned} 1 &\Leftrightarrow \delta(t) \\ \frac{1}{s} &\Leftrightarrow U(t) \\ \frac{1}{s + \omega_0} &\Leftrightarrow e^{-\omega_0 t} U(t) \end{aligned}$$

$$h(t) = \eta \cdot 2\xi\omega_0 \left(e^{-\xi\omega_0 t} \cos \sqrt{1-\xi^2} \omega_0 t - \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin \sqrt{1-\xi^2} \omega_0 t \right) \cdot U(t)$$

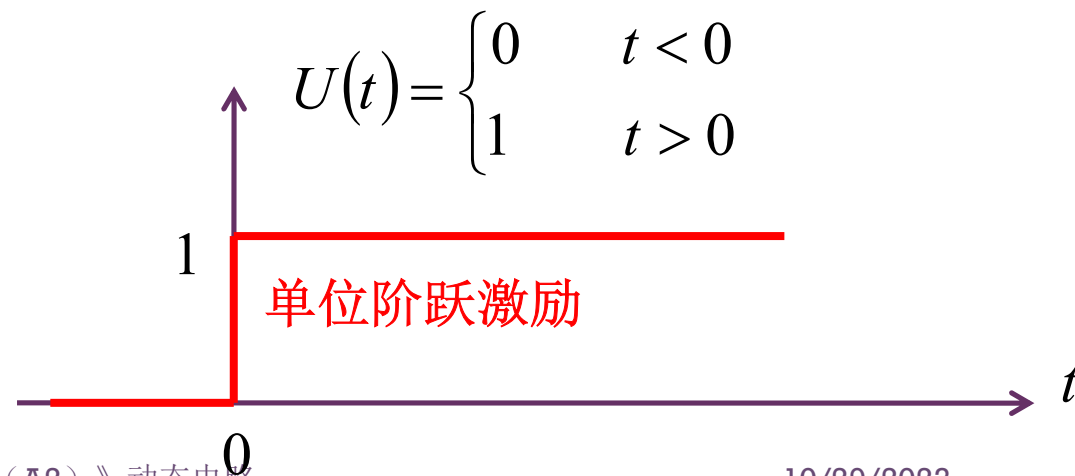
$$\frac{s + \xi\omega_0}{s^2 + 2\xi\omega_0 s + \omega_0^2} \Leftrightarrow e^{-\xi\omega_0 t} \cos \sqrt{1-\xi^2} \omega_0 t \cdot U(t)$$

作业4.6 一阶RC电路都可用三要素法分析

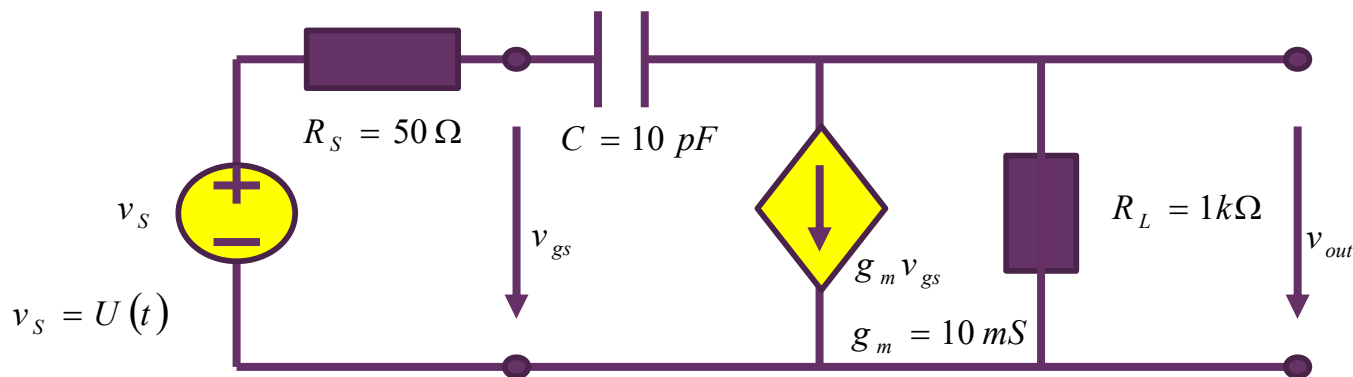


1/用三要素法获得输出电压单位阶跃响应

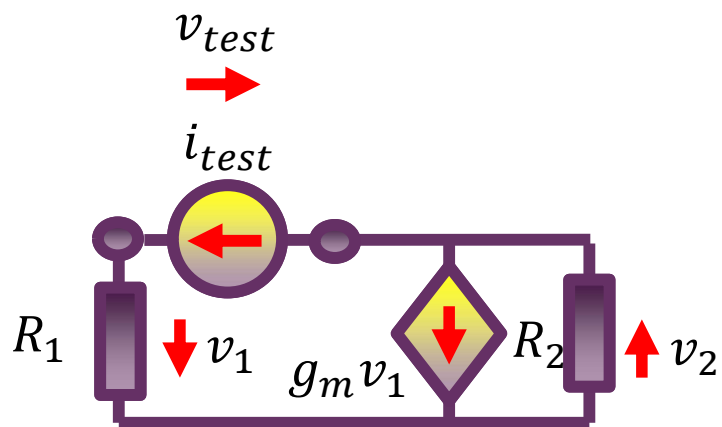
2/用任意方法在相量域获得电压增益传递函数



三要素法：时间常数



$$v_o(t) = v_{o\infty}(t) + (v_o(0^+) - v_{o\infty}(0^+))e^{-\frac{t}{\tau}} \quad (t \geq 0)$$



$$\tau = R_{eq}C = ?$$

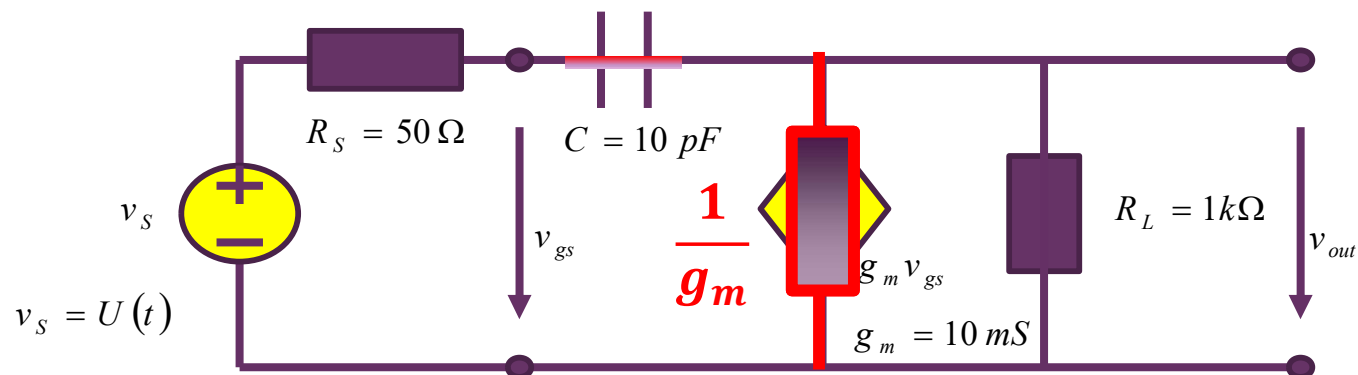
$$v_{test} = v_1 + v_2 = i_{test}R_1 + (i_{test} + g_m i_{test}R_1)R_2$$

$$R_{eq} = \frac{v_{test}}{i_{test}} = R_1 + R_2 + g_m R_1 R_2$$

求等效电阻时，独立源不起作用，但受控源的作用必须保留

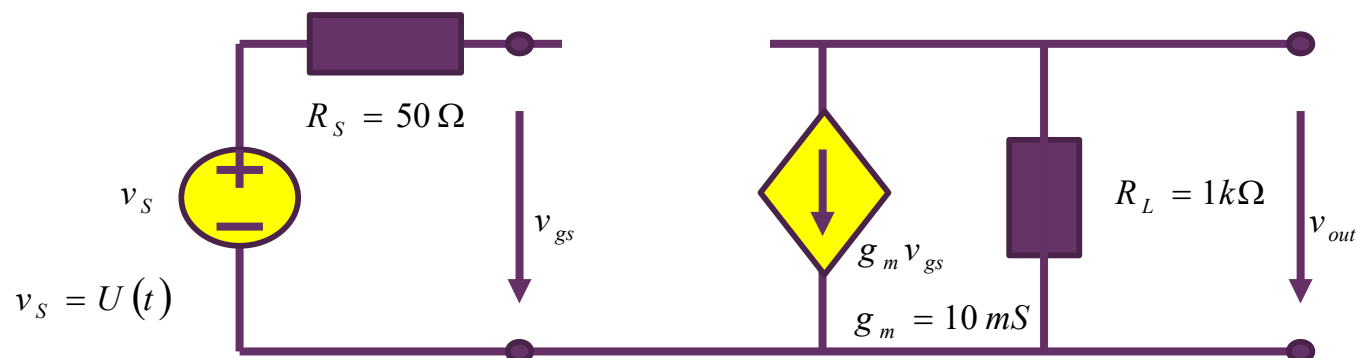
$$\begin{aligned} \tau &= R_{eq}C = (R_S + R_L + g_m R_S R_L)C \\ &= (50 + 1000 + 0.01 \times 1000 \times 50) \times 10p \\ &= 1550 \times 10p = 15.5ns \end{aligned}$$

三要素法：初值



$$\begin{aligned}
 v_o(0^+) &= \frac{R_L \parallel \frac{1}{g_m}}{R_S + R_L \parallel \frac{1}{g_m}} v_s(0^+) = \frac{\frac{R_L}{1 + g_m R_L}}{R_S + \frac{R_L}{1 + g_m R_L}} v_s(0^+) \\
 &= \frac{R_L}{R_S + R_L + g_m R_S R_L} v_s(0^+) = \frac{1000}{1550} \times 1 = 0.645V
 \end{aligned}$$

三要素法：稳态响应



$$v_{o\infty}(t) = -g_m R_L v_{S\infty}(t) = -10\text{m} \times 1\text{k} \times 1 = -10\text{V}$$

$$v_o(0^+) = \frac{R_L}{R_S + R_L + g_m R_S R_L} v_S(0^+) = 0.645\text{V}$$

$$\tau = R_{eq}C = (R_S + R_L + g_m R_S R_L)C = 15.5\text{ns}$$

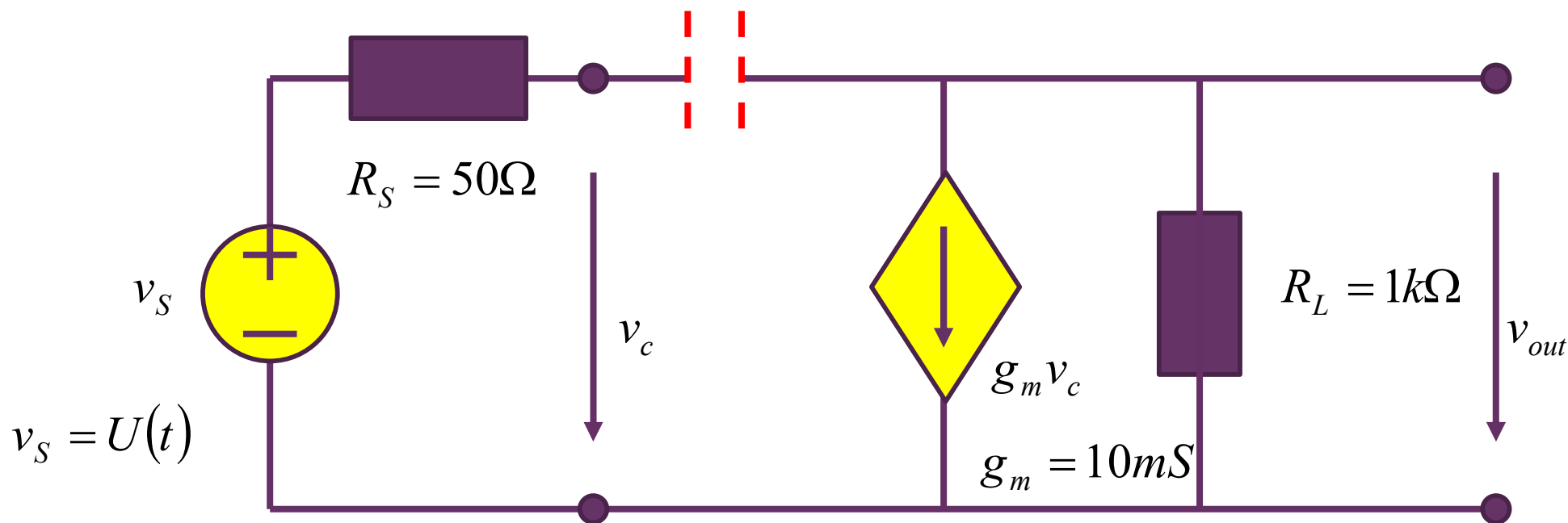
晶体管放大器的阶跃响应

$$v_o(t) = v_{o\infty}(t) + (v_o(0^+) - v_{o\infty}(0^+))e^{-\frac{t}{\tau}} \quad (t \geq 0)$$

$$= -10 + 10.645e^{-\frac{t}{15.5\text{n}}} \quad (\text{V})$$

电容的影响

上学期结论：理想晶体管模型就是理想压控流源



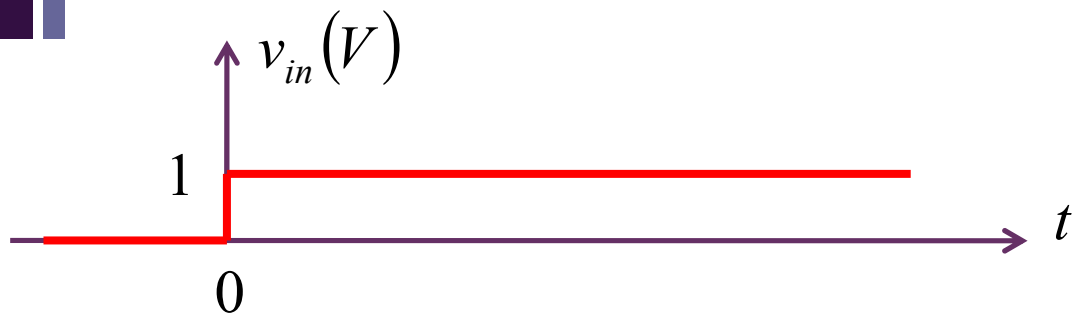
$$v_{out}(t) = -g_m R_L v_S(t) = -10U(t)$$

没有电容，简单的晶体管反相放大器

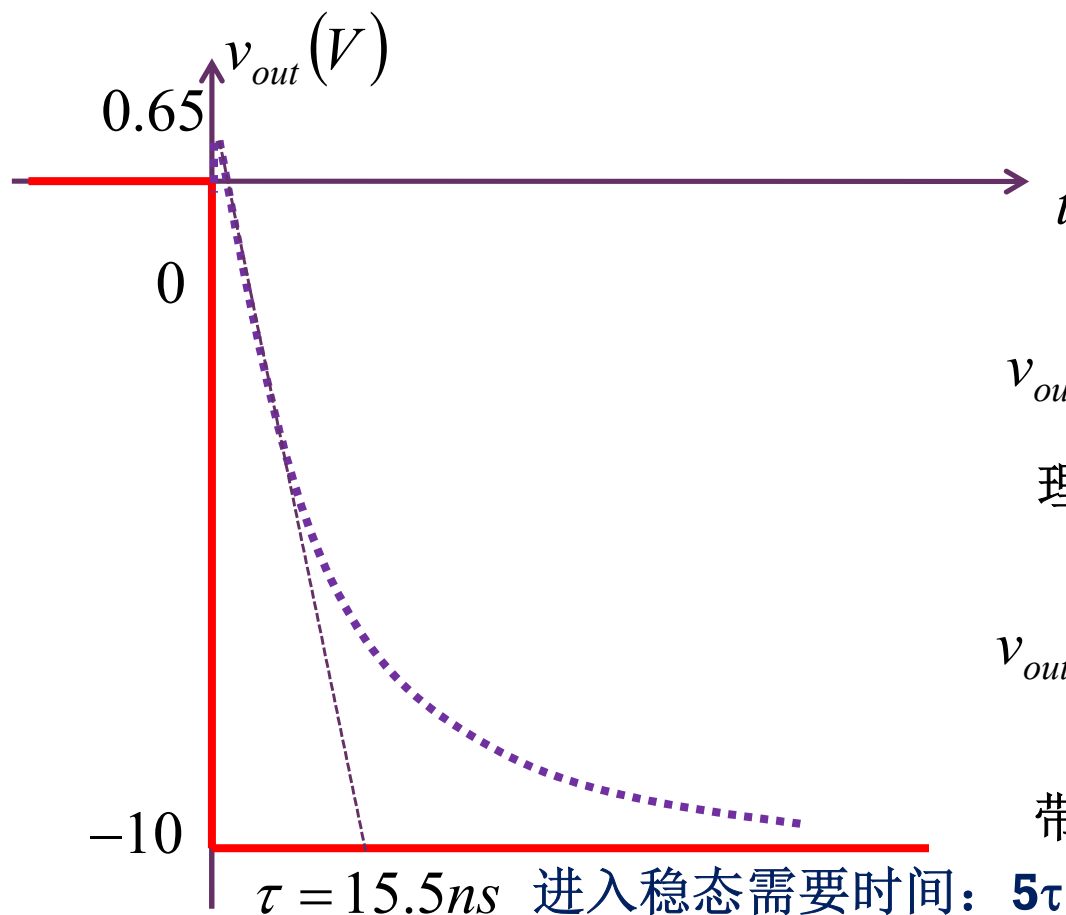
$$v_{out}(t) = \left(-10 + 10.65e^{-\frac{t}{15.5 \times 10^{-9}}} \right) U(t)$$

考虑晶体管寄生电容，不再是简单的反相放大电路，还需考虑电容充放电形成的过渡过程

电容导致信号延时传输



$$v_{in}(t) = U(t)$$



输出想要达到理想的稳态，
需要等待一段时间：这个等
待时间一般称为延时

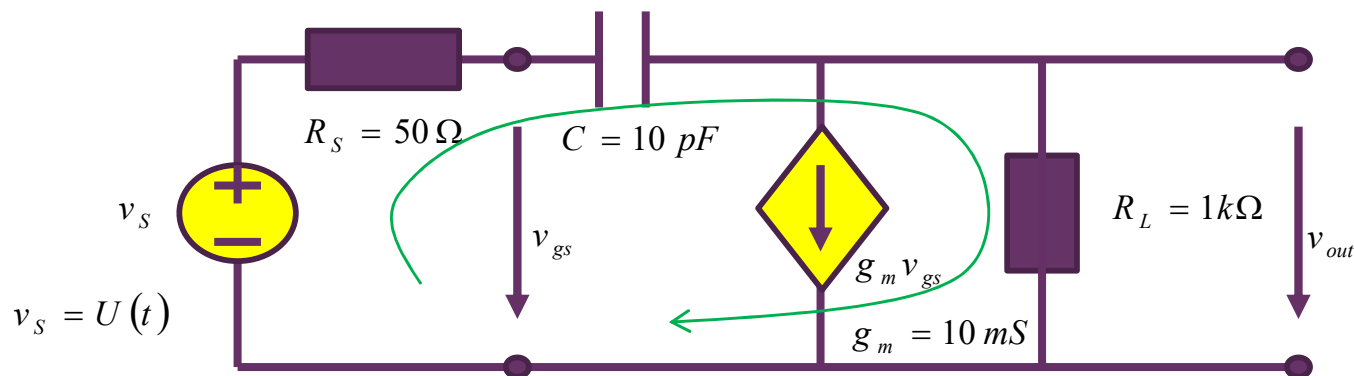
$$v_{out}(t) = -10U(t)$$

理想晶体管的反相放大输出响应

$$v_{out}(t) = \left(-10 + 10.65e^{-\frac{t}{15.5 \times 10^{-9}}} \right) U(t)$$

带跨接寄生电容的反相放大器输出响应

$\tau = 15.5 \text{ ns}$ 进入稳态需要时间： 5τ



$$\left(R_S + \frac{1}{sC} + R_L\right) \dot{I}_l = \dot{V}_s + g_m \dot{V}_{gs} R_L = \dot{V}_s + g_m (\dot{V}_s - R_S \dot{I}_l) R_L$$

$$\left(\frac{1}{sC} + R_S + R_L + g_m R_S R_L\right) \dot{I}_l = (g_m R_L + 1) \dot{V}_s \quad \dot{I}_l = \frac{g_m R_L + 1}{\frac{1}{sC} + R_S + R_L + g_m R_S R_L} \dot{V}_s$$

$$\dot{V}_{out} = R_L \dot{I}_l - g_m \dot{V}_{gs} R_L = R_L \dot{I}_l - g_m (\dot{V}_s - R_S \dot{I}_l) R_L = (1 + g_m R_S) R_L \dot{I}_l - g_m R_L \dot{V}_s$$

$$= (1 + g_m R_S) R_L \frac{g_m R_L + 1}{\frac{1}{sC} + R_S + R_L + g_m R_S R_L} \dot{V}_s - g_m R_L \dot{V}_s$$

$$= -g_m R_L \frac{1 - s \frac{C}{g_m}}{1 + s(R_S + R_L + g_m R_S R_L)C} \dot{V}_s$$

结点电压法自行练习

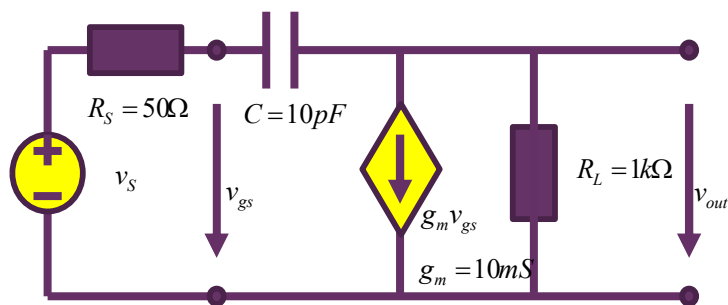
频率特性

$$H(s) = -g_m R_L \frac{1 - s \frac{C}{g_m}}{1 + s(R_S + R_L + g_m R_S R_L)C} = H_0 \frac{1 - \frac{j\omega}{\omega_z}}{1 + \frac{j\omega}{\omega_p}}$$

$$H_0 = -g_m R_L$$

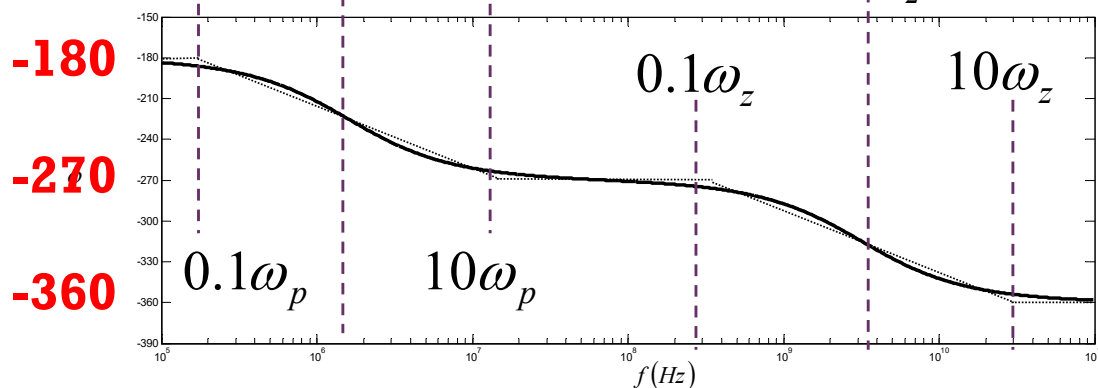
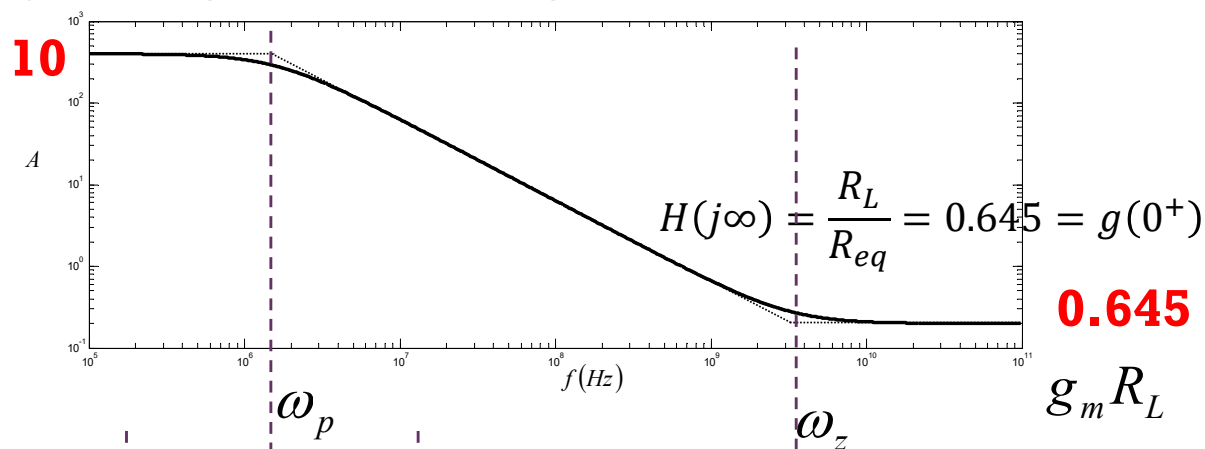
$$\omega_z = \frac{g_m}{C}$$

$$\omega_p = \frac{1}{(R_S + R_L + g_m R_S R_L)C}$$



产生右半平面零点的原因是信号有两条不同属性路径到达输出端，且导致不同方向的输出电流

$$H(j0) = -g_m R_L = -10 = g(\infty)$$

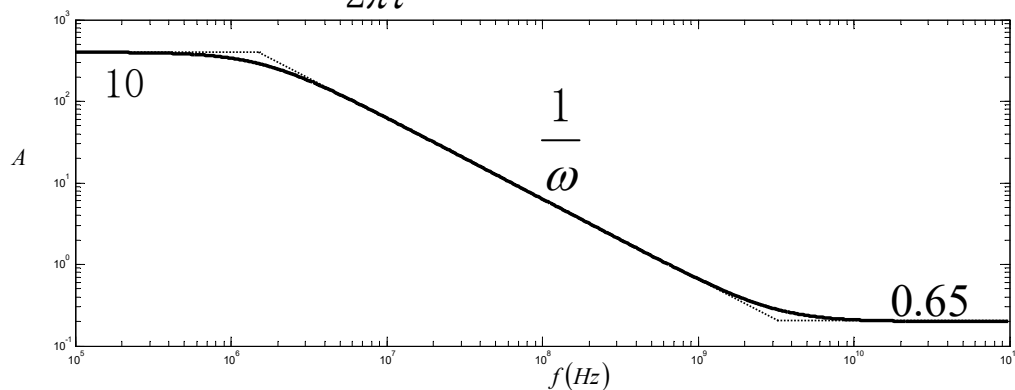


时频是对应的

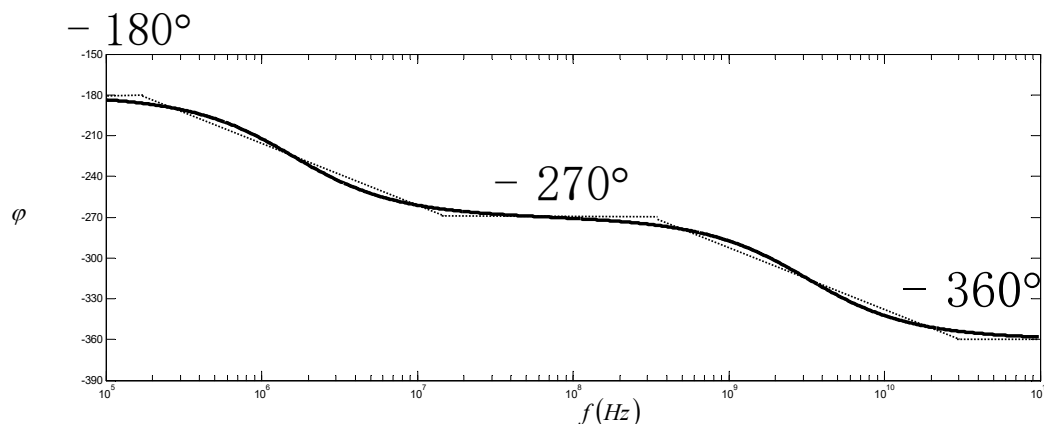
$$H(s) = -g_m R_L \frac{1 - \frac{s}{g_m/C}}{1 + sR_{eq}C} = -g_m R_L \frac{1}{1 + sR_{eq}C} + \frac{R_L}{R_{eq}} \frac{sR_{eq}C}{1 + sR_{eq}C}$$

$$R_{eq} = R_S + R_L + g_m R_S R_L$$

$$f_0 = \frac{1}{2\pi\tau} = 10.27 \text{ MHz}$$

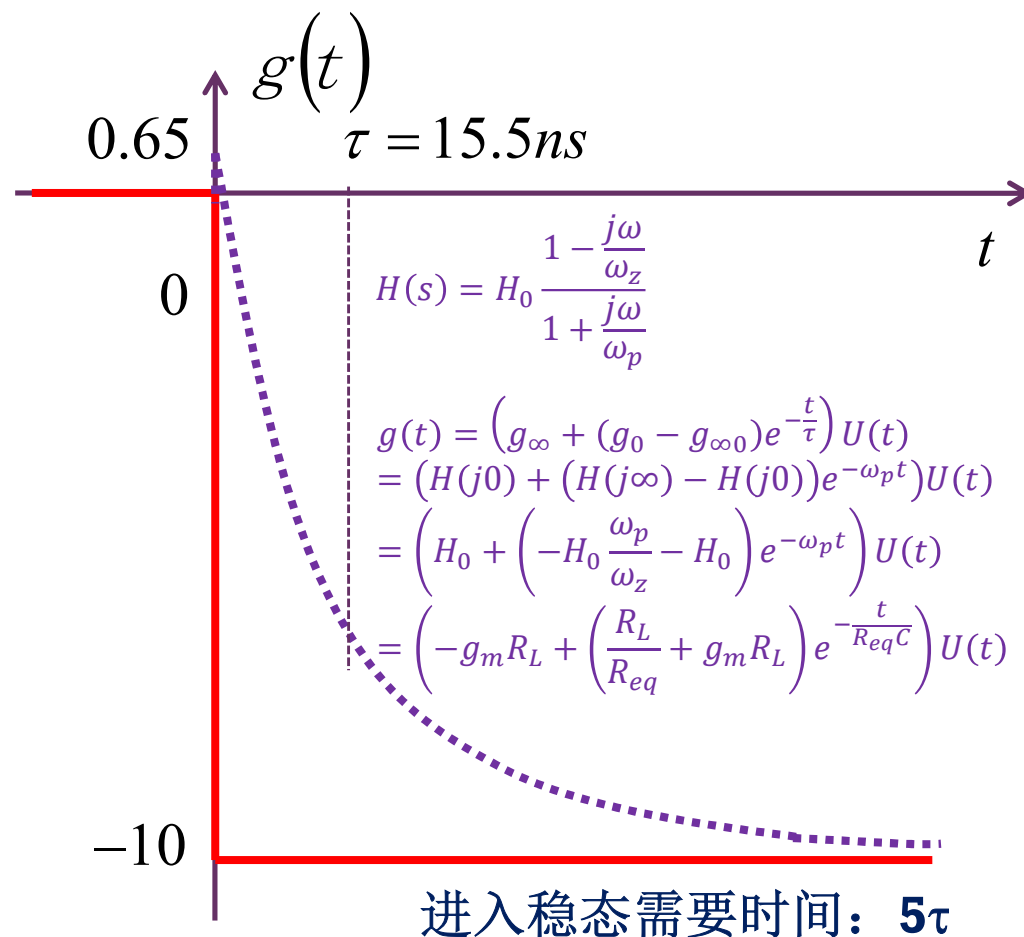


$$10 \times f_0 = 0.65 \times f_z$$



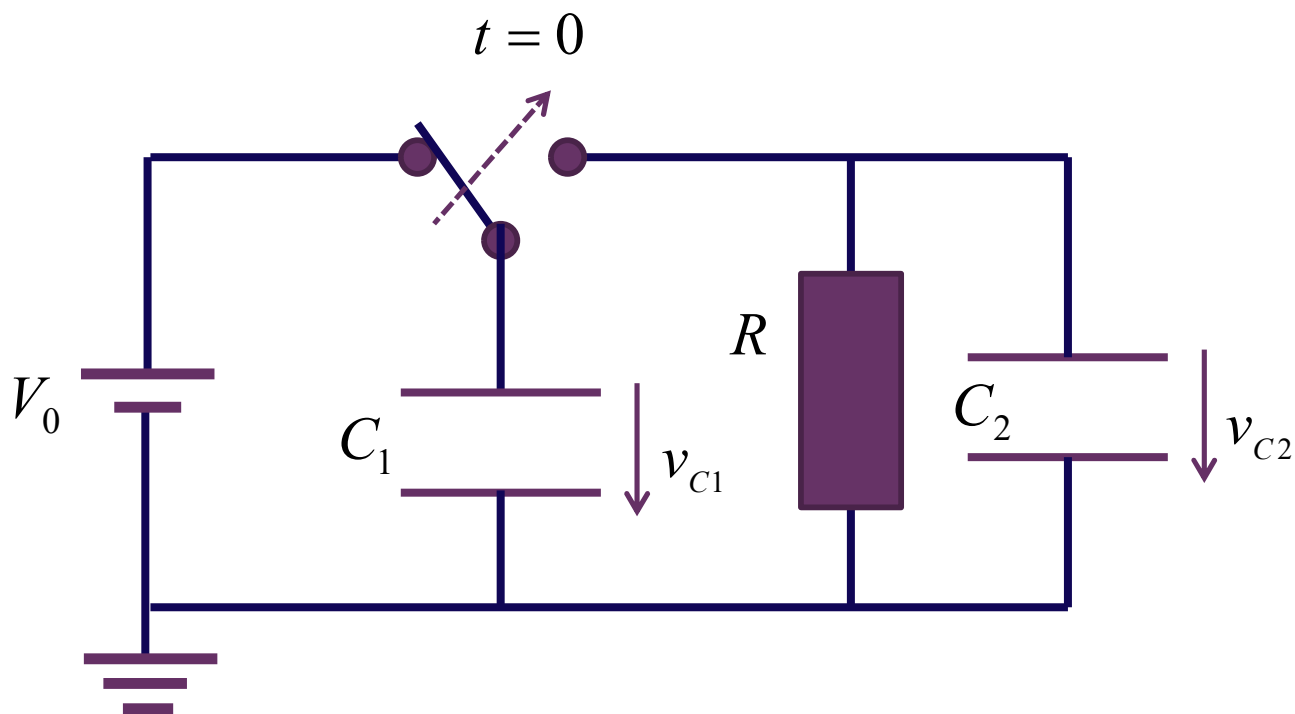
$$g(t) = \left(-g_m R_L + \left(\frac{R_L}{R_{eq}} + g_m R_L \right) e^{-\frac{t}{R_{eq}C}} \right) U(t)$$

$$= -g_m R_L \left(1 - e^{-\frac{t}{R_{eq}C}} \right) U(t) + \frac{R_L}{R_{eq}} e^{-\frac{t}{R_{eq}C}} U(t)$$



作业4.4 电容电压出现跳变

- 在 $t=0$ 时刻，将开关拨向右侧电路，求电容 C_1 、 C_2 两端电压变化规律，写出表达式，画出时域波形

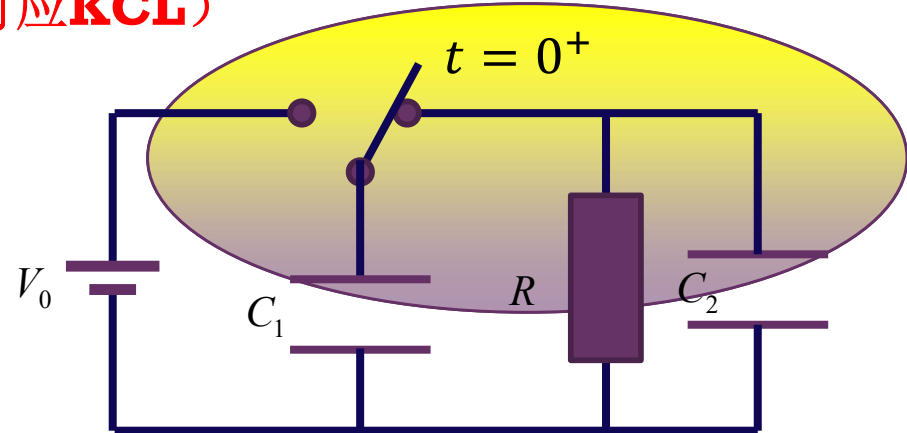
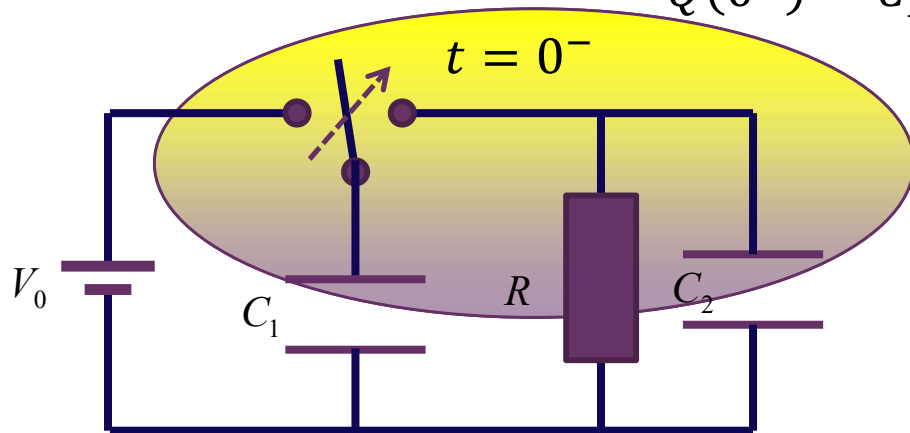


跳变初值的确定

$$Q(0^-) = C_1 V_0$$

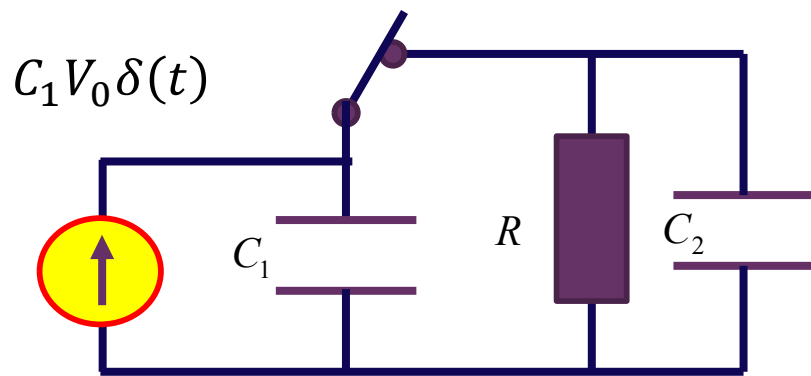
电荷守恒定律
(对应**KCL**)

$$Q(0^+) = (C_1 + C_2)v_o(0^+) = Q(0^-) = C_1 V_0$$



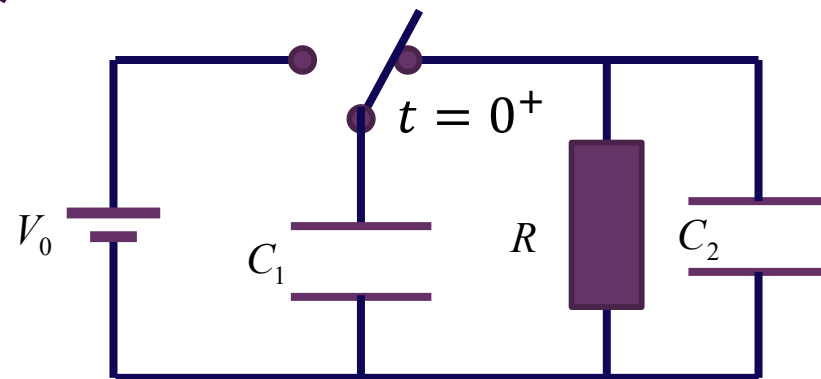
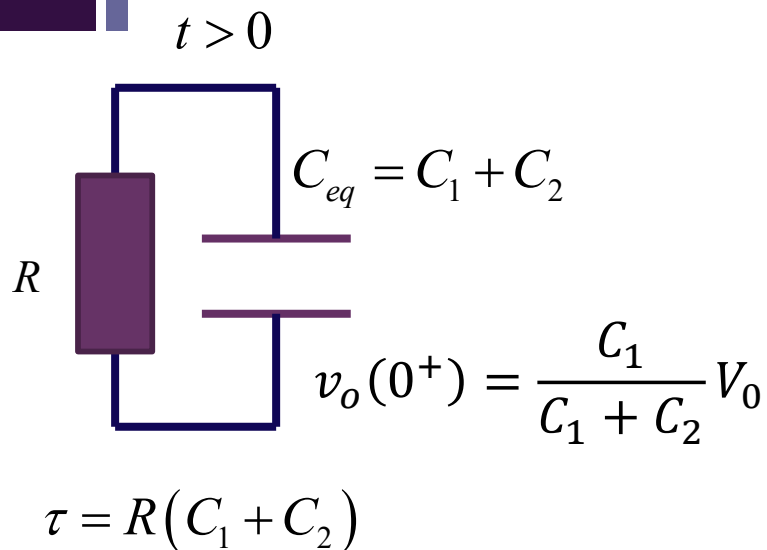
$$v_o(0^+) = \frac{C_1}{C_1 + C_2} V_0$$

也可用等效电路法

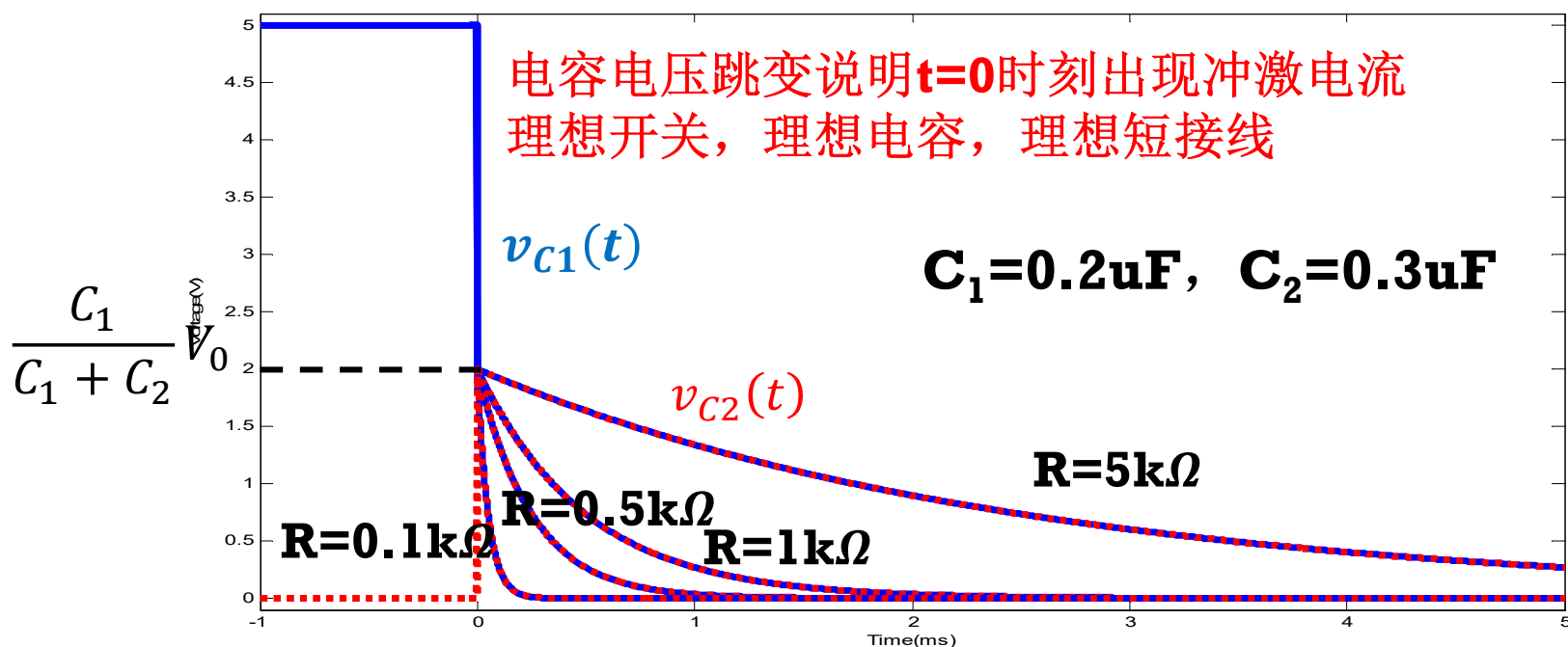


$$\begin{aligned} v_o(0^+) &= \frac{1}{C_1 + C_2} \int_{0^-}^{0^+} C_1 V_0 \delta(t) dt \\ &= \frac{C_1 V_0}{C_1 + C_2} \int_{0^-}^{0^+} \delta(t) dt = \frac{C_1 V_0}{C_1 + C_2} \end{aligned}$$

其后就是简单的电容放电

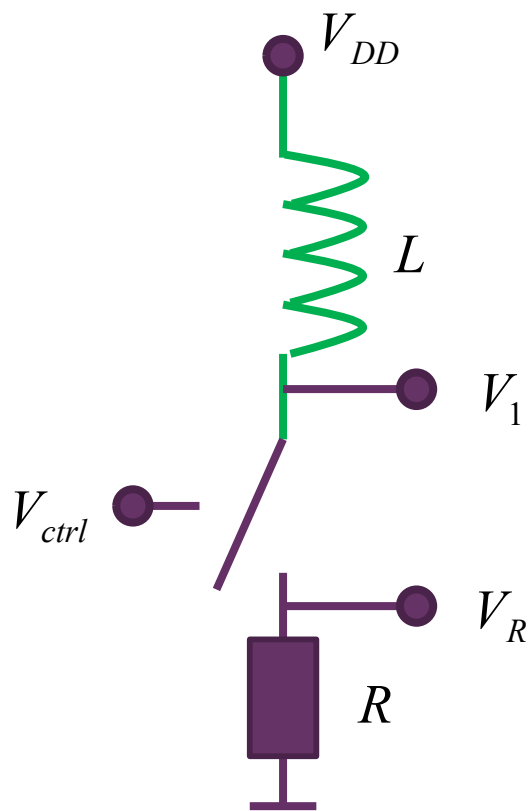


$$v_{C1}(t) = v_{C2}(t) = \frac{C_1}{C_1 + C_2} V_0 e^{-\frac{t}{\tau}} \quad (t > 0)$$



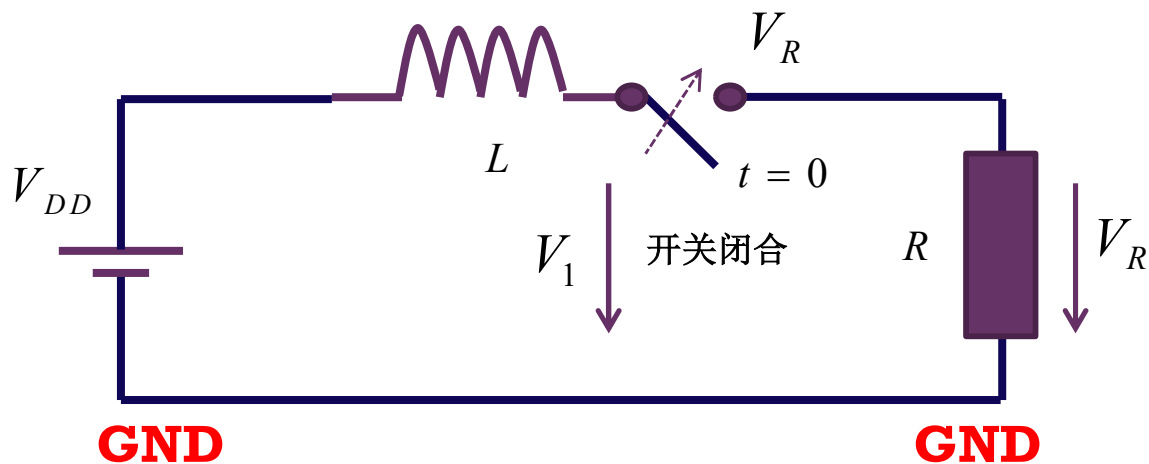
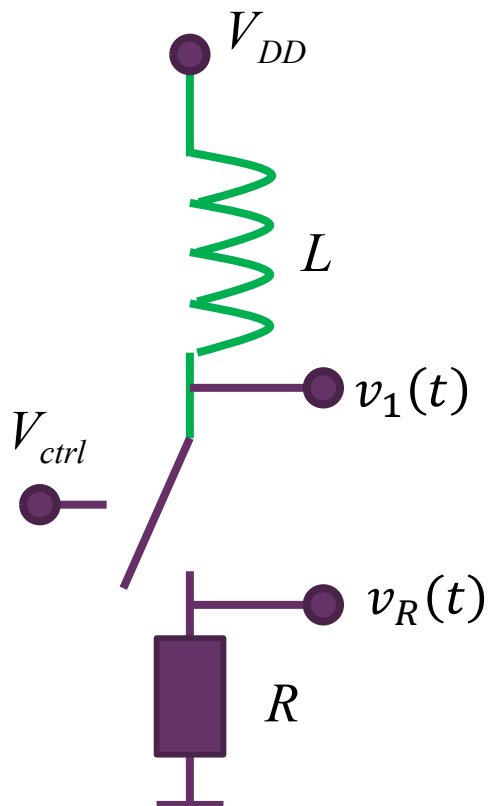
电容两端电压波形

作业5.2 电火花产生



- 这是一个继电器等效电路，晶体管开关可以接通电路，为负载电阻供电
 - 假设开关是理想开关
- 请分析开关闭合瞬间，负载电阻上的电压变化情况
- 请分析开关断开瞬间，开关两端电压变化情况
 - 机械开关则产生电火花，晶体管开关则击穿，给出解决方案

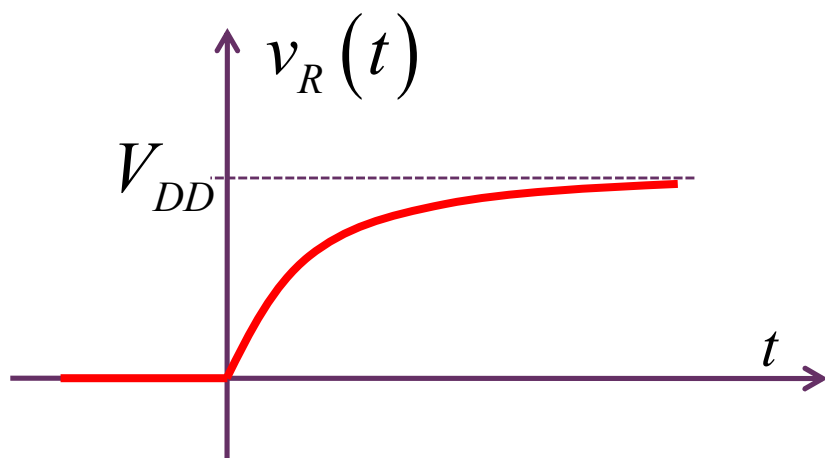
开关闭合



开关闭合：三要素法

$$i_L(0^+) = i_L(0^-) = 0 \quad i_{L\infty}(t) = \frac{V_{DD}}{R} \quad \tau = GL = \frac{L}{R}$$

$$i_L(t) = i_{L\infty}(t) + (i_L(0^+) - i_{L\infty}(0^+))e^{-\frac{t}{\tau}} \quad (t > 0)$$

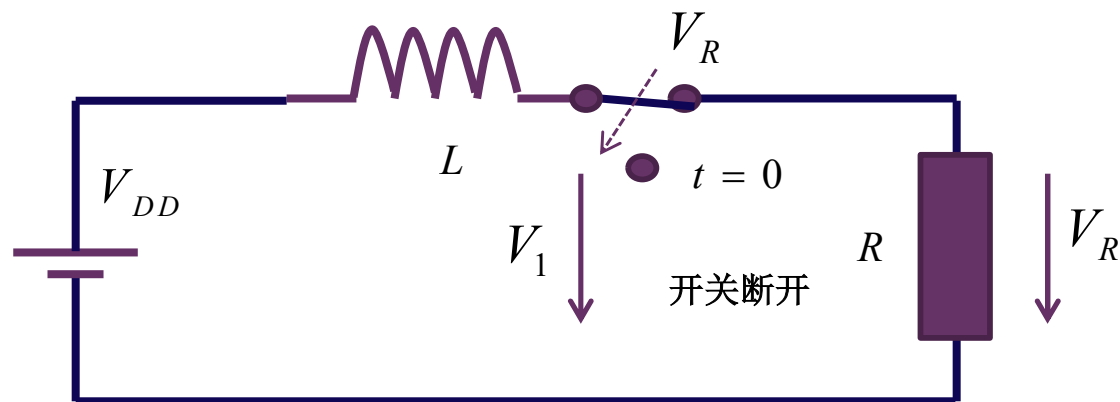
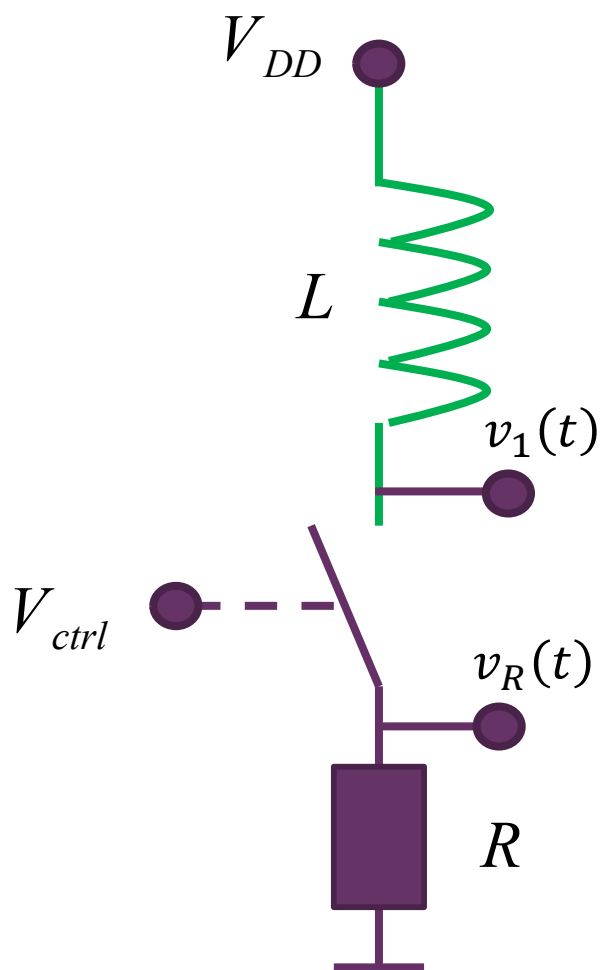


$$i_L(t) = \frac{V_{DD}}{R} \left(1 - e^{-\frac{t}{GL}} \right) U(t)$$

$$v_R(t) = i_L(t)R = V_{DD} \left(1 - e^{-\frac{t}{GL}} \right) U(t)$$

一阶低通系统：通直流阻交流

开关断开



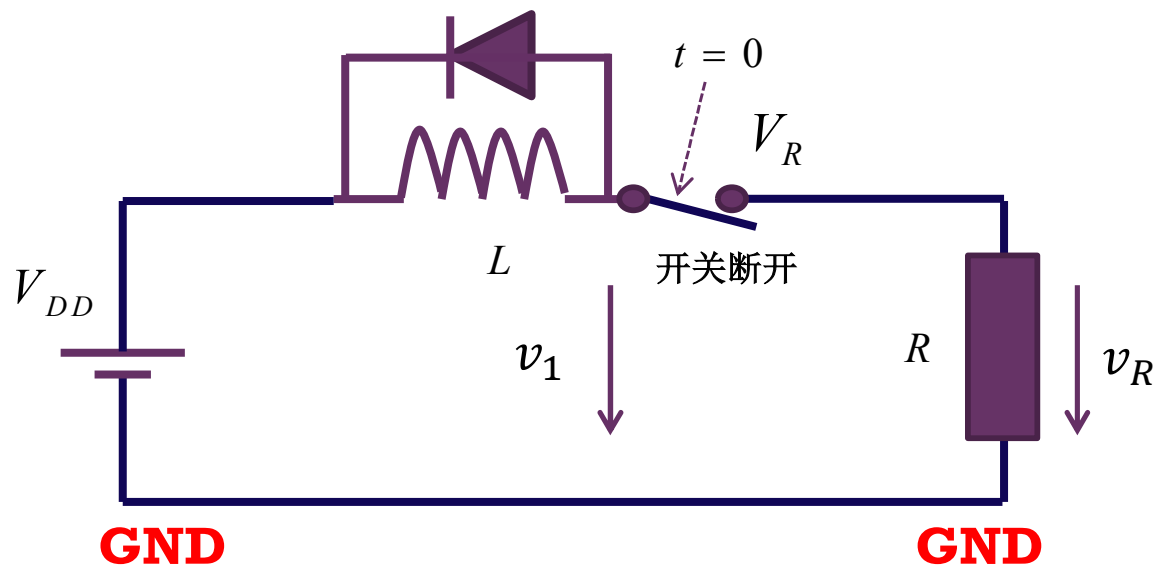
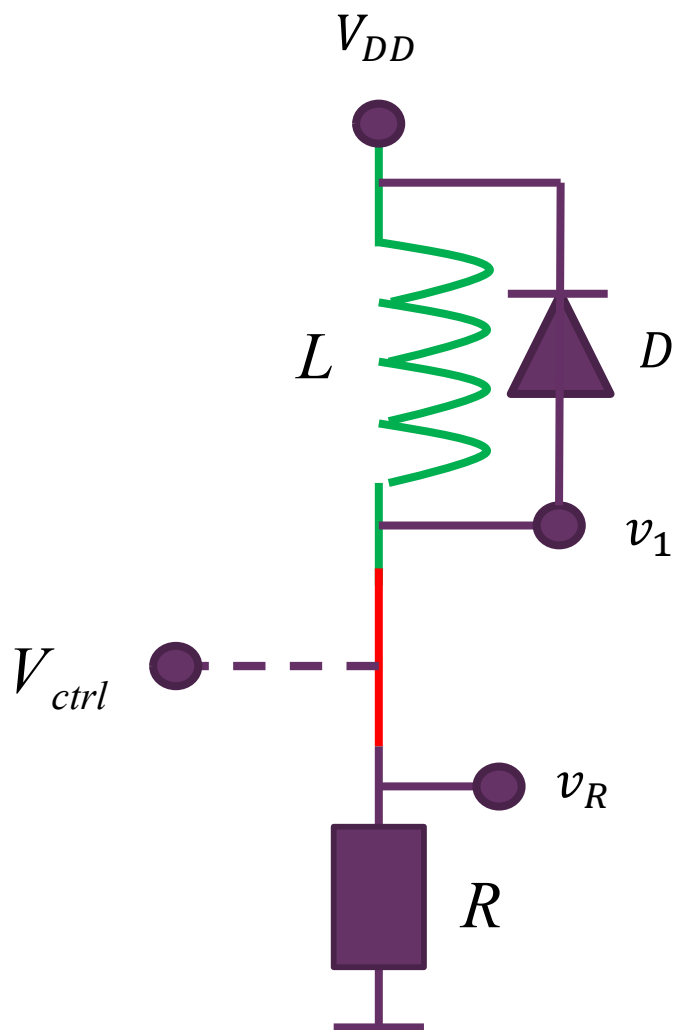
$$i_L(0^-) = \frac{V_{DD}}{R} \quad i_L(0^+) = 0 \quad \text{强行断流}$$

$$v_1(0^-) = V_{DD}, \quad v_1(0^+) = V_{DD} - L \frac{di_L(0^+)}{dt} = V_{DD} + L \frac{V_{DD}}{R} \delta(t)$$

开关断开瞬间，开关上端点产生冲激电压

晶体管开关瞬间击穿损毁
(机械开关，空气击穿，产生电火花)

续流二极管



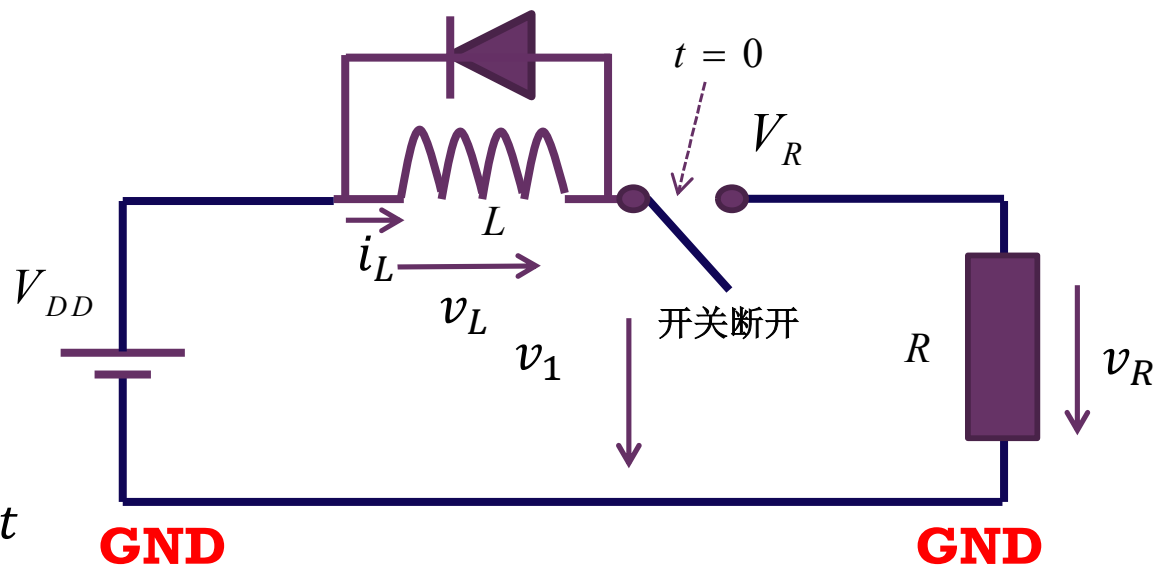
加续流二极管后断开

$$i_L(0^-) = \frac{V_{DD}}{R}$$

$$i_L(0^+) = i_L(0^-) = \frac{V_{DD}}{R}$$

$$v_1(0^-) = V_{DD} \quad v_1(0^+) = V_{DD} + 0.7$$

恒压放磁



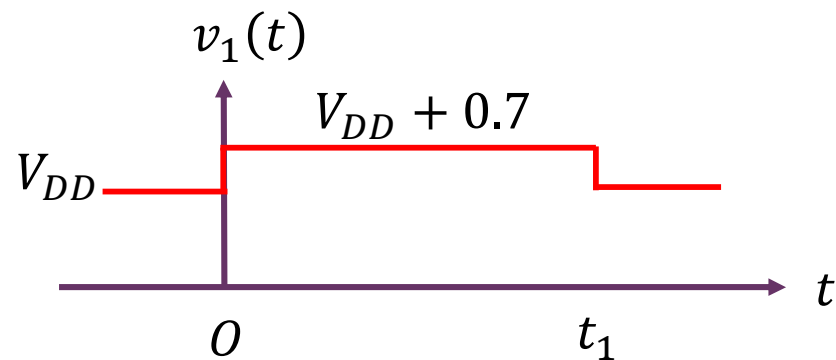
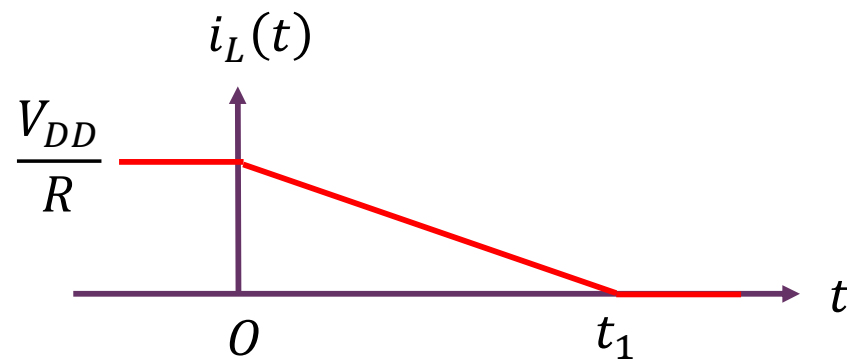
$$i_L(t) = i_L(0) + \frac{1}{L} \int_0^t v_L(t) dt$$

$$= \frac{V_{DD}}{R} + \frac{1}{L} \int_0^t (-0.7) dt = \frac{V_{DD}}{R} - \frac{V_{Don}}{L} t$$

$$i_L(t_1) = \frac{V_{DD}}{R} - \frac{V_{Don}}{L} t_1 = 0$$

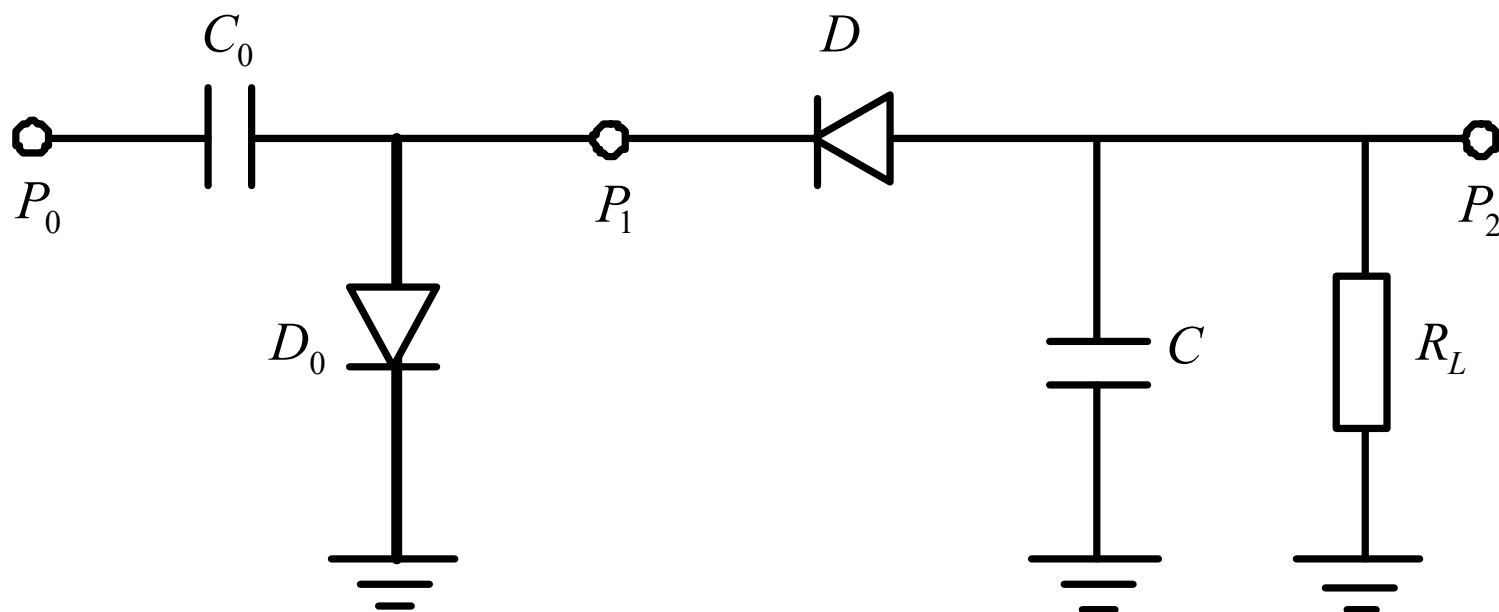
$$t_1 = \frac{V_{DD}}{V_{Don}} \frac{L}{R} = \frac{V_{DD}}{V_{Don}} \tau$$

$$v_1(t) = \begin{cases} V_{DD} & t < 0 \\ V_{DD} + 0.7 & 0 < t < t_1 \\ V_{DD} & t > t_1 \end{cases}$$

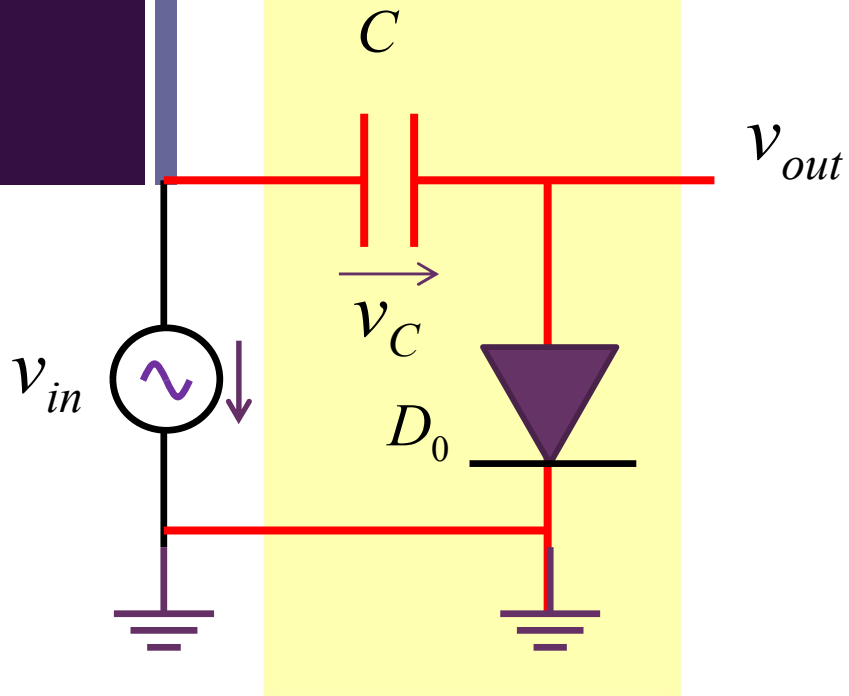


作业5.1 电路功能分析

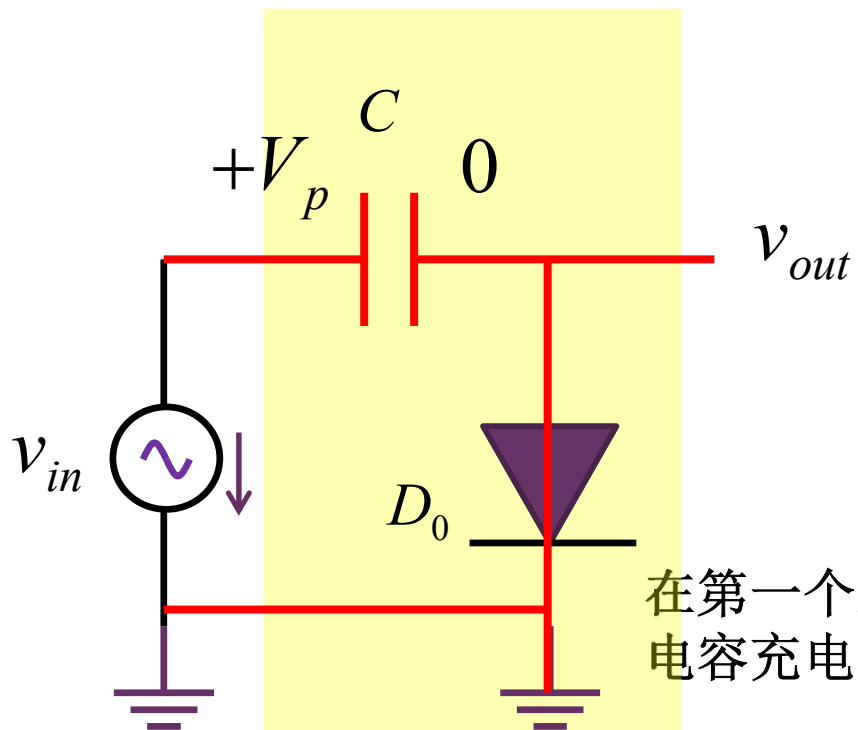
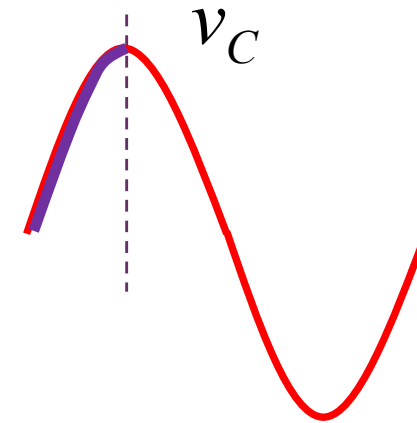
- 如图所示电路是否有错误，如果有，如何修正，修正后完成什么功能？如果没有错误，它可完成什么功能？



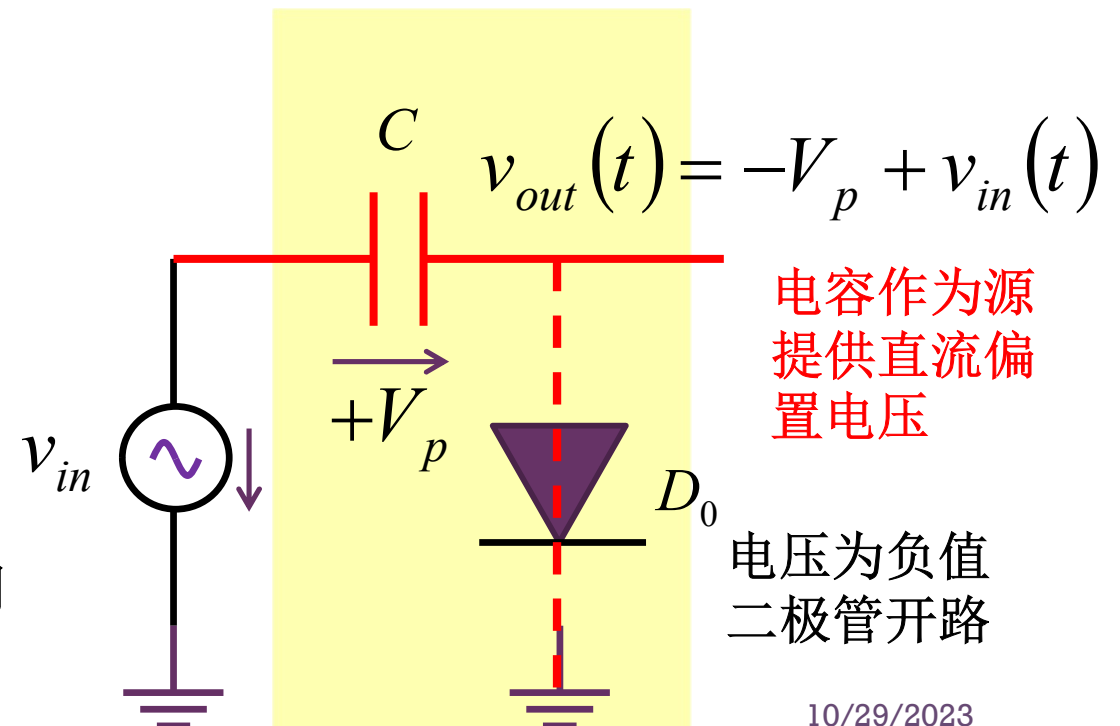
嵌位器电路



电容充电



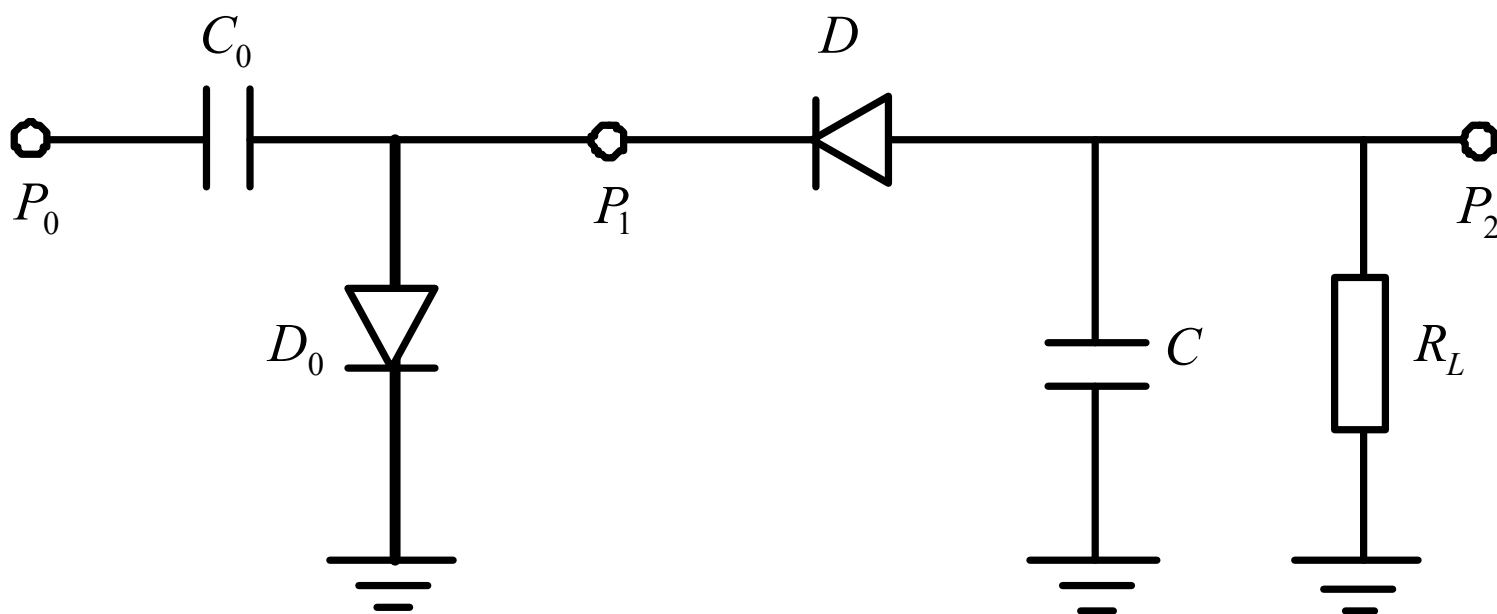
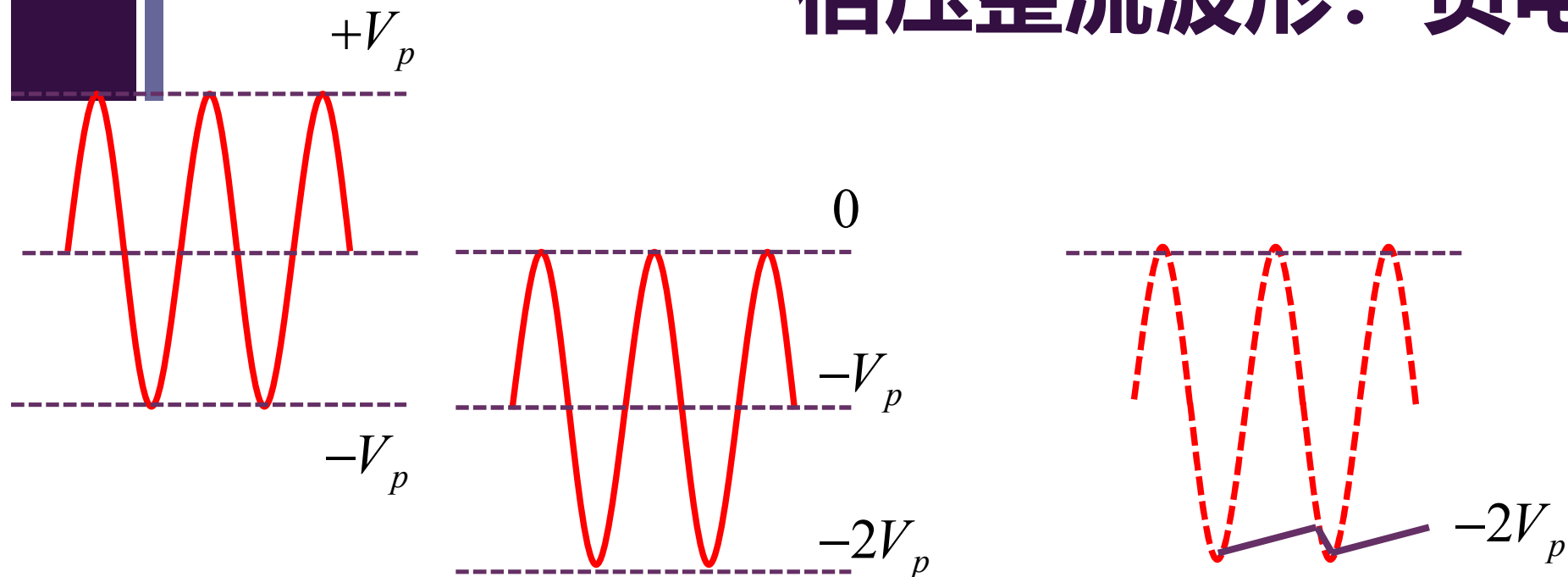
在第一个正半周
电容充电



电容作为源
提供直流偏
置电压

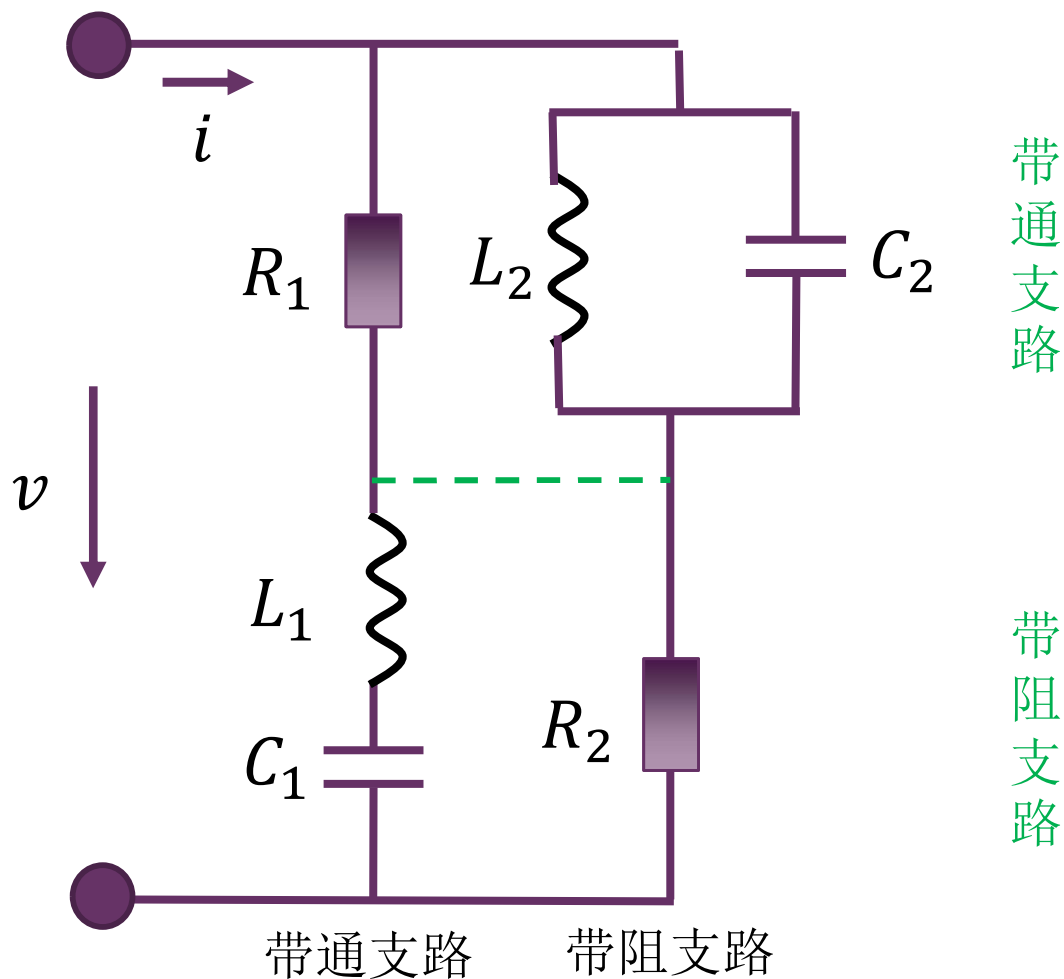
电压为负值
二极管开路

倍压整流波形：负电压



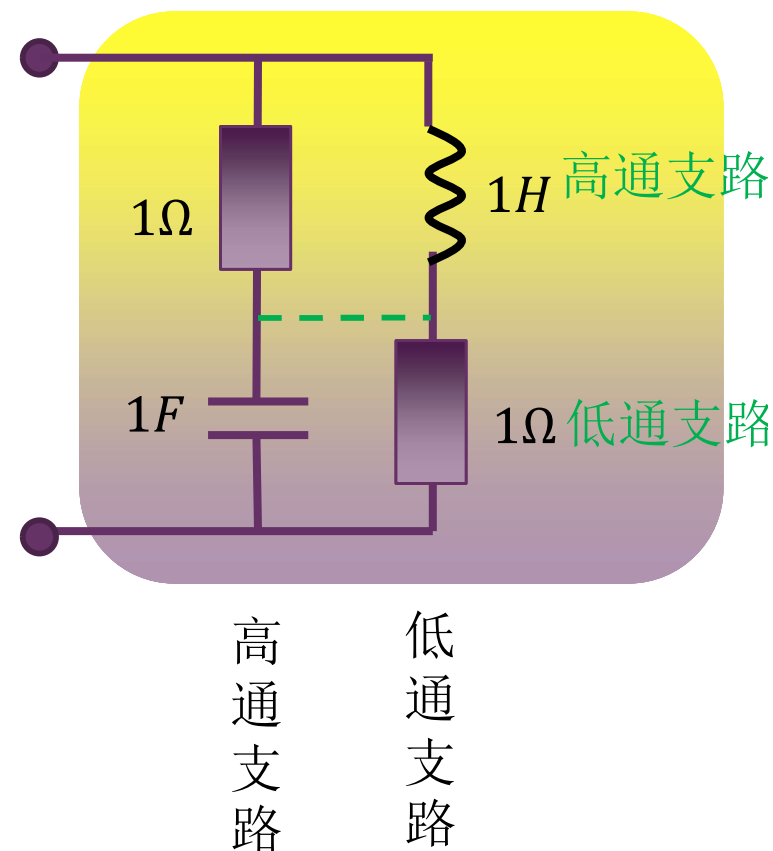
第07讲 有源滤波 作业

作业1 带通带阻互补为直通



$$R \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2} + R \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = R$$

二阶带通和二阶带阻互补



低通+高通=直通

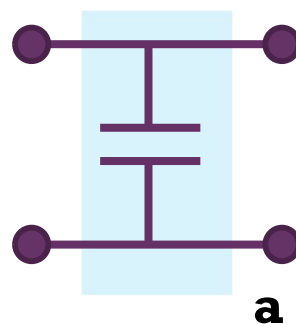
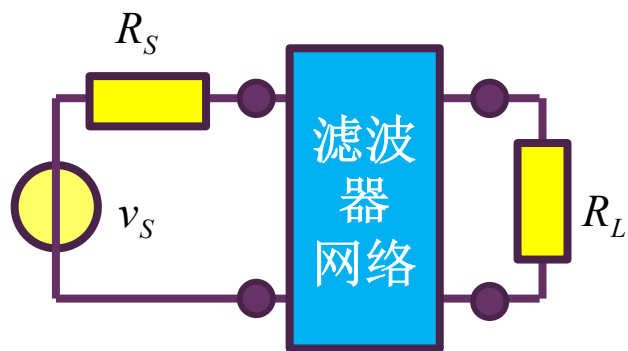
$$R \frac{\omega_0}{s + \omega_0} + R \frac{s}{s + \omega_0} = R$$

一阶低通和一阶高通互补

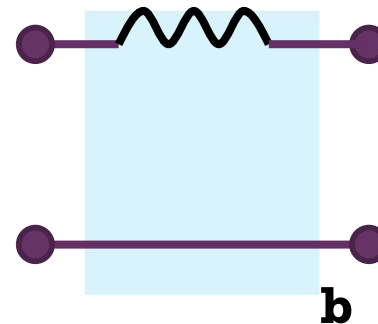
- 求电路中6个元件值满足什么关系时，总端口看入阻抗为纯阻？

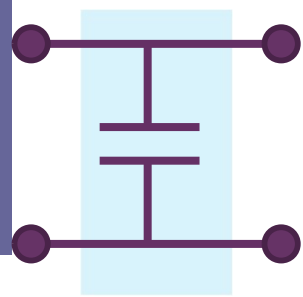
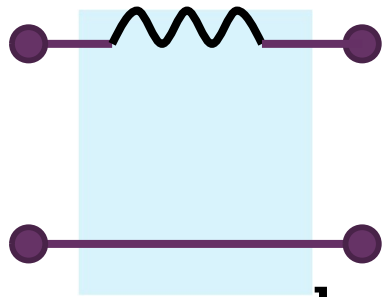
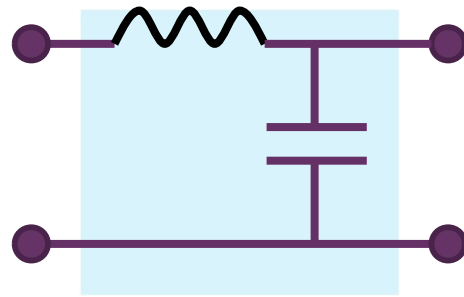
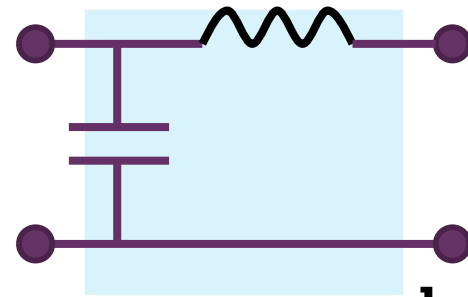
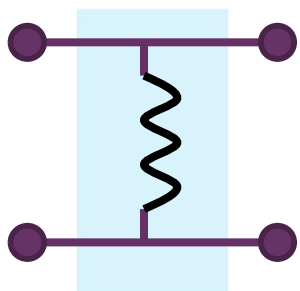
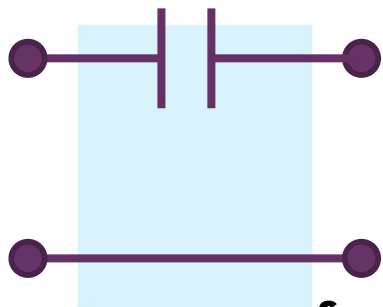
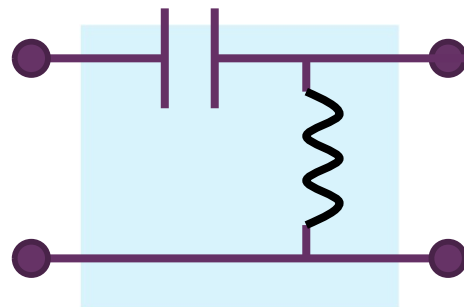
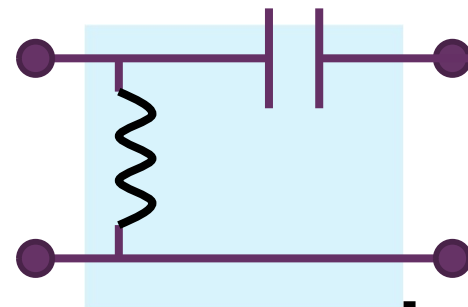
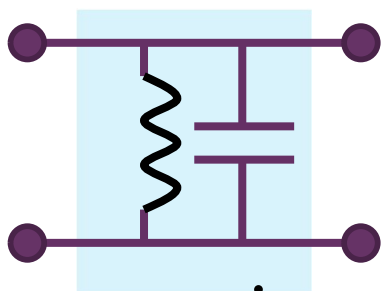
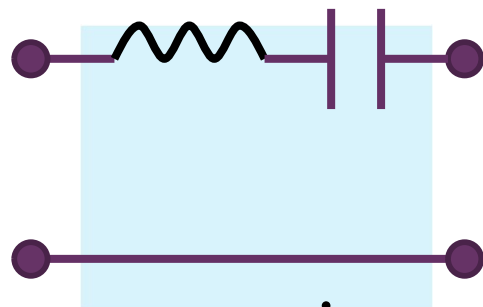
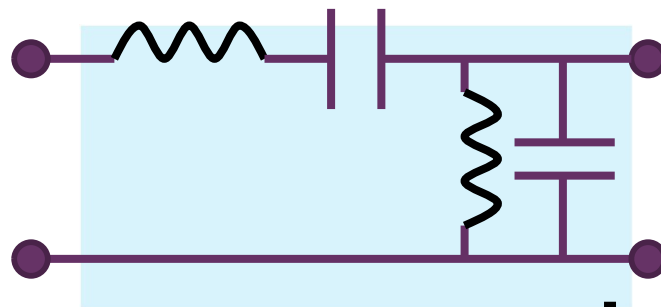
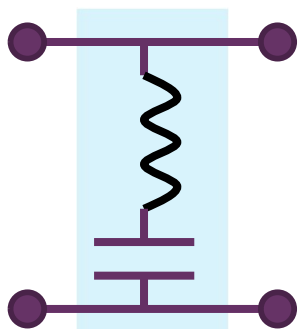
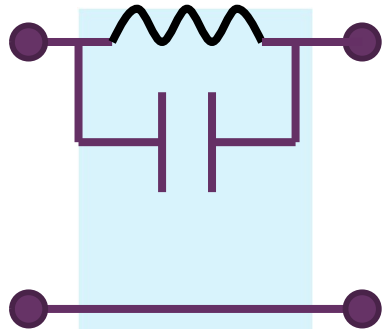
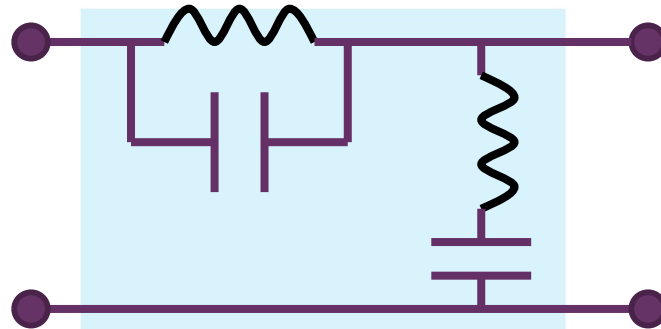
作业2 滤波器类型判定

- 电容和电感的记忆能力或者积分效应，导致时域上的延时和频域上的选频特性
- 常见滤波器分类
 - 低通、高通、带通、带阻
 - 请给出正确的滤波器分类



一阶低通



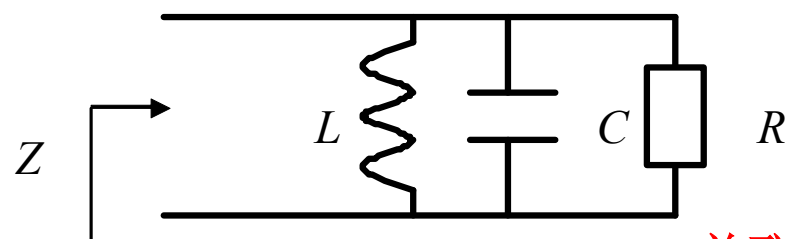
**a****b****c****d****e****f****g****h****i****j****k****l****m****n**

作业3 RLC串联谐振和并联谐振腔阻抗

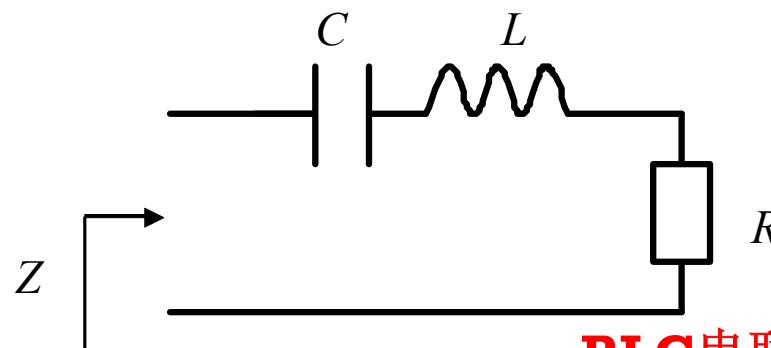
- 研究教材P655页“习题8.2”，仿照该习题，
 - 写出RLC并联谐振回路和RLC串联谐振回路的端口输入阻抗表达式
 - 自编matlab代码，画出端口输入电阻、输入电抗、输入阻抗幅度、输入阻抗相位随频率变化的特性曲线
 - 取 $Q=5, 0.5, 0.05$ 三种情况

$$Z(j\omega) = R(\omega) + jX(\omega)$$

$$= |Z(\omega)|e^{j\varphi(\omega)}$$



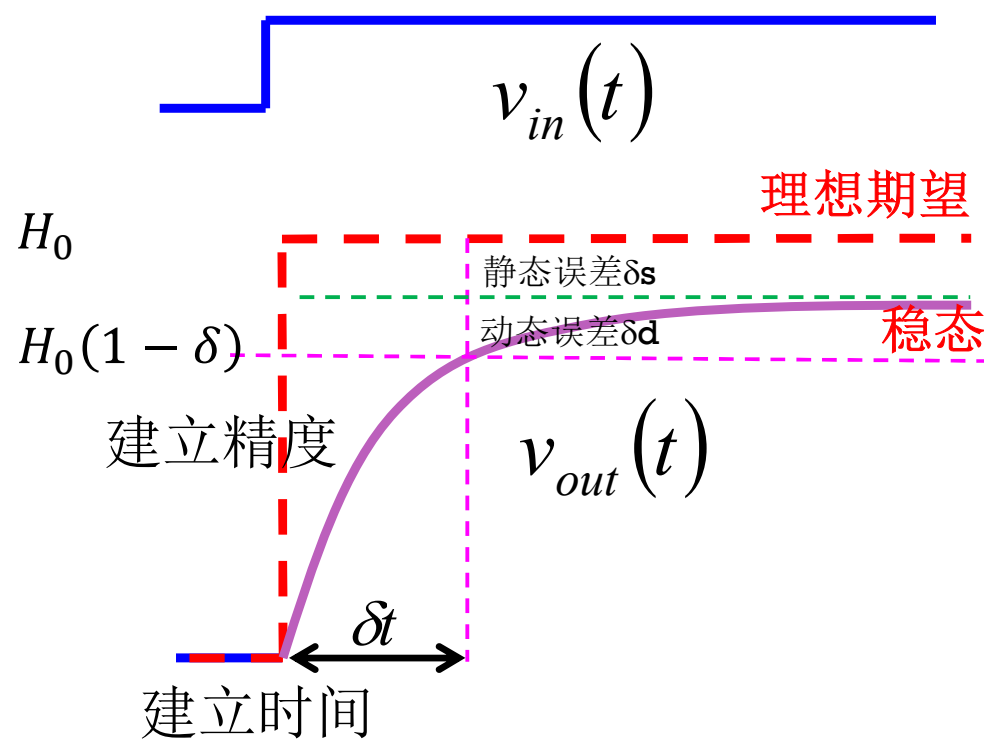
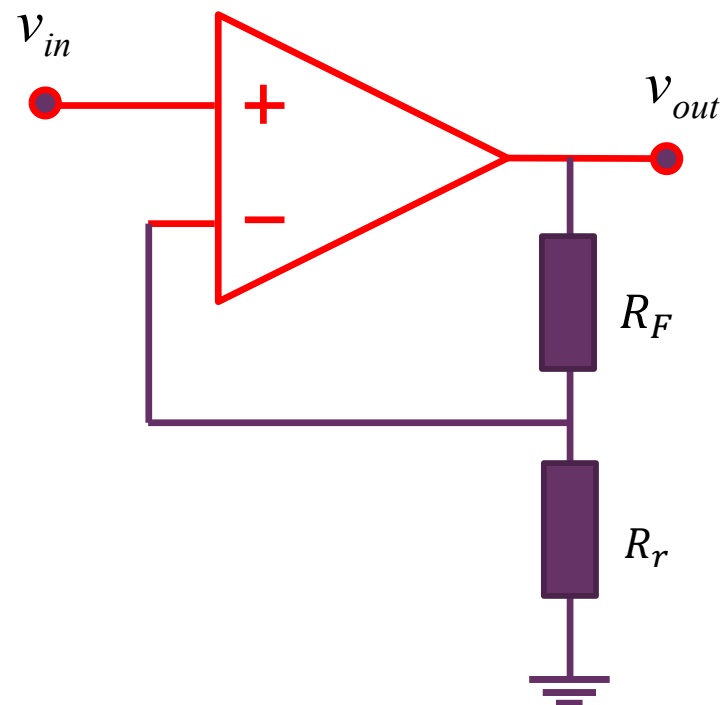
RLC并联谐振回路



RLC串联谐振回路

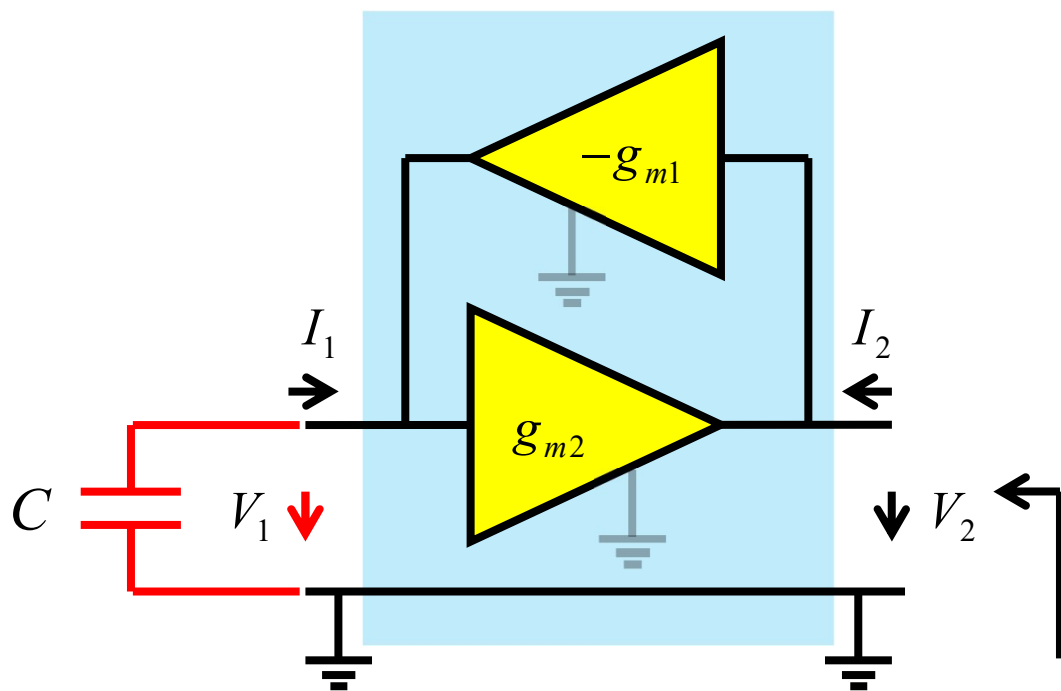
作业4 一阶运放

- (1) 图示同相放大电路中，假设运放为理想运放，请分析确认同相放大倍数表达式， $H(s) = \frac{V_{out}}{V_{in}} = H_0 = \dots$
- (2) 实际运放有很多非理想因素，这里只考虑运放增益非无穷大，运放带宽非无穷大两个非理想效应，并假设运放本身是一阶系统， $A(s) = \frac{A_0}{1 + \frac{s}{\omega_0}}$ （对于理想运放， $A_0 \rightarrow \infty$ ， $\omega \rightarrow \infty$ ），请分析图示同相电压放大器，说明放大器是一阶低通系统，请确认该一阶低通系统的增益带宽积就是运放的增益带宽积
- 增益带宽积为低通系统的增益和3dB带宽之积，例如上述一阶运放的增益带宽积为 $GBW = A_0 \cdot BW_{3dB} = A_0 \frac{\omega_0}{2\pi}$
- 对该一阶系统施加阶跃激励，希望在1ns的建立时间内，建立精度（误差）小于1%，在理论设计增益 H_0 取值为10的假设下，对一阶运放的增益 A_0 和增益带宽积GBW有何要求？

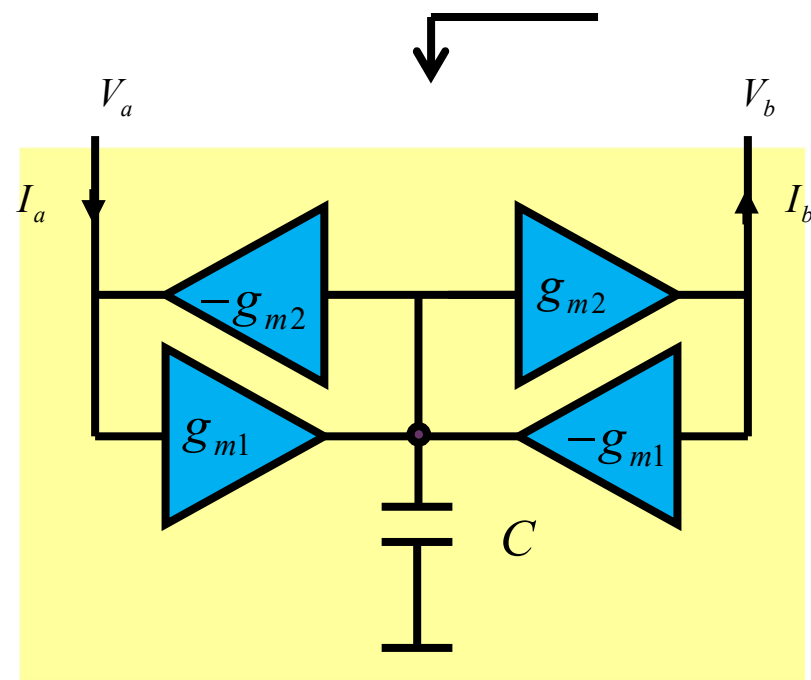


作业5 用回旋器和电容实现电感功能

- 请给出从单端口看入的等效电感感值



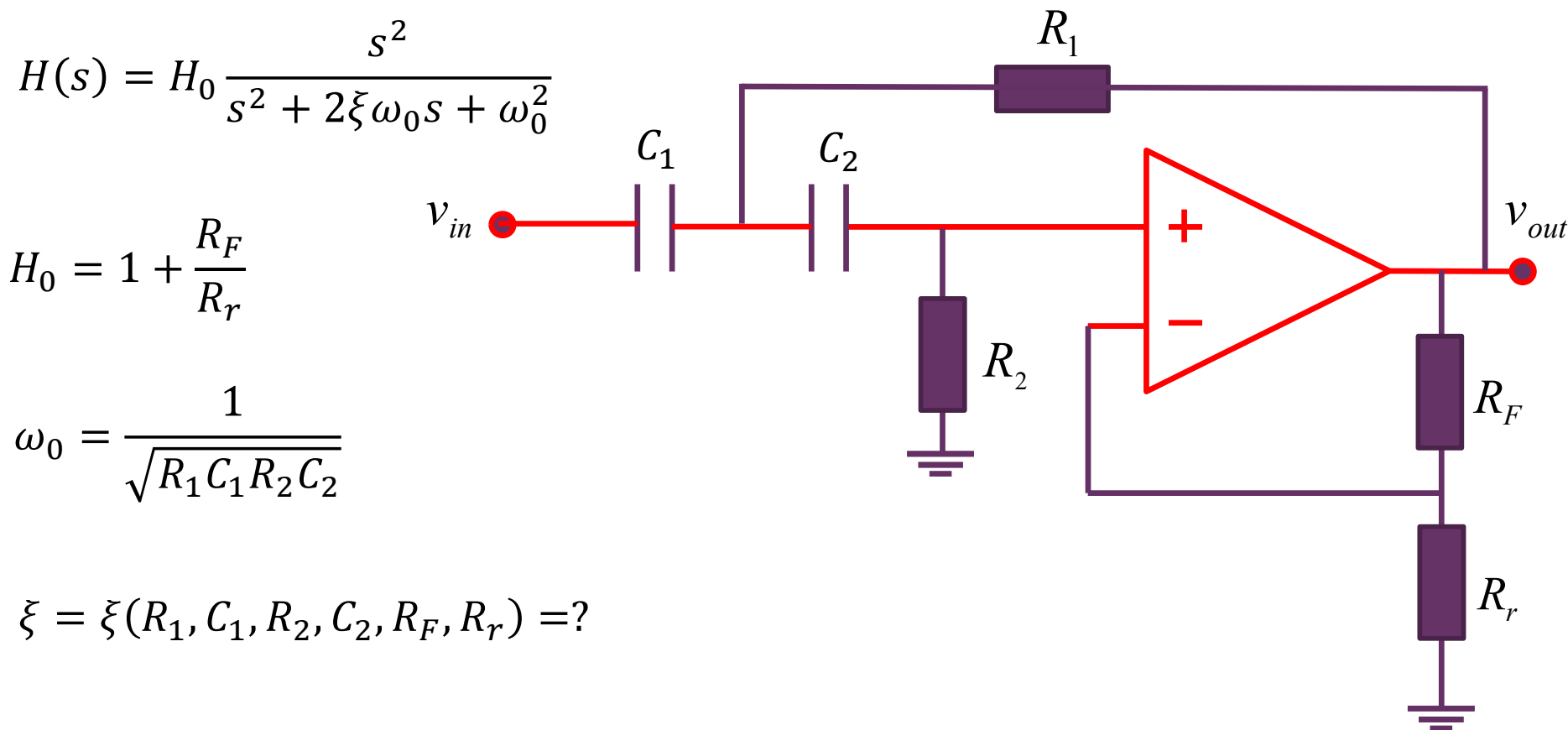
单端接地电感



悬浮电感

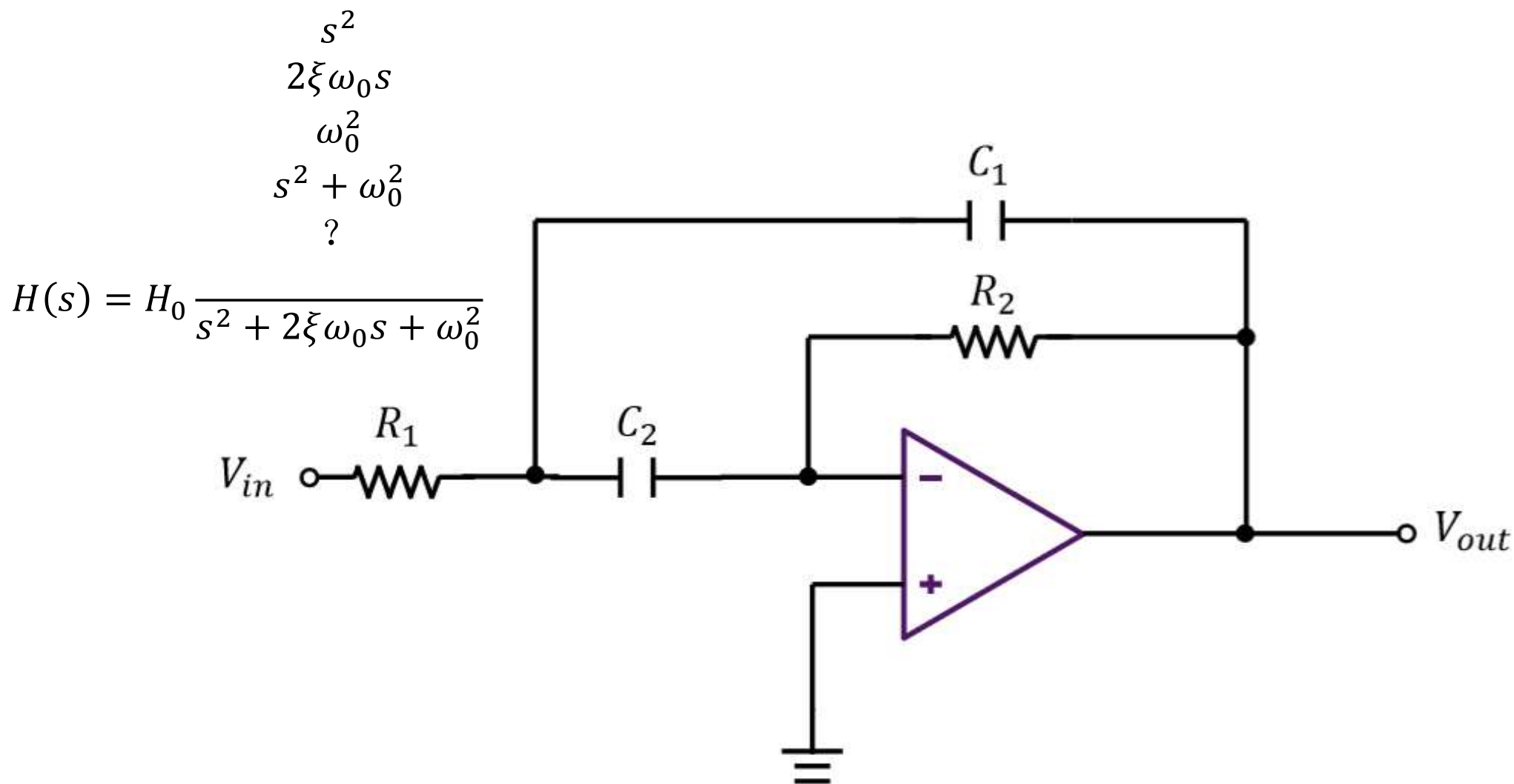
作业6 二阶有源RC高通滤波电路分析

确认该滤波器的阻尼系数表达式，请问 R_F 和 R_r 如何设置可使得阻尼系数为**0.707**



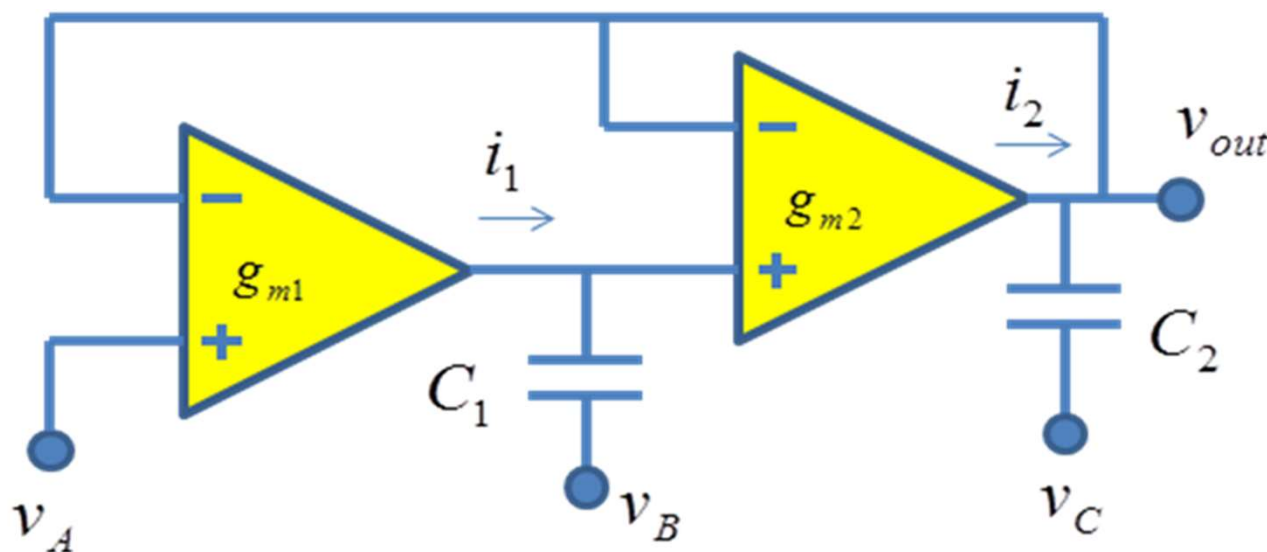
作业7 确认滤波器类型

- 请给出图示有源RC滤波电路的传递函数，确认其滤波类型



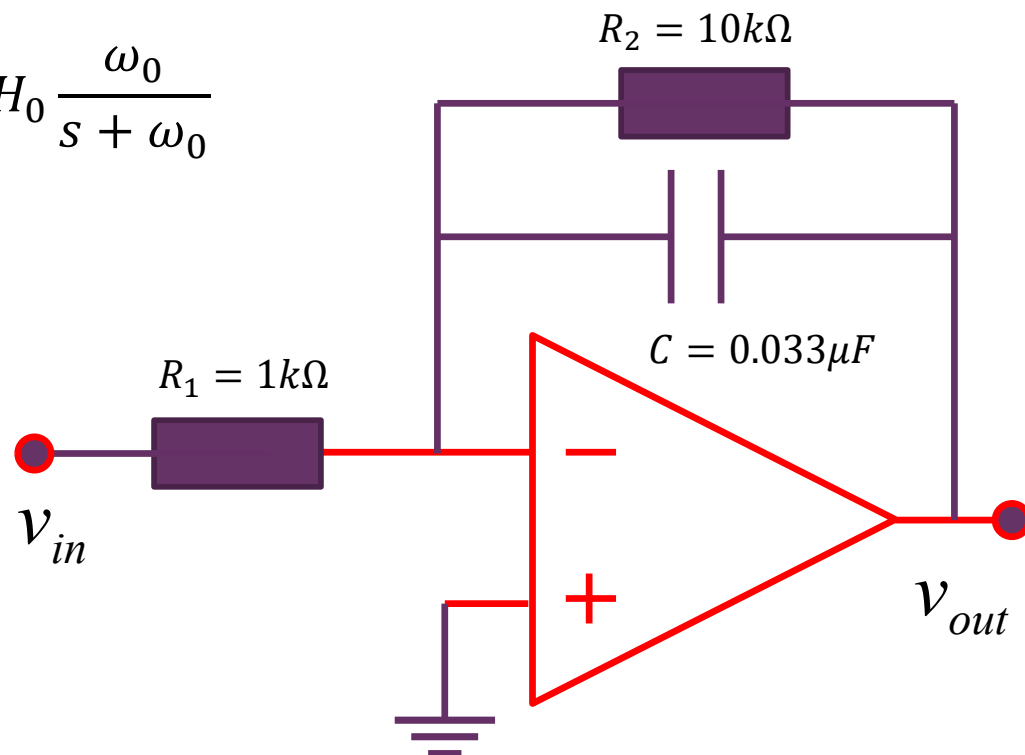
作业8 Gm-C二阶节

- 如图所示，这是一个三输入（ v_A 、 v_B 、 v_C ）单输出（ v_{out} ）的二阶Gm-C滤波器，其中，两个理想跨导器的跨导值为 g_{m1} 和 g_{m2}
- 请写出输出 v_{out} 和三个输入之间的传递函数
- 判断三个输入电压到输出电压之间是什么类型的滤波器，给出二阶滤波器的自由振荡频率，阻尼系数参量



$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = H_0 \frac{\omega_0}{s + \omega_0}$$

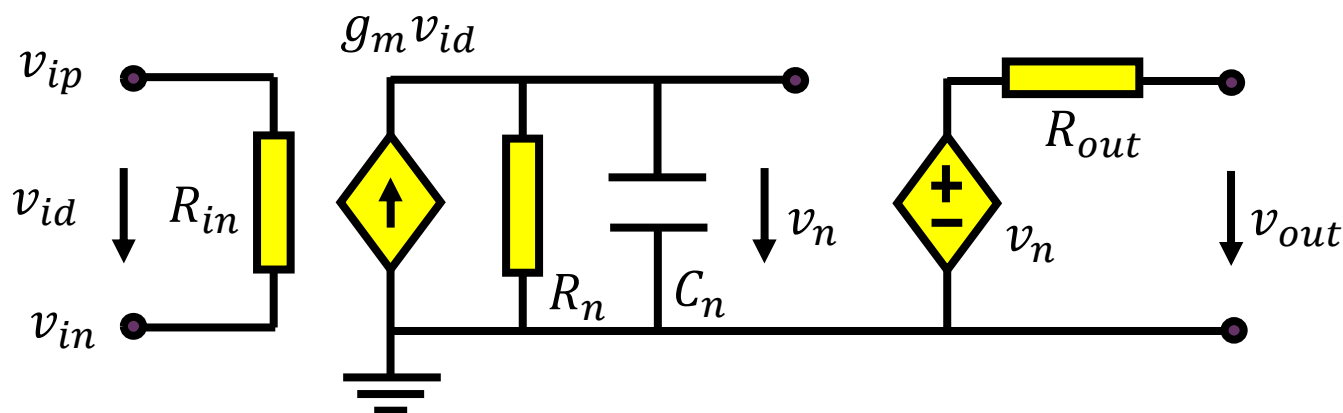
- 如果运放是理想运放，则可实现期望的一阶低通滤波特性，但是实际的工作在线性区的运放存在输入电阻、输出电阻，有限增益，有限带宽，可以建立如图所示的宏模型，请扫描运放宏模型的输入电阻 R_{in} 、输出电阻 R_{out} 、增益（通过扫描 R_n 扫描电压增益）、增益带宽积（通过扫描 C_n 扫描运放的增益带宽积），说明这些值取多大时，设计的一阶滤波器的增益 H_0 和带宽 BW_{3dB} 改变不超过1%



- 如是在选择运放时，可以选用价格最低的商用运放实现期望的滤波特性
- 如是在设计运放时，可给出符合要求的低成本实现方案的运放设计方案
- 系统设计前应做必要的系统宏模型分析

$$A_{v0} = g_m R_n$$

$$GBW = A_{v0} BW_{3dB} = g_m R_n \frac{1}{2\pi R_n C_n} = \frac{g_m}{2\pi C_n}$$



下讲内容在教材中的章节位置 预习用

- P634: 最大功率传输匹配
- P588: 互感变压器
- P817: 互感变压器阻抗匹配网络
- P828: LC阻抗匹配网络
- P836: 阻抗变换原理